AMU

APPLIED II MODULE

Content **page no.**

Chapter One: Sequence and Series 2

1.1 Definition and type of Sequences 2
1.2 Types of Sequences 6
1.3 Convergence properties of Sequences 8
1.4 Infinite series 19
1.5 Convergence and Divergence properties of Infinite Series. 21
1.6 Convergence and Divergence Tests 26
1.7 Alternating Series and Alternating Series Test 46
1.8 Generalized convergence tests 57

Chapter Two: Power series 58

2.1 Definition of power series at any point. 58
2.2 Convergence and divergence, radius and interval of convergence 59
2.3 Representations of Functions as Power Series 64
2.4 Differentiation and integration of power series 66
2.5 Taylor series; Taylor polynomial and application 70

Chapter Three: Orthogonal functions and Fourier series 76

3.1 Periodic Function 76
3.2 Even and Odd Functions 77
3.3 Orthogonal Functions 78
3.4 Fourier sine series and Fourier cosine series 82
3.5 Fourier series 96

Chapter Four: Differential calculus of Functions of several variables 102

4.1 Notations, Examples, Level curves and Graphs 102
4.2 Limit and Continuity 106
4.3 Partial Derivatives, Higher order partial derivatives 112
4.4 Directional derivatives and gradient 116
4.5 Total differential and Tangent planes 120
4.6 Application, Tangent plane approximation of values of a function 124
4.7 The Chain Rule and Implicit differentiation 125
4.8 Relative extreme of function of two variables 131
4.9 Largest and smallest values of a function on a given set 137
4.10 Extreme Values under constraint Condition: Lagrange's multiplier 144

Chapter Five: Multiple Integral 151

5.1 Double Integrals and their Evaluation by Iterated integrals 151
5.2 Double Integrals in Polar coordinates 158
5.3 Surface Area 162
5.4 Triple Integrals in Cartesian, Cylindrical and Spherical Coordinates 164

Chapter One: Sequence and Series

1.1 Definition and type of Sequences

Let us start off this section with a discussion of just what a sequence is. A sequence is nothing more than a list of numbers written in a specific order. The list may or may not have an infinite number of terms in them although we will be dealing exclusively with infinite sequences in this class.

A sequence can be thought of as a list of numbers written in a definite order:

$a_1, a_2, a_3, a_4, ...a_n, ...$

The numbers

a_1 is called the first term,

a_2 is the second term,

.

.

.

a_n is the nth term.

We will deal exclusively with infinite sequences and so each term a_n will have a successor a_{n+1}.

Notice that for every positive integer n there is a corresponding number a_n and so a Sequence can be defined as a function whose domain is the set of positive integers. But we usually write a_n instead of the function notation $f(n)$ for the value of the function at the number n.

Notation: The sequence $a_1, a_2, a_3, a_4, ...a_n, ...$ is also denoted by

$$\{a_n\} \quad or \quad \{a_n\}_{n=1}^{\infty}$$

A Sequence is like a set, except:

- the terms are in order (with Sets the order does not matter)
- the same value can appear many times (only once in Sets)

Example 1: {0, 1, 0, 1, 0, 1 ...} is the sequence of alternating 0s and 1s.

The first term is 0, the second term is 1, the third term is 0, and so on, where as the set would be just {0, 1}

A Sequence usually has a Rule, which is a way to find the value of each term.

Example 2: the sequence {3, 5, 7, 9 ...} starts at 3 and jumps 2 every time:

As a formula, Saying "*starts at 3 and jumps 2 every time*" is fine, but it doesn't help us to calculate the:

- 10th term,
- 100th term, or
- n^{th} term, where n could be any term number we want.

So, we want a formula with "n" in it (where n is any term number).

So, What Would A Rule For {3, 5, 7, 9 ...} Be?

Firstly, we can see the sequence goes up 2 every time, so we can guess that a Rule will be something like "2 times n" (where "n" is the term number). Let's test it out:

Test Rule: 2n

n	Term	Test Rule
1	3	$2n = 2\times1 = 2$
2	5	$2n = 2\times2 = 4$
3	7	$2n = 2\times3 = 6$

That nearly worked ... but it is too low by 1 every time, so let us try changing it to:

Test Rule: 2n+1

n	Term	Test Rule
1	3	2n+1 = 2×1 + 1 = 3
2	5	2n+1 = 2×2 + 1 = 5
3	7	2n+1 = 2×3 + 1 = 7

It Works!

So instead of saying "starts at 3 and jumps 2 every time" we write this:

2n+1

Now we can calculate, for example, the 100th term by inserting 100 in place of n:

I.e. 2 × 100 + 1 = 201

Since mathematics is so powerful we can find more than one Rule that works for any sequence.

Example 3: the sequence {3, 5, 7, 9 ...}

We have just shown a Rule for {3, 5, 7, 9 ...} is: 2n+1

And so we get: {3, 5, 7, 9, 11, 13 ...}

But can we find another rule?

How about "odd numbers without a 1 in them":

And we would get: {3, 5, 7, 9, 23, 25 ...}

This is completely different sequence!

And we could find more rules that match {3, 5, 7, 9 ...}. Really we could.

So it is best to say "A Rule" rather then "The Rule" (unless you know it is the right Rule).

Notation:

To make it easier to use rules, we often use this special style:

- X_n , where $\mathbf{x_n}$ is the term and

  n is the term number

Example 4: to mention the "5th term" you just write: x_5

So a rule for {3, 5, 7, 9 ...} can be written as an equation like this:

$$x_n = 2n+1$$

And to calculate the 10th term we can write:

$$x_{10} = 2n+1 = 2\times10+1 = 21$$

Can you calculate the following?

a) x_{60} b) x_{100} c) x_{150}

Here is another example:

Example 5: Calculate the first four terms of this sequence:

$\{a_n\} = \{(-1/n)^n\}$

Calculations:

- $a_1 = (-1/1)^1 = -1$
- $a_2 = (-1/2)^2 = 1/4$
- $a_3 = (-1/3)^3 = -1/27$
- $a_4 = (-1/4)^4 = 1/256$

Answer: $\{a_n\} = \{ -1, 1/4, -1/27, 1/256, ... \}$

Exercise: Calculate the first five terms of the following sequences

a) $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$ b) $\left\{\frac{(-1)^n(n+1)}{3^n}\right\}_{n=1}^{\infty}$ c) $\left\{\cos\frac{n\pi}{6}\right\}_{n=0}^{\infty}$

1.2 Types of Sequences

Now let's look at some special sequences, and their rules.

1.2.1 Infinite or Finite sequence

If the sequence goes on forever it is called an infinite sequence,
otherwise it is a finite sequence

Examples:

a) {1, 2, 3, 4...} is a very simple sequence (and it is an infinite sequence)

b) {20, 25, 30, 35 ...} is also an infinite sequence

c) {1, 3, 5, 7} is the sequence of the first 4 odd numbers (and is a finite sequence)

d) {4, 3, 2, 1} is 4 to 1 backwards

e) {1, 2, 4, 8, 16, 32 ...} is an infinite sequence where every term doubles

f) {a, b, c, d, e} is the sequence of the first 5 letters alphabetically

g) {f, r, e, d} is the sequence of letters in the name "fred"

h) {0, 1, 0, 1, 0, 1 ...} is the sequence of alternating 0s and 1s (yes they are in order, it is an alternating order in this case)

1.2.2 Arithmetic Sequences

In an Arithmetic sequence the difference between one term and the next is a constant.

In other words, you just add some value each time ... on to infinity.

Example: Consider the sequence 1, 4, 7, 10, 13, 16, 19, 22, 25...

This sequence has a difference of 3 between each number.

Its Rule is $x_n = 3n-2$

In General you could write an arithmetic sequence like this:

$$\{a, a+d, a+2d, a+3d, ...\}$$

Where:

- *a* is the first term, and
- *d* is the difference between the terms (Usually called the "common difference")

And you can make the rule by:

$$x_n = a + d(n-1)$$

(We use "n-1" because **d** is not used in the 1st term).

1.2.3 Geometric Sequence

In a geometric sequence each term is found by multiplying the previous term by a constant.

Example: Consider the sequence 2, 4, 8, 16, 32, 64, 128, 256...

This sequence has a factor of 2 between each number.

Its Rule is $x_n = 2^n$

In General you could write an arithmetic sequence like this:

$$\{a, ar, ar^2, ar^3, ... \}$$

Where:

- a is the first term, and
- *r* is the factor between the terms (called the "common ratio")

Note that: *r* should not be 0.

- Because when $r = 0$, you get the sequence {a,0,0,...} which is not geometric

And the rule is: $x_n = ar^{(n-1)}$

(We use "n-1" because ar^0 is the 1st term)

1.3 Convergence properties of Sequences

For any sequence $\{a_n\}_{n=m}^{\infty}$ we may consider the behavior of a_n as n increases without bound.

What happens for the sequence $a_n = \left\{\frac{n}{n+1}\right\}$ as the value of n increases without bound?

Check by listing few terms. In this case, as the value of n increases without bound, we write it as $\lim_{n\to\infty} \frac{n}{n+1}$. In genera, the notation $\lim_{n\to\infty} a_n = L$ means that the terms of the sequence a_n approach L as n becomes large. Notice that the following definition of the limit of a sequence is very similar to the definition of a limit of a function at infinity.

Definition: Let $\{a_n\}_{n=m}^{\infty}$ be a sequence. A number L is the limit of $\{a_n\}_{n=m}^{\infty}$ if for every $\varepsilon > 0$ there is an integer N such that if $n \geq N$, *then* $|a_n - L| < \varepsilon$.

In this case we write $\lim_{n\to\infty} a_n = L$, if such a number L exists, we say that $\{a_n\}_{n=m}^{\infty}$ converges to L or that $\lim_{n\to\infty} a_n = L$ exists. If such a number L does not exist, we say that $\{a_n\}_{n=m}^{\infty}$ diverges or that $\lim_{n\to\infty} a_n = L$ does not exist.

Note that as limit of functions, the limit of a sequence is unique if it exists.

In order to conclude that weather a given sequence is convergent or divergent it is suffices to compute the limit of the sequence as the value of n increases without bound, and if this limit exists then the sequence converges otherwise diverge.

Example1: Compute $\lim_{n\to\infty} \frac{n}{n+1}$ and conclude the sequence as convergent or divergent.

Solution: The method computing the limit is similar to the one we used in case of finding limit of functions: Divide numerator and denominator by the highest power of n and then use the Limit Laws.

$$\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = \frac{\lim_{n\to\infty} 1}{\lim_{n\to\infty} 1 + \lim_{n\to\infty} \frac{1}{n}}$$

$$= \frac{1}{1+0} = 1$$

Hence the limit of the sequence exists and we conclude that the sequence converges (converges to 1)

Example 2: Check weather the sequence $\left\{\left(1+\frac{1}{n}\right)^n\right\}_{n=1}^{\infty}$ is convergent or divergent

Solution: First let's compute the limit

i.e. $\underset{n\to\infty}{Lim}\left(1+\frac{1}{n}\right)^n$

$$\underset{n\to\infty}{Lim}\left(1+\frac{1}{n}\right)^n = \underset{n\to\infty}{Lim}\, e^{\left(1+\frac{1}{n}\right)n}$$

$$= e^{\underset{n\to\infty}{Lim}\left(1+\frac{1}{n}\right)n} = e^{\underset{n\to\infty}{Lim}\frac{\left(1+\frac{1}{n}\right)}{\frac{1}{n}}}$$

$$= e^1 = e$$

Since the limit of the sequence exists it is convergent, more over it converges to e .

Example 3: Check weather the sequence $\{(-1)^n\}$ is convergent or divergent

Solution: First let's compute the limit

$\underset{n\to\infty}{Lim}\ (-1)^n = 1$ if n is even and -1 if n is odd

i.e. $\underset{n\to\infty}{Lim}\ (-1)^n = \begin{cases} 1 & \text{if n is even} \\ -1 & \text{if n is odd} \end{cases}$

Or if we write out the terms of the sequence, we obtain

$\{-1, 1, -1, 1, -1, 1, -1, \ldots\ldots\ldots\}$

Since the terms oscillate between 1 and -1 infinitely often, $a_n = (-1)^n$ does not approach any number. Thus, $\underset{n\to\infty}{Lim}\, a_n$ does not exist;

that is, the sequence $a_n = (-1)^n$ is divergent.

This implies the limit does not converge to a unique number, hence we conclude that the limit doesn't exist, so this sequence is a divergent sequence.

Exercise: by computing the limit of the following sequences decide weather the given sequence is convergent or divergent

a) $\{-4\}^{2n}$ b) $\left\{\frac{-1}{3}\right\}^{n}$

c) $\left\{\frac{6n^2+1}{4n^2}\right\}$ d) $\left\{1,2,3,\frac{1}{2},\frac{1}{4},\frac{1}{6},\frac{1}{8},.....\frac{1}{2n},...\right\}$

e) Verify that the sequence $\left\{r^n\right\}$ is convergent if $-1<r\leq 1$ and divergent for all other values of r.

In this section we continue our analysis of the convergence and divergence of sequences. Since sequences are functions, we may add, subtract, multiply, and divide sequences just as we do for functions on Applied Mathematics I. Rules for computing the limits of combination of sequences are analogous to the rules for limits of combinations of functions. We present these rules now.

Let $\{a_n\}_{n=m}^{\infty}$ and $\{b_n\}_{n=m}^{\infty}$ be convergent sequences such that $\{a_n\}_{n=m}^{\infty}$ and $\{b_n\}_{n=m}^{\infty}$ converges to L and K respectively. Then the sum $\{a_n+b_n\}_{n=m}^{\infty}$, any scalar multiple $\{ca_n\}_{n=m}^{\infty}$, the product $\{a_nb_n\}_{n=m}^{\infty}$, and the quotient $\left\{\frac{a_n}{b_n}\right\}_{n=m}^{\infty}$ (provided that $\lim_{n\to\infty} b_n \neq 0$) all converges, with

1) $\lim_{n\to\infty}\left(a_n+b_n\right)=\lim_{n\to\infty}a_n+\lim_{n\to\infty}b_n=L+K$

2) $\lim_{n\to\infty}\left(a_n-b_n\right)=\lim_{n\to\infty}a_n-\lim_{n\to\infty}b_n=L-K$

3) $\lim_{n\to\infty}\left(ca_n\right)=c\lim_{n\to\infty}a_n=cL$

4) $\lim_{n\to\infty}\left(a_nb_n\right)=\lim_{n\to\infty}a_n\lim_{n\to\infty}b_n=LK$

5) $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)=\frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}=\frac{L}{K}$

6) $\lim_{n\to\infty}\left(a_n{}^p\right)=[\lim_{n\to\infty} a_n]^p=[L]^p$

In order to compute the limit of sequences we can use the following theorem without proof.

Theorem 1: Given the sequence $\{a_n\}$ if we have a function *f(x)* such that $f(x)=a_n$ and $\lim_{x\to\infty} f(x)=L$ then $\lim_{n\to\infty} a_n=L$

This theorem is basically telling us that we take the limits of sequences much like we take the limit of functions. In fact, in most cases we'll not even really use this theorem by explicitly writing down a function. We will more often just treat the limit as if it were a limit of a function.

Theorem 2: If $\lim_{n\to\infty}|a_n|=0$ *then* $\lim_{n\to\infty} a_n=0$

This theorem is convenient for sequences that alternate in signs and note that it will only work if the sequence has a limit of zero.

Note that before moving on we need to give a warning about misusing Theorem 2. Theorem 2 only works if the limit is zero. If the limit of the absolute value of the sequence terms is not zero then the theorem will not hold. See example *(d)* below

Example 1: Determine if the sequences $\left\{\frac{3n^2-1}{10n+5n^2}\right\}_{n=2}^{\infty}$ converge or diverge. If the sequence converges determine its limit.

Solution:

In this case all we need to do is recall the method that was developed in Applied Mathematics I to deal with the limits of rational functions.[Please if you do not remember go to Applied Mathematics I and read about Limit]

To do a limit in this form all we need to do is factor from the numerator and denominator the largest power of *n*, cancel and then take the limit.

$$\lim_{n\to\infty}\frac{3n^2-1}{10n+5n^2}=\lim_{n\to\infty}\frac{n^2(3-\frac{1}{n^2})}{n^2(\frac{10}{n}+5)}=\lim_{n\to\infty}\frac{3-\frac{1}{n^2}}{\frac{10}{n}+5}=\frac{\lim_{n\to\infty}(3-\frac{1}{n^2})}{\lim_{n\to\infty}(\frac{10}{n}+5)}=\frac{3-0}{0+5}=\frac{3}{5}$$

So the sequence converges and its limit is $\frac{3}{5}$

Example 2: Determine if the sequences $\left\{\frac{e^{2n}}{n}\right\}_{n=1}^{\infty}$ converge or diverge. If the sequence Converges determine its limit.

Solution:

We will need to be careful with this one. We will need to use L'Hospital's Rule on this sequence. The problem is that L'Hospital's Rule only works on functions and not on sequences. Normally this would be a problem, but we've got Theorem 1 from above to help us out. Let's define $f(x)=\frac{e^{2x}}{x}$ and note that, $f(n)=\frac{e^{2n}}{n}$

Theorem 1 says that all we need to do is take the limit of the function.

$$\lim_{n\to\infty}\frac{e^{2n}}{n}=\lim_{x\to\infty}f(x)=\lim_{x\to\infty}\frac{e^{2x}}{x}$$

$$=\lim_{x\to\infty}\frac{2e^{2x}}{1}=\infty$$

So, the sequence in this part diverges (to ∞).

More o4/29/2013ften than not we just do L'Hospital's Rule on the sequence terms without first

converting to *x*'s since the work will be identical regardless of whether we use *x* or *n*. However, we really should remember that technically we can't do the derivatives while dealing with sequence terms.

Example 3: Determine if the sequences $\left\{\frac{(-1)}{n}\right\}_{n=1}^{\infty}$ converge or diverge. If the sequence Converges determine its limit.

Solution:

We will also need to be careful with this sequence. We might be tempted to just say that the limit of the sequence terms is zero (and we'd be correct).However, technically

we can't take the limit of sequences whose terms alternate, because we don't know how to do limits of functions that exhibit that same behavior. Also, we want to be very careful to not rely too much on intuition with these problems. As we will see in this section, and in later sections, our intuition can lead us astray in these problem if we aren't careful.

We will need to use Theorem 2 on this problem.

$$\lim_{n\to\infty}\left|\frac{(-1)^n}{n}\right| = \lim_{n\to\infty}\frac{1}{n} = 0$$

Therefore, since the limit of the sequence terms with absolute value bars on them goes to zero we know by Theorem 2 that,

$$\lim_{n\to\infty}\frac{(-1)^n}{n} = 0$$

Which also means that the sequence converges to a value of zero.

Example 4: Determine if the sequences $\left\{(-1)^{2n}\right\}_{n=0}^{\infty}$ converge or diverge. If the sequence Converges determine its limit.

Solution:

For this sequence all that we need to do is acknowledge that $\lim_{n\to\infty}(-1)^{\infty}$ doesn't exist to get that the sequence is divergent. If you're not convinced that this limit doesn't exist write down the first few terms of the sequence.

$$\left\{(-1)^n\right\}_{n=0}^{\infty} = \{1,-1,1,-1,1,-1,1,-1,\ldots\}$$

In order for a limit to exist the terms *must* be settling down towards a specific value and these clearly will never do that.

We now need to take a look at some more terminology and definitions for sequences.

Definition: Given any sequence $\{a_n\}$ we have the following.

1. We call the sequence $\{a_n\}$ is *increasing* if $a_n \le a_{n+1}$ for every *n*.
2. We call the sequence $\{a_n\}$ is *decreasing* if $a_n \ge a_{n+1}$ for every *n*.
3. If $\{a_n\}$ is an increasing sequence or $\{a_n\}$ is a decreasing sequence we call it is *monotonic* sequence.

4. If there exists a number *m* such that $m \le a_n$ for every *n* we say the sequence is *bounded below*. The number *m* is sometimes called a *lower bound* for the sequence.
5. If there exists a number *M* such that $M \ge a_n$ for every *n* we say the sequence is *bounded above*. The number *M* is sometimes called an *upper bound* for the sequence.
6. If the sequence is both bounded below and bounded above we call the sequence is *bounded*.

Note that in order for a sequence to be increasing or decreasing it must be increasing/decreasing for every *n*. In other words, a sequence that increases for three terms and then decreases for the rest of the terms is not a decreasing sequence!

Before moving on we should make a quick point about the bounds for a sequence that is bounded above and/or below. We'll make the point about lower bounds, but we could just as easily make it about upper bounds.

A sequence is bounded below if we can find any number *m* such that $m \le a_n$ for every *n*.

Note however that if we find one number *m* to use for a lower bound then any number smaller than *m* will also be a lower bound. Also, just because we find one lower bound that doesn't mean there won't be a better lower bound for the sequence than the one we found. In other words, there are an infinite number of lower bounds for a sequence that is bounded below, some will be better than others. In my class all that I'm after will be a lower bound. I don't necessarily need the best lower bound, just a number that will be a lower bound for the sequence.

We also have the following theorem about bounded and monotonic sequences.

Theorem 3: If $\{a_n\}$ is bounded and monotonic then $\{a_n\}$ is convergent.

Proof: Suppose $\{a_n\}$ is an increasing sequence. Since $\{a_n\}$ is bounded, the set $S = \{a_a : n \ge 1\}$ has an upper bound. By the Completeness Axiom **[**which says that if S is a nonempty set of real numbers that has an upper bound M ($x \le M$ for all x in S), then S has a *least upper bound say* b. (This means that b is an upper bound for S, but if M is any other upper bound, then $b \le M$.) The Completeness Axiom is an expression of

the fact that there is no gap or hole in the real number line**]** it has a least upper bound say L.

Given $\varepsilon > 0, L - \varepsilon$, is not an upper bound for S (since L is the least upper bound).

Therefore,

$a_N > L - \varepsilon$ for some integer *N*.

But the sequence is increasing so $a_n \geq a_N$ for every $n > N$.

Thus, if $n > N$ we have $a_n > L - \varepsilon$

So $0 \leq L - a_n < \varepsilon$ Since $a_n \leq L$.

Thus $|L - a_n| < \varepsilon$ whenever $n > N$

So $\lim_{n\to\infty} a_n = L$

Note that a similar proof (using the greatest lower bound) works if $\{a_n\}$ is decreasing. The proof of this Theorem shows that a sequence that is increasing and bounded above is convergent. (Likewise, a decreasing sequence that is bounded below is convergent.) This fact is used many times in dealing with infinite series.

Example 1: Determine if the sequence $\{-n^2\}_{n=0}^{\infty}$ is monotonic and/or bounded.

Solution:

This sequence is a decreasing sequence (and hence monotonic) because, $-n^2 > -(n+1)^2$ for every *n*. Also, since the sequence terms will be either zero or negative this sequence is bounded above. We can use any positive number or zero as the bound, *M*, however, it's standard to choose the smallest possible bound if we can and it's a nice number,

So we will choose *M* = 0 since $-n^2 \leq 0$ for every *n*.

This sequence is not bounded below however since we can always get below any Potential bound by taking *n* large enough. Therefore, while the sequence is bounded above it is not bounded.

As a side note we can also note that this sequence diverges (to $-\infty$ if we want to be Specific.)

Example 2: Determine if the sequence $\left\{(-1)^{n+1}\right\}_{n=1}^{\infty}$ is monotonic and/or bounded.

Solution:

The sequence terms in this sequence alternate between 1 and -1 and so the sequence is not monotonic as it is neither increasing nor decreasing. It is bounded however since it is bounded above by 1 and bounded below by -1.

This sequence is also divergent.

Example 3: Determine if the sequence $\left\{\frac{2}{n^2}\right\}_{n=5}^{\infty}$ is monotonic and/or bounded.

Solution:

This sequence is a decreasing sequence (and hence monotonic) since, $\frac{2}{n^2} > \frac{2}{(n+1)^2}$

The terms in this sequence are all positive and so it is bounded below by zero. Also, since the sequence is a decreasing sequence the first sequence term will be the largest and so we can see that the sequence will also be bounded above by $\frac{2}{25}$. therefore, this sequence is bounded. Theorem 3 above says that this sequence is then convergent since it is both bounded and monotonic. A quick limit can verify that this sequence is convergent and its value is zero.

Example 4: Determine if the sequence $\left\{\frac{n}{n+2}\right\}_{n=0}^{\infty}$ is monotonic and/or bounded.

Exercise!!

Now, let's work a couple more examples that are designed to make sure that we do not get too used to relying on our intuition with these problems.

Example 5: Determine if the sequence $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$ monotonic and/or bounded.

Solution:

We will start with the bounded part of this example first and then come back and deal with the increasing/decreasing question since that is where students often make mistakes with problems of this type.

First, *n* is positive and so the sequence terms are all positive. The sequence is therefore bounded below by zero. Likewise each sequence term is the quotient of a number divided by a larger number and so is guaranteed to be less that one. The sequence is then bounded above by one. So, this sequence is bounded.

Now let's think about the monotonic question. First, students will often make the mistake of assuming that because the denominator is larger the quotient must be decreasing. This will not always be the case and in this case we would be wrong. This sequence is increasing.

To see this we will need to resort to Applied Mathematics I techniques. First consider the following function and its derivative.

$$f(x) = \frac{x}{x+1} \qquad\qquad f'(x) = \frac{1}{(x+1)^2}$$

We can see that the first derivative is always positive and so from Calculus I we know that the function must then be an increasing function. So, how does this help us?

Notice that,

$$f(n) = \frac{n}{n+1} = a_n$$

Therefore,

$$a_n = \frac{n}{n+1} = f(n) < f(n+1) = \frac{n+1}{n+2} = a_{n+1}$$

$$\Rightarrow a_n < a_{n+1}$$

In other words, the sequence must be increasing. Note that now that we know the sequence is an increasing sequence we can get a better lower bound for the sequence. Since the sequence is increasing the first term in the sequence must be the smallest term and so since we are starting at n =1 we could also use a lower bound of $\frac{1}{2}$ for this sequence. It is important to remember that any number that is always less than or equal to all the sequence terms can be a lower bound. Some are better than others however. Also, the sequence converges to 1, which again is a verification of Theorem 3.

Before moving on to the next part there is a natural question that many students will have at this point. That is this. Why did we use Calculus to determine the increasing/decreasing nature of the sequence when we could have just plugged in a couple of *n*'s and quickly determined the say thing?

The answer to this question is the next part (which we consider later)

Example 6: Determine if the sequence $\left\{\frac{n^3}{n^4+10000}\right\}_{n=0}^{\infty}$ monotonic and/or bounded.

Let's leave this as an exercise for the reader.

1.4 Infinite series

In this section we will introduce the topic that we will be discussing for the rest of this chapter. That topic is series. So just what is a series? Well, let's start with a sequence $\{a_n\}_{n=1}^{\infty}$ (note the *n=1* is for convenience, it can be anything) and then add up all the terms of the sequence.

$$a_1 + a_2 + a_3 + a_4 + \ldots\ldots\ldots + a_n + \ldots\ldots$$

This is not an easy thing to write down on a regular basis and so we introduce the following notation.

$$a_1 + a_2 + a_3 + a_4 + \ldots\ldots\ldots + a_n + \ldots\ldots = \sum_{n=1}^{\infty} a_n$$

For example consider

$$2+4+6+8+10+\ldots\ldots\ldots$$

It is the sum of even numbers starting from 2, now rather than writing this we use the denotation $\sum_{n=1}^{\infty} 2n$ which means

$$2+4+6+8+10+\ldots\ldots\ldots = \sum_{n=1}^{\infty} 2n$$

We have just defined informally what a series is. A series is nothing more than the summation of a list of numbers or sequence. Since we started out with an infinite sequence we will be summing up an infinite list of numbers. Because of this the series

above is sometimes called an infinite series. The *n* is often called an *index of summation* or just *index* for short.

Note as well that we will use $\sum a_n$ to represent an infinite series in which the starting point for the index is not important. We will do this in quite a few facts and theorems that we will be seeing throughout this chapter. In these facts the starting point of the series will not affect the result and so to simplify the notation and to avoid giving the impression that the starting point is important we will drop the index from the notation.

Do not forget however, that there is a starting point and that this will be an infinite series.

The notation used for series is called a number of things. The most common names are: *Series notation, summation notation,* and *sigma notation*.

The series notation (or summation notation or sigma notation, which ever you prefer) tells us to add all the items from the sequence starting at the value of the index that is below the sigma. Also note that the letter that we use for the index is not important. The following two series are identical.

$$\sum_{n=0}^{\infty} \frac{n}{n^2+1} \quad = \quad \sum_{i=0}^{\infty} \frac{i}{i^2+1}$$

As long as we are starting at the same spot (*n=0* or *i=0* respectively) and we're adding the same terms so the series will be identical.

The point of this section is to cover the basic ideas, concepts and manipulations involved in series. We need to be somewhat familiar with these before we get into the later sections.

It would be impossible to find a finite sum for the series

$$1+2+3+4+....+n+....$$

Because if we start adding the terms we get the cumulative sums 1, 3, 6, 10, 15, 21 . . . and, after the n*th* term, we get $\frac{n(n+1)}{2}$, which becomes very large as n increases.

However, if we start to add the terms of the series

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+.....+\frac{1}{2^n}+....$$

However, if we start to add the terms of the series

$\frac{1}{2},\frac{3}{4},\frac{7}{8},\frac{15}{16},.....,1-\frac{1}{2^n},....$ This shows that as we add more and more terms, these *partial sums* become closer and closer to 1. In fact, by adding sufficiently many terms of the series we can make the partial sums as close as we like to 1. So it seems reasonable to say that the sum of this infinite series is 1 and to write

$$\sum_{n=1}^{\infty}\frac{1}{2^n}=\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+.....+\frac{1}{2^n}+....=1$$

1.5 Convergence and Divergence properties of Infinite Series.

In the previous section we spent some time getting familiar with series, but to be honest, most of what we did in that section won't be used on a regular basis in this chapter. We covered that material because we need to be aware of how series work and can be manipulated, but we also covered it so that we could start getting our feet wet in the subject of series.

It is now time to start talking about an idea involved in series that we will deal with to one extent of another in almost all of the remaining sections of this chapter.

There was a very important question about series that was only mentioned in passing towards the very end of the previous section. The question is simply this : Does it even make sense to add up an infinite sequence of numbers? Technically we can always write down an infinite series or summation, but that doesn't mean that it makes sense to do it.

The real question that we'll be asking here is does the (infinite) series/summation yield a finite value or infinite value? Of course that also assumes that the series yields a value at all! As we will see it will be possible for a series to not even have a value.

To answer this question we'll need some more terminology out of the way. Let's start with the following series.

$$\sum_{n\to 1}^{\infty}a_n$$

Note that we're starting at *n=1* only for convenience. We could start the series any where, but the following notation and terminology demands that we start somewhere and so for the sake of the work we choose to start at *n=1*.

Now, instead of adding all the terms out to infinity, let's look at the following finite summations/series.

$$S_1 = a_1$$
$$S_2 = a_1 + a_2$$
$$S_3 = a_1 + a_2 + a_3$$
$$\vdots$$
$$S_n = a_1 + a_2 + a_3 + \ldots\ldots + a_n$$

In general, we can write

$$S_n = a_1 + a_2 + a_3 + \ldots\ldots + a_n = \sum_{i=1}^{\infty} a_i$$

These are called *partial sums*. Notice that the partial sums will form an infinite sequence, $\{S_n\}_{n=1}^{\infty}$, and that while it might not make sense to perform a summation of an infinite list of numbers these are all summations of a finite list of numbers and so are guaranteed to be finite numbers. Well they will be finite numbers provided we don't end up with a division by zero error somewhere in the list. In all of the work that we'll be doing in this chapter we will assume that all the sequence terms exist and are finite numbers.

These partial sums form a new sequence $\{S_n\}$, which may or may not have a limit. If $\lim_{n\to\infty} S_n = s$ exists (as a finite number), then, as in the preceding example, we call it the sum of the infinite series $\sum a_n$.

From the section on sequences we know how to determine if the sequence of partial sums converges or diverges. Also notice that as $n \to \infty$ the sequence terms, s_n, should start looking more and more like the infinite series. In fact it can be shown that if the sequence of partial sums is convergent and if we define,

$$\lim_{n\to\infty} S_n = s$$

Then,

$$\sum_{n=1}^{\infty} a_n = s$$

In these cases we call the series *convergent* and we call *s* the *sum* or *value* of the series.
If the sequence of partial sums is divergent (*i.e.* either the limit doesn't exist or is infinite) then we call the series *divergent*.
In other words, the series is convergent if the sequence of partial sums is convergent and hence has a finite value. Likewise the series will be divergent if the sequence of partial sums is divergent. In the case of a divergent series, either the series will have an infinite value or won't have a value at all depending on whether or not the limit of the sequence of partial sums exists or is infinite.
Let's take a look at some series and see if we can determine if they are convergent or divergent.

Definition: Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots,$, let S_n denote its nth

Partial sum:

$$S_n = a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_n = \sum_{i=1}^{\infty} a_i$$

If the sequence $\{S_n\}$ is convergent and $\lim_{n\to\infty} S_n = s$ exists as a real number, then the series is called convergent and we write

$$a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_n = s \quad \text{or} \quad \sum_{n=1}^{\infty} a_n = s$$

The number s is called the *sum* of the series. Otherwise, the series is called *divergent***.**

Example1: Determine if the following series is convergent or divergent. If it converges determine its sum.

$$\sum_{n=1}^{\infty} n$$

Solution:

To determine if the series is convergent we first need to get our hands on a formula for the general term in the sequence of partial sums.

$$S_n = \sum_{i=1}^{n} i$$

This is a known series and its value can be shown to be,

$$S_n = \sum_{i=1}^{n} i = \frac{n(n+1)}{n}$$

So, to determine if the series is convergent we will first need to see the sequence of partial sums,

$$\left\{ \frac{n(n+1)}{n} \right\}_{n=1}^{\infty}$$

is convergent or divergent. That's not terribly difficult in this case.

$$\lim_{n\to\infty} \frac{n(n+1)}{n}$$

Therefore, the sequence of partial sums diverges to ∞ and so the series also diverges.

Note that we can say that the series has the value of ∞ in these cases, although the series is still called divergent.

So, as we saw in this example we had to know a fairly obscure formula in order to determine the convergence of this series. In general finding a formula for the general term in the sequence of partial sums is a very difficult process. In fact after the next section we will not be doing much with the partial sums of series due to the extreme difficulty faced in find the general formula.

We will continue with a few more examples however, since this is technically how we determine convergence of a series. Also, the remaining examples we will be looking at in this section will lead us to a very important fact about the convergence of series.

Example 2: Determine if the following series converges or diverges. If it converges determine its sum.

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$$

Solution:

This is actually one of the few series in which we are able to determine a formula for the general term in the sequence of partial fractions. However, in this section we are more interested in the general idea of convergence and divergence and so we'll put off discussing the process for finding the formula until the next section.

The general formula for the partial sums is,

$$S_n = \sum_{i=2}^{n} \frac{1}{i^2 - 1} = \frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)}$$

and in this case we have,

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} [\frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)}] = \frac{3}{4}$$

The sequence of partial sums converges and so the series converges as well and its value is,

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - 1} = \frac{3}{4}$$

Example 3: Determine if the following series converges or diverges. If it converges determine its sum.

$$\sum_{n=0}^{\infty} (-1)^n$$

Solution:

In this case we really do not need a general formula for the partial sums to determine the convergence of this series. Let's just write down the first few partial sums.

$$S_0 = 1$$
$$S_1 = 1 - 1 = 0$$
$$S_3 = 1 - 1 + 1 = 1$$
$$S_4 = 1 - 1 + 1 - 1 = 0$$
$$\vdots$$
$$etc$$

So, it looks like the sequence of partial sums is,

$$\{S_n\}_{n=0}^{\infty} = \{1, 0, 1, 0, 1, 0, 1, ...\}$$

and this sequence diverges since $\lim_{n\to\infty} S_n$ doesn't exist. Therefore, the series also diverges this series doesn't even have a value.

Theorem: If $\sum a_n$ converges then $\lim_{n\to\infty} a_n = 0$

Be careful to not misuse this theorem however! This theorem gives us a requirement for Convergence but not a guarantee of convergence. In other words, the converse is NOT true. If $\lim_{n\to\infty} a_n = 0$ the series may actually diverge! Consider the following series.

$$\sum_{n=1}^{\infty} \frac{1}{n} \qquad\qquad \sum_{n=1}^{\infty} \frac{1}{n^2}$$

In both cases the series terms are zero in the limit as *n* goes to infinity,

i.e. $\lim_{n\to\infty} \frac{1}{n} = 0 \quad and \lim_{n\to\infty} \frac{1}{n^2} = 0$

But only the second series converges. The first series diverges. It will be a couple of sections before we can prove this, so at this point please believe this and know that you'll be able to prove the convergence of these two series in a couple of sections.

Again, as noted above, all this theorem does is give us a requirement for a series to converge. In order for a series to converge the series terms must go to zero in the limit.
If the series terms do not go to zero in the limit then there is no way the series can converge since this would violate the theorem.
Now it time to talk about the divergence and convergences of sequences

1.6 Convergence and Divergence Tests

1.6.1 Divergence Test

Theorem: a) If $\lim_{n\to\infty} a_n \neq 0$ then $\sum a_n$ diverges

b) If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$

Again, do NOT misuse this test. This test only says that a series is guaranteed to diverge if the series terms don't go to zero in the limit. If the series terms do happen to go to zero the series may or may not converge! Again, recall the following two series,

$$\sum_{n=1}^{\infty} \frac{1}{n}$$ diverges

$$\sum_{n=1}^{\infty}\frac{1}{n^2}$$ converges

One of the more common mistakes that students make when the first get into series is to assume that if $\lim_{n\to\infty} a_n = 0$, then $\sum a_n$ will converge. There is just no way to guarantee this so be careful!

Example : Determine if the series below is convergent or divergent

$$\sum_{n=0}^{\infty}\frac{4n^2 - n^3}{10 + 2n^3}$$

Solution:

With almost every series the first thing that we should do is take a look at the series terms and see if they go to zero of not. If it's clear that the terms don't go to zero use the divergence Test and be done with the problem.

$$\lim_{n\to\infty}\frac{4n^2 - n^3}{10 + 2n^3} = \frac{-1}{2} \neq 0$$

The limit of the series terms isn't zero and so by the Divergence Test the series diverges.

Geometric Series

Theorem: The Geometric series $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + ...$

is convergent if $|r| < 1$ and its sum is $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$, $|r| < 1$

if $|r| \geq 1$, the geometric series is divergent. Here r is called the common ratio.

Example: Is the series $\sum_{n=1}^{\infty} 2^{2n}3^{1-n}$ convergent or divergent?

Solution:

Let's rewrite the *n*th term of the series in the form ar^{n-1}:

$$\sum_{n=1}^{\infty} 2^{2n}3^{1-n} = \sum_{n=1}^{\infty} \frac{4^n}{3^{n-1}} = \sum_{n=1}^{\infty} 4(\frac{4}{3})^{n-1}$$

We recognize this series as a geometric series with $a = 4$ and $r = \frac{4}{3}$. Since $r > 1$, the series diverges

Telescoping Series

It is now time to look at the second of the three series in this section. In this portion we are going to look at a series that is called a telescoping series. The name in this case comes from what happens with the partial sums and is best shown in an example.

Example: Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent, and find its sum.

Solution: This is not a geometric series, so we go back to the definition of a convergent series and compute the partial sums.

$$S_n = \sum_{i=1}^{\infty} \frac{1}{i(i+1)} = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + ... + \frac{1}{n(n+1)}$$

We can simplify this expression if we use the partial fraction decomposition

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

Thus we have $S_n = \sum_{i=1}^{\infty} \frac{1}{i(i+1)} = \sum_{i=1}^{\infty} (\frac{1}{i} - \frac{1}{i+1})$

$$S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + ... + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$= 1 - \frac{1}{n+1}$$

And here $\lim_{n\to\infty} S_n = \lim_{n\to\infty} 1 - \frac{1}{n+1} = 1 - 0 = 0$

Therefore, the given series is convergent and $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

Exercise: Determine if the following series converges or diverges. If it converges find its value.

a) $\sum_{n=0}^{\infty} \frac{1}{n^2+3n+2}$

b) $\sum_{n=1}^{\infty} \frac{1}{n^2+4n+3}$

Note that it's not always obvious if a series is telescoping or not until you try to get the partial sums and then see if they are in fact telescoping. There is no test that will tell us that we have got a telescoping series right off the bat. Also note that just because you can do partial fractions on a series term does not mean that the series will be a telescoping series. The following series, for example, is not a telescoping series despite the fact that we can partial fraction the series terms.

$$\sum_{n=1}^{\infty} \frac{3+2n}{n^2+3n+2} = \sum_{n=1}^{\infty} \left(\frac{1}{n+1} + \frac{1}{n+2}\right)$$

In order for a series to be a telescoping we must get terms to cancel and all of these terms are positive and so none will cancel.

Harmonic Series

This is the third and final series that we're going to look at in this chapter. Here is the harmonic series.

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

The harmonic series is divergent and we'll need to wait until the next section to show that. This series is here because it's got a name and so I wanted to put it here with the other two named series that we looked at in this section. We are also going to use the harmonic series to illustrate a couple of ideas about divergent series that we've already discussed for convergent series. We will do that with the following example.

Example : Show that the series $\sum_{n=1}^{\infty}\frac{5}{n}$ is divergent.

Solution:

To see that this series is divergent all we need to do is use the fact that we can factor a constant out of a series as follows,

$$\sum_{n=1}^{\infty}\frac{5}{n}=5\sum_{n=1}^{\infty}\frac{1}{n}$$

Now, $\sum_{n=1}^{\infty}\frac{1}{n}$ is divergent and so five times this will still not be a finite number and so the series has to be divergent. In other words, if we multiply a divergent series by a constant it will still be divergent.

In this case we'll start with the harmonic series and strip out the first three terms.

$$\sum_{n=1}^{\infty}\frac{1}{n}=1+\frac{1}{2}+\frac{1}{3}+\sum_{n=4}^{\infty}\frac{1}{n} \qquad \sum_{n=4}^{\infty}\frac{1}{n}=(\sum_{n=1}^{\infty}\frac{1}{n})-(\frac{11}{6})$$

In this case we are subtracting a finite number from a divergent series. This subtraction will not change the divergence of the series. We will either have infinity minus a finite number, which is still infinity, or a series with no value minus a finite number, which will still have no value.

Therefore, this series is divergent. Just like with convergent series, adding/subtracting a finite number from a divergent series is not going to change the fact the convergence of the series.

So, some general rules about the convergence/divergence of a series are now in order.

Multiplying a series by a constant will not change the convergence/divergence of the series and adding or subtracting a constant from a series will not change the convergence/divergence of the series. These are nice ideas to keep in mind.

Exercise: Show that the series $\sum_{n=4}^{\infty}\frac{1}{n}$ is divergent

Combination of Series

If $\sum_{n=1}^{\infty}a_n$ and $\sum_{n=1}^{\infty}b_n$ are two series, we can add term by term; we can also multiply all the terms of either series by a single number c .These operations generate two new series, $\sum_{n=1}^{\infty}(a_n+b_n)$ and $\sum_{n=1}^{\infty}ca_n$,whose convergence is generated by the convergence of original series, as now we state below

Theorem: 1.If $\sum_{n=1}^{\infty}a_n$ and $\sum_{n=1}^{\infty}b_n$ converge, then $\sum_{n=1}^{\infty}(a_n+b_n)$ also converges, and

$$\sum_{n=1}^{\infty}(a_n+b_n)=\sum_{n=1}^{\infty}a_n+\sum_{n=1}^{\infty}b_n$$

2. If $\sum_{n=1}^{\infty}a_n$ converges and c is any number, then $\sum_{n=1}^{\infty}ca_n$ also converges and

$$\sum_{n=1}^{\infty}ca_n=c\sum_{n=1}^{\infty}a_n$$

Example: Show that the series $\sum_{n=1}^{\infty}\left(\frac{4}{2^n}-\frac{2}{n(n+1)}\right)$ converges, and find its sum.

Solution: Let's consider $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\frac{4}{2^n}$ and

$$\sum_{n=1}^{\infty}b_n=\sum_{n=1}^{\infty}-\frac{2}{n(n+1)}$$

The geometric series implies that $\sum_{n=1}^{\infty}\frac{4}{2^n}=4\frac{1/2}{1-1/2}=4$

This means the series $\sum_{n=1}^{\infty}\frac{4}{2^n}$ converges

Again, $\sum_{n=1}^{\infty}-\frac{2}{n(n+1)}=(-2)\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$ =-2 from the previous example

Hence this series converges to -2

There fore by the theorem the series $\sum_{n=1}^{\infty}\left(\frac{4}{2^n}-\frac{2}{n(n+1)}\right)$ converges as it is the sum of two convergent series.

More over this series converges to 4+(-2) = 2

1.6.2 Test of convergence of a series

1.6.2.1 Integral Test

In general, it is difficult to find the exact sum of a series. We were able to accomplish this for geometric series and the series $\sum\frac{1}{n(n+1)}$ because in each of those cases we could find a simple formula for the *n*th partial sum. But usually it isn't easy to compute $\underset{n\to\infty}{Lim}\ S_n$.

Therefore, in the next few sections we develop several tests that enable us to determine whether a series is convergent or divergent without explicitly finding its sum.

(In some cases, however, our methods will enable us to find good estimates of the sum.)

Our first test involves improper integrals.

The last topic that we discussed in the previous section was the harmonic series. In that discussion we stated that the harmonic series was a divergent series. It is now time to prove that statement. This proof will also get us started on the way to our next test for convergence that we'll be looking at.

So, we will be trying to prove that the harmonic series

$$\sum_{n=1}^{\infty}\frac{1}{n}$$ diverges.

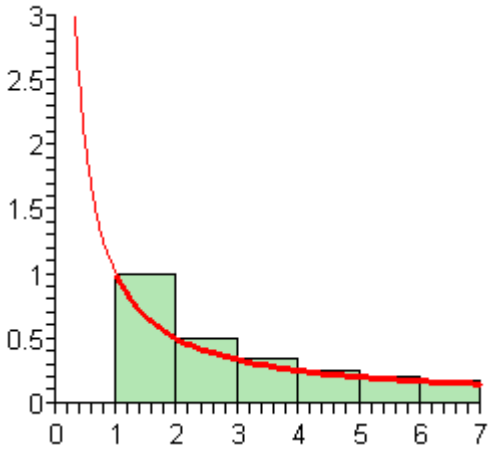
We will start this off by looking at an apparently unrelated problem. Let's start off by asking what the area under $f(x)=\frac{1}{x}$ is on the interval $[1,\ \infty)$. From the section on Improper Integral we know that this is,

$$\int_1^{\infty}\frac{1}{x}dx=\infty$$

and so we called this integral divergent (yes, that's the same term we are using here with series….).

So, just how does that help us to prove that the harmonic series diverges? Well, recall that we can always estimate the area by breaking up the interval into segments and then sketching in rectangles and using the sum of the area all of the rectangles as an estimate of the actual area. Let's do that for this problem as well and see what we get.

We will break up the interval into subintervals of width 1 and we will take the function value at the left endpoint as the height of the rectangle. The image below shows the first few rectangles for this area.



So, the area under the curve is approximately,

$$A \approx \left(\frac{1}{1}\right)(1)+\left(\frac{1}{2}\right)(1)+\left(\frac{1}{3}\right)(1)+\left(\frac{1}{4}\right)(1)+\left(\frac{1}{5}\right)(1)+\ldots\ldots\ldots\ldots$$
$$=\left(\frac{1}{1}\right)+\left(\frac{1}{2}\right)+\left(\frac{1}{3}\right)+\left(\frac{1}{4}\right)+\left(\frac{1}{5}\right)+\ldots\ldots\ldots\ldots$$

Now note a couple of things about this approximation. First, each of the rectangles over estimates the actual area and secondly the formula for the area is exactly the harmonic series!

Putting these two facts together gives the following

$$A \approx \sum_{n=1}^{\infty} \frac{1}{n} > \int_{1}^{\infty} \frac{1}{x}\,dx = \infty$$

Notice that this tells us that we must have

$$A \approx \sum_{n=1}^{\infty} \frac{1}{n} > \infty$$

This implies $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$

Since we can't really be larger than infinity the harmonic series must also be infinite in value. In other words, the harmonic series is in fact divergent.

So, we have managed to relate a series to an improper integral that we could compute and it turns out that the improper integral and the series have exactly the same convergence.

Let's see if this will also be true for a series that converges. When discussing the divergence test we made the claim that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \qquad \text{converges.}$$

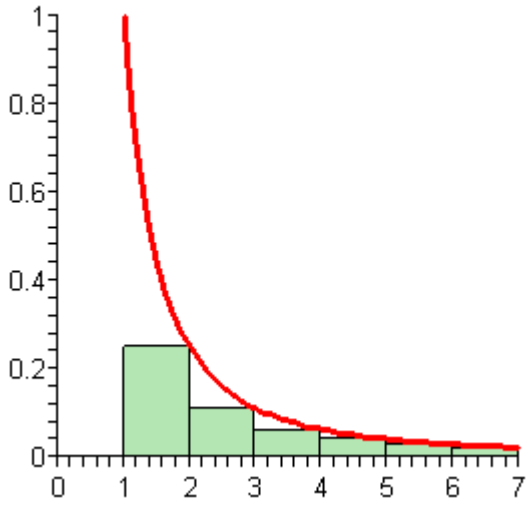
Let's see if we can do something similar to the above process to prove this.

We will try to relate this to the area under $f(x) = \frac{1}{x^2}$ is on the interval $[1, \infty)$. Again, from the Improper Integral section we know that,

$$\int_{1}^{\infty} \frac{1}{x^2}\,dx = 1 \quad \text{and so this integral converges.}$$

We will once again try to estimate the area under this curve. We will do this in an almost identical manner as the previous part with the exception that we will instead of using the

left end points for the height of our rectangles we will use the right end points. Here is a sketch of this case,



In this case the area estimation is,

$$\begin{aligned} A &\approx \left(\frac{1}{2^2}\right)(1)+\left(\frac{1}{3^2}\right)(1)+\left(\frac{1}{4^2}\right)(1)+\left(\frac{1}{5^2}\right)(1)+\left(\frac{1}{6^2}\right)(1)+\ldots\ldots\ldots\ldots \\ &= \left(\frac{1}{2^2}\right)+\left(\frac{1}{3^2}\right)+\left(\frac{1}{4^2}\right)+\left(\frac{1}{5^2}\right)+\left(\frac{1}{6^2}\right)+\ldots\ldots\ldots\ldots \\ &= \left(\frac{1}{2}\right)^2+\left(\frac{1}{3}\right)^2+\left(\frac{1}{4}\right)^2+\left(\frac{1}{5}\right)^2+\left(\frac{1}{6}\right)^2+\ldots\ldots\ldots\ldots \end{aligned}$$

This time, unlike the first case, the area will be an underestimation of the actual area and the estimation is not quite the series that we are working with. Notice however that the only difference is that we're missing the first term. This means we can do the following,

$$\begin{aligned} \sum_{n=1}^{\infty}\frac{1}{n^2} &= \left(\frac{1}{1^2}\right)+\left(\frac{1}{2^2}\right)+\left(\frac{1}{3^2}\right)+\left(\frac{1}{4^2}\right)+\left(\frac{1}{5^2}\right)+\ldots\ldots\ldots \\ &< 1+\int_1^{\infty}\frac{1}{x^2}\,dx = 1+1=2 \end{aligned}$$

Or, putting all this together we see that

$$\sum_{n=1}^{\infty}\frac{1}{n^2} < 2$$

Integral Test:

Suppose that *f(x)* is a positive, decreasing function on the interval $[k, \infty)$ and that $a_n = f(n)$ then,

1. If $\int_k^\infty f(x)dx$ is convergent so is $\sum_{n=k}^\infty a_n$

2. If $\int_k^\infty f(x)dx$ is divergent so is $\sum_{n=k}^\infty a_n$.

There are a couple of things to note about the integral test. First, the lower limit on the improper integral must be the same value that starts the series. Second, the function does not actually need to be decreasing everywhere in the interval. All that's really required is that eventually the function is decreasing. In other words, it is okay if the function increases for a while, but eventually the function must start decreasing and then continue to decrease from that point on.

There is one more very important point that must be made about this test. This test does NOT give the value of a series. It will only give the convergence/divergence of the series. That's it. No value. We can use the above series as a perfect example of this.
All that the test gave us was that,

$$\sum_{n=1}^\infty \frac{1}{n^2} < 2$$

So, we got an upper bound on the value of the series, but not an actual value for the series. In fact, from this point on we will not be asking for the value of a series we will only be asking whether a series converges or diverges. In a later section we look at estimating values of series, but even in that section still won't actually be getting values of series

Example1: Determine if the following series is convergent or divergent

a) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

b) $\sum_{n=0}^{\infty} n e^{-n^2}$

c) $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$

Solution:

Let's do only (a) and leave the others for the readers

In this case the function we will use

$$f(x) = \frac{1}{x \ln x}$$

This function is clearly positive and if we make *x* larger the denominator will get larger and so the function is also decreasing. Therefore, all we need to do is determine the convergence of the following integral.

$$\int_2^{\infty} \frac{1}{x \ln x} dx = \lim_{k \to \infty} \int_2^k \frac{1}{x \ln x} dx$$ to complete this integral use the substitution $u = \ln x$

$$= \lim_{k \to \infty} \left[\ln(\ln x) \Big|_2^k \right]$$

$$= \lim_{k \to \infty} \left[\ln(\ln k - \ln 2) \right]$$

$$= \infty$$

The integral is divergent and so the series is also divergent by the Integral Test.

1.6.2.2 P-Series Test:

If *k>0* then $\sum_{n=k}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Sometimes the series in this fact are called *p*-series and so this fact is sometimes called the *p*-series test.

Example: Determine if the following series are convergent or divergent

$$a)\ \sum_{n=4}^{\infty}\frac{1}{n^{7}}$$

$$b)\ \sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$$

$$c)\ \sum_{n=1}^{\infty}\frac{1}{n^{1/3}}$$

Solution:

a) In this case $p=7>1$ and so by this fact the series is convergent.

b) For this series $p=\frac{1}{2}\le 1$ and so the series is divergent by the fact.

c) Similarly decide for convergence or divergence

It is important to note before leaving this section that in order to use the Integral Test the series terms MUST be positive. If they are negative then the test doesn't work. Also remember that the test only determines the convergence of a series and does NOT give the value of the series.

1.6.2.3 The comparison test

In the previous section we saw how to relate a series to an improper integral to determine the convergence of a series. While the integral test is a nice test, it does force us to do improper integrals which are not always easy and in some cases may be impossible to evaluate.

In the comparison tests the idea is to compare a given series with a series that is known to be convergent or divergent. For instance, the series

$$\sum_{n=1}^{\infty}\frac{1}{2^{n}+1}$$

reminds us of the series $\sum_{n=1}^{\infty}\frac{1}{2^{n}}$, which is a geometric series with $r=\frac{1}{2}$ and $a=\frac{1}{2}$ and

is therefore convergent. Because the series $\sum_{n=1}^{\infty}\frac{1}{2^n+1}$ is so similar to a convergent series, we have the feeling that it too must be convergent. Indeed, it is. The inequality

$$\frac{1}{2^n+1}<\frac{1}{2^2} \quad \text{as } 2^{n+1}>2^n \text{ (to prove use Mathematical induction)}$$

shows that our given series $\sum_{n=1}^{\infty}\frac{1}{2^n+1}$ has smaller terms than those of the geometric series and therefore all its partial sums are also smaller than 1 (the sum of the geometric series). This means that its partial sums form a bounded increasing sequence, which is convergent.

It also follows that the sum of the series is less than the sum of the geometric series:

$$\sum_{n=1}^{\infty}\frac{1}{2^n+1}<1$$

Similar reasoning can be used to prove the following test, which applies only to series whose terms are positive. The first part says that if we have a series whose terms are smaller than those of a known convergent series, then our series is also convergent. The second part says that if we start with a series whose terms are larger than those of a known divergent series, then it too is divergent.

The comparison test

Suppose that we have two series $\sum a_n$ and $\sum b_n$ with, $a_n, b_n \geq 0$ for all n and $a_n \leq b_n$ for all n. Then,

1. If $\sum b_n$ is convergent then so is $\sum a_n$.

2. If $\sum a_n$ is divergent then so is $\sum a_n$.

In other words, we have two series of positive terms and the terms of one of the series is always larger than the terms of the other series. Then if the larger series is convergent the

smaller series must also be convergent. Likewise, if the smaller series is divergent then the larger series must also be divergent.

Do not misuse this test. Just because the smaller of the two series converges does not say anything about the larger series. The larger series may still diverge. Likewise, just because we know that the larger of two series diverges we can't say that the smaller series will also diverge! Be very careful in using this test

Recall that we had a similar test for Improper integrals back when we were looking at integration techniques. So, if you could use the comparison test for improper integrals you can use the comparison test for series as they are pretty much the same idea.

Example1: Determine if the series $\sum_{n=1}^{\infty} \frac{n}{n^2 - \cos^2(n)}$ is convergent or divergent.

Solution:

Since the cosine term in the denominator doesn't get too large we can assume that the series terms will behave like,

$$\frac{n}{n^2} = \frac{1}{n}$$

which, as a series, will diverge. So, from this we can guess that the series will probably diverge and so we'll need to find a smaller series that will also diverge.

Recall that from the comparison test with improper integrals that we determined that we can make a fraction smaller by either making the numerator smaller or the denominator larger. In this case the two terms in the denominator are both positive. So, if we drop the cosine term we will in fact be making the denominator larger since we will no longer be subtracting off a positive quantity. Therefore,

$$\frac{n}{n^2 - \cos^2(n)} > \frac{n}{n^2} = \frac{1}{n}$$

Then, since

$$\sum_{n=1}^{\infty}\frac{1}{n}$$

diverges (it's harmonic and the *p*-series test) by the Comparison Test our original series must also diverge.

Example 2: Determine if the series $\sum_{n=1}^{\infty}\frac{n^2+2}{n^4+5}$ is convergent or divergent.

Solution:

In this case the "+2" and the "+5" don't really add anything to the series and so the series terms should behave pretty much like

$$\frac{n^2}{n^4}=\frac{1}{n^2}$$

which will converge as a series. Therefore, we will need to find a larger series which also converges.

This means that we'll either have to make the numerator larger or the denominator smaller. We can make the denominator smaller by dropping the "+5". Doing this gives,

$$\frac{n^2+2}{n^4+5}<\frac{n^2+2}{n^4}$$

At this point, notice that we can't drop the "+2" from the numerator since this would make the term smaller and that's not what we want. However, this is actually all the further that we need to go. Let's take a look at the following series.

$$\sum_{n=1}^{\infty}\frac{n^2+2}{n^4}=\sum_{n=1}^{\infty}\frac{n^2}{n^4}+\sum_{n=1}^{\infty}\frac{2}{n^4}$$

$$=\sum_{n=1}^{\infty}\frac{1}{n^2}+\sum_{n=1}^{\infty}\frac{2}{n^4}$$

As shown, we can write the series as a sum of two series and both of these series are convergent by the *p*-series test. Therefore, since each of these series are convergent we know that the sum,

$$\sum_{n=1}^{\infty}\frac{n^2+2}{n^4}$$

is also a convergent series. Recall that the sum of two convergent series will also be convergent.

Now, since the terms of this series are larger than the terms of the original series we know that the original series must also be convergent by the Comparison Test.

Example 3: Determine if the series $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ is convergent or divergent

Solution:

The large dominant term in the denominator is $2n^2$ so we compare the given series with the series $\sum \frac{5}{2n^2}$.

Observe that $\frac{5}{2n^2+4n+3} < \frac{5}{2n^2}$

Because the left side has a bigger denominator. (In the notation of the Comparison Test, a_n is the left side and b_n is the right side.) We know that

$$\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2}\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is convergent because it's a constant times a p -series with $p = 2 > 1$. Therefore

$$\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$$

is convergent by part (1) of the Comparison Test.

The comparison test is a nice test that allows us to do problems that either we couldn't have done with the integral test or at the best would have been very difficult to do with the integral test. That doesn't mean that it doesn't have problems of its own.

Consider the following series.

$$\sum_{n=0}^{\infty} \frac{1}{3^n - n}$$

This is not much different from the first series that we looked at. The original series converged because the 3^n gets very large very fast and will be significantly larger than the *n*. Therefore, the *n* doesn't really affect the convergence of the series in that case. The fact that we are now subtracting the *n* off now instead of adding the *n* on really shouldn't change the convergence. We can say this because the 3^n gets very large very fast and the fact that we're subtracting *n* off won't really change the size of this term for all sufficiently large value of *n*.

So, we would expect this series to converge. However, the comparison test won't work with this series. To use the comparison test on this series we would need to find a larger series that we could easily determine the convergence of. In this case we can't do what we did with the original series. If we drop the *n* we will make the denominator larger (Since the *n* was subtracted off) and so the fraction will get smaller and just like when we looked at the comparison test for improper integrals knowing that the smaller of two series converges does not mean that the larger of the two will also converge.

So, we will need something else to do help us determine the convergence of this series. The following variant of the comparison test will allow us to determine the convergence of this series.

1.6.2.4 Limit Comparison Test

Suppose that we have two series $\sum a_n$ and $\sum b_n$ with $a_n, b_n \ge 0$ for all *n*. Define,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

If *c* is positive (*i.e.* $c > 0$) and is finite (*i.e.* $c < \infty$) then either both series converge or both series diverge.

Note that it doesn't really matter which series term is in the numerator for this test, we could just have easily defined *c* as,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

and we would get the same results.

Example1: Determine if the series $\sum_{n=0}^{\infty} \frac{1}{3^n - n}$ converges or diverges.

Solution:

To use the limit comparison test we need to find a second series that we can determine the convergence of easily and has what we assume is the same convergence as the givenseries. On top of that we will need to choose the new series in such a way as to give usan easy limit to compute for *c*.

We've already guessed that this series converges and since it's vaguely geometric let's use

$$\sum_{n=0}^{\infty} \frac{1}{3^n}$$

as the second series. We know that this series converges and there is a chance that since both series have the 3^n in it the limit won't be too bad.

Here's the limit

$$c = \lim_{n\to\infty} \frac{1}{3^n} \frac{3^n - n}{n}$$

$$= \lim_{n\to\infty} 1 - \frac{n}{3^n}$$

Now, we'll need to use L'Hospital's Rule on the second term in order to actually evaluate this limit.

$$c = 1 - \lim_{n\to\infty} \frac{1}{3^n \ln(3)}$$

$$= 1$$

So, *c* is positive and finite so by the Comparison Test both series must converge since

$$\sum_{n=0}^{\infty} \frac{1}{3^n}$$

converges.

We conclude that the series $\sum_{n=0}^{\infty}\frac{1}{3^n-n}$ converges.

Example 2: Determine if the series $\sum_{n=2}^{\infty}\frac{4n^2+n}{\sqrt[3]{n^7+n^3}}$ converges or diverges.

Solution:

Fractions involving only polynomials or polynomials under radicals will behave in the same way as the largest power of *n* will behave in the limit. So, the terms in this series should behave as,

$$\frac{n^2}{\sqrt[3]{n^7}}=\frac{n^2}{n^{\frac{7}{3}}}=\frac{1}{n^{\frac{1}{3}}}$$

and as a series this will diverge by the *p*-series test. In fact, this would make a nice choice for our second series in the limit comparison test so let's use it.

$$\lim_{n\to\infty}\frac{4n^2+n}{\sqrt[3]{n^7+n^3}}\frac{n^{\frac{1}{3}}}{1}=\lim_{n\to\infty}\frac{4n^{\frac{7}{3}}+n^{\frac{4}{3}}}{\sqrt[3]{n^7(1+\frac{1}{n^4})}}$$

$$=\lim_{n\to\infty}\frac{n^{\frac{7}{3}}(4+\frac{1}{n})}{n^{\frac{7}{3}}\sqrt[3]{1+\frac{1}{n^4}}}$$

$$=\frac{4}{\sqrt[3]{1}}=1=c$$

So, *c* is positive and finite and so both limits will diverge since

$\sum_{n=2}^{\infty}\frac{1}{n^{\frac{1}{3}}}$ diverges.

Example 3: Determine if the series $\sum_{n=1}^{\infty}\frac{4n-3}{n^3-5n-7}$ converges or diverges.

Let's leave it for the readers

1.7 Alternating Series and Alternating Series Test

The convergence tests that we have looked at so far apply only to series with positive terms. In this section and the next we learn how to deal with series whose terms are not necessarily positive. Of particular importance are *alternating series*, whose terms alternate in sign.

The last two tests that we looked at for series convergence have required that all the terms in the series be positive. Of course there are many series out there that have negative terms in them and so we now need to start looking at tests for these kinds of series.

Definition: An alternating series is a series whose terms are alternately positive and negative.

Example: $a)\ \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}-.....$

$$b)\ \sum_{n=1}^{\infty}(-1)^{n}\frac{n}{n+1}=-\frac{1}{2}+\frac{2}{3}-\frac{3}{4}+\frac{4}{5}-.....$$

We see from these examples that the nth term of an alternating series is of the form

$$a_n=(-1)^{n-1}b_n \qquad or \quad a_n=(-1)^{n}b_n$$

Where b_n is a positive number (In fact, $b_n=|a_n|$.)

The following test says that if the terms of an alternating series decrease toward 0 in absolute value, then the series converges.

1.7.1 Alternating Series Test

Suppose that we have a series $\sum a_n$ and either $a_n=(-1)^{n}b_n \quad or \quad a_n=(-1)^{n+1}b_n$ where

$b_n \geq o$ for all *n*. Then if,

1. $\lim_{n\to\infty} b_n=0$ and,

2. $\{b_n\}$ is eventually a decreasing sequence

then, the series $\sum a_n$ is convergent.

There are a couple of things to note about this test. First, unlike the Integral Test and the Comparison/Limit Comparison Test, this test will only tell us when a series converges and not if a series will diverge.

Secondly, in the second condition all that we need to require is that the series terms, b_n will eventually decreasing. It is possible for the first few terms of a series to increase and still have the test be valid. All that is required is that eventually we will have $b_n \geq b_{n+1}$ for

all *n* after some point.

Example1: Determine if the following series is convergent or divergent

$$a)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}1}{n} \qquad b)\sum_{n=1}^{\infty}\frac{(-1)^n n^2}{n^2+5}$$

$$c)\sum_{n=0}^{\infty}\frac{(-1)^{n-3}\sqrt{n}}{n+4} \qquad d)\sum_{n=2}^{\infty}\frac{\cos(n\pi)}{\sqrt{n}}$$

Solution: $a)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}1}{n}$

First, identify the b_n for the test.

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}=\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} \qquad b_n=\frac{1}{n}$$

Now, all that we need to do is run through the two conditions in the test.

$$\lim_{n\to\infty} b_n=\lim_{n\to\infty}\frac{1}{n}=0$$

$$b_n=\frac{1}{n}>\frac{1}{n+1}=b_{n+1}$$

Both conditions are met and so by the Alternating Series Test the series must converge.

The series from the previous example is sometimes called the *Alternating Harmonic Series.*

In the previous example it was easy to see that the series terms decreased since increasing *n* only increased the denominator for the term and hence made the term smaller. In general however we will need to resort to Calculus I techniques to prove the series terms decrease.

Solution: *b*) $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2+5}$

First, identify the b_n for the test.

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2+5} = \sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^2+5} \qquad b_n = \frac{n^2}{n^2+5}$$

Let's check the conditions.

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n^2}{n^2+5} = 1$$

So, the first condition isn't met and so there is no reason to check the second. Since this condition isn't met we'll need to use another test to check convergence. In these cases where the first condition isn't met it is usually best to use the divergence test.

$$\lim_{n\to\infty} \frac{(-1)^n n^2}{n^2+5} = \left(\lim_{n\to\infty} (-1) \right) \left(\lim_{n\to\infty} \frac{n^2}{n^2+5} \right)$$

$$= \left(\lim_{n\to\infty} (-1)^n \right) (-1)$$

$$= \lim_{n\to\infty} (-1)^n \qquad \text{doesn't exist}$$

This limit doesn't exist and so by the Divergence Test this series diverges.

Solution: *c*) $\sum_{n=0}^{\infty} \frac{(-1)^{n-3} \sqrt{n}}{n+4}$

Notice that in this case the exponent on the "-1" isn't *n* or *n+1*. That won't change how the test works however so we won't worry about that. In this case we have,

$$b_n = \frac{\sqrt{n}}{n+4}$$

so let's check the conditions.

The first is easy enough to check.

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{\sqrt{n}}{n+4} = 0$$

The second condition requires some work however. It is not immediately clear that these terms will decrease. Increasing *n* to *n+1* will increase both the numerator and the denominator. Increasing the numerator says the term should also increase while increasing the denominator says that the term should decrease. Since its not clear which of these will win out we will need to resort to Calculus I techniques to show that the terms decrease.

Let's start with the following function and its derivative.

$$f(x) = \frac{\sqrt{x}}{x+4} \qquad f'(x) = \frac{4-x}{2\sqrt{x}(x+4)^2}$$

Now, there are three critical points for this function, *x=-4*, *x=0*, and *x=4*. The first is outside the bound of our series so we won't need to worry about that one. Using the test points,

$$f'(1) = \frac{3}{50} \qquad f'(5) = \frac{\sqrt{5}}{810}$$

and so we can see that the function in increasing on $0 \le x \le 4$ and decreasing on $x \ge 4$. Therefore, since $f(n) = b_n$ we know as well that the b_n are also increasing on $0 \le n \le 4$ and decreasing on $n \ge 4$.

The b_n are then eventually decreasing and so the second condition is met.

Both conditions are met and so by the Alternating Series Test the series must be converging.

Solution: $d)\ \sum_{n=2}^{\infty}\frac{\cos(n\pi)}{\sqrt{n}}$

To check for the convergence or divergence of this series use the fact $\cos(n\pi)=(-1)^n$ (*Exercise*) and then $\sum_{n=2}^{\infty}\frac{\cos(n\pi)}{\sqrt{n}}=\sum_{n=2}^{\infty}\frac{(-1)^n}{\sqrt{n}}$

1.7.2 Absolute and Conditional convergence

When we first talked about series convergence we briefly mentioned a stronger type of convergence but didn't do anything with it because we didn't have any tools at our disposal that we could use to work problems involving it. We now have some of those tools so it's now time to talk about absolute convergence in detail.

We have convergence tests for series with positive terms and for alternating series. But what if the signs of the terms switch back and forth irregularly? We will see that the idea of absolute convergence sometimes helps in such cases.

Definition 1: A series $\sum a_n$ is called *absolutely convergent* if the series of absolute values $\sum|a_n|$ is convergent.

Notice that if $\sum a_n$ is a series with positive terms, then $|a_n|=a_n$ and so absolute convergence is the same as convergence in this case.

Example 1: The series

$$\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^2}=1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+.....$$

is absolutely convergent because

$$\sum_{n=1}^{\infty}\left|\frac{(-1)^{n-1}}{n^2}\right|=\sum_{n=1}^{\infty}\frac{1}{n^2}=1+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+.....$$

is a convergent *p*-series ($p=2$).

Example 2: We know that the alternating harmonic series

$$\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+.....$$

is convergent, but it is not absolutely convergent because the corresponding series of absolute values is

$$\sum_{n=1}^{\infty}\left|\frac{(-1)^{n-1}}{n}\right|=\sum_{n=1}^{\infty}\frac{1}{n}=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+.....$$

which is the harmonic series (p-series with $p=1$) and is therefore divergent.

Definition 2: A series $\sum a_n$ is called *conditionally convergent* if it is convergent but not absolutely convergent.

Example 2 above shows that the alternating harmonic series is conditionally convergent. Thus, it is possible for a series to be convergent but not absolutely convergent. However, the next theorem shows that absolute convergence implies convergence.

We also have the following fact about absolute convergence.

If $\sum a_n$ is absolutely convergent, then it is also convergent

It is this fact that makes absolute convergence a "stronger" type of convergence. Series that are absolutely convergent are guaranteed to be convergent. However, series that are convergent may or may not be absolutely convergent.

1.7.3 Ratio Test

In this section we are going to take a look at a test that we can use to see if a series is absolutely convergent or not. Recall that if a series is absolutely convergent then we will also know that it's convergent and so we will more often than not use it to simply determine the convergence of a series. We may as well acknowledge that it will give us absolute convergence as well however.

Before proceeding with the test let's do a quick reminder of factorials. This test will be particularly useful for series that contain factorials (and we will see some in the applications) so let's make sure we can deal with them before we run into them in an example.

If n is an integer such that $n \ge 0$, then n factorial is defined as,

$$n! = n(n-1)(n-2)(n-3).....4.3.2.1 \qquad if\ n \ge 1$$
$$0! = 1! = 1 \qquad \text{by definition}$$

Example: 5!=5.4.3.2.1=120

6!=6.5.4.3.2.1=6.5!=720

Also, when dealing with factorials we need to be very careful with parenthesis. For instance, $(2n)! \neq 2n!$ as we can see if we write each of the following factorials out.

$$(2n)! = (2n)(2n-1)(2n-2)(2n-3).....(3)(2)(1)$$

$$2n! = 2[n(n-1)(n-2)(n-3).....(3)(2)(1)]$$

Again, we will run across factorials with parenthesis so don't drop them. This is often one of the more common mistakes that students make when the first run across factorials.

Okay, we are now ready for the test.

Ratio Test:

Suppose we have the series $\sum a_n$ .Define,

$$r = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Then,

1. If r <1 the series is absolutely convergent (and hence convergent).
2. If r >1 the series is divergent.

3. If r =1 the series may be divergent, convergent, or absolutely convergent.
I.e. from this test alone we cannot draw any conclusion about the convergence or divergence of the sequence $\sum a_n$

Notice that in the case of r =1 the ratio test is pretty much worthless and we would need to resort to a different test to determine the convergence of the series.

Also, the absolute value bars in the definition of r are absolutely required. If they are not there it will be possible for us to get the incorrect answer.

Example1: Determine if the series $\sum_{n=1}^{\infty}(-1)^n \frac{n^3}{3^n}$ is convergent or divergent.

Solution:

We use the Ratio Test with $a_n = (-1)^n \frac{n^3}{3^n}$

$$\Rightarrow a_{n+1} = (-1)^{n+1} \frac{(n+1)^3}{3^{n+1}} :$$

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{\frac{(-1)^n n^3}{3^n}}\right| = \frac{(n+1)^3}{3^{n+1}} \frac{3^n}{n^3}$$

Now taking the limit we have $\lim_{n\to\infty} \frac{(n+1)^3}{3^{n+1}} \frac{3^n}{n^3}$

$$= \lim_{n\to\infty} \frac{1}{3}\left(\frac{n+1}{n}\right)^3 = \frac{1}{3} \text{ which is less than 1}$$

Thus, by the Ratio Test, the given series is absolutely convergent and therefore convergent.

Example 2: Determine if the series $\sum_{n=0}^{\infty} \frac{n!}{5^n}$ is convergent or divergent.

Solution:

Now that we've worked one in detail we won't go into quite the detail with the rest of these. Here is the limit.

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)!5^n}{5^{n+1}n!}\right|$$

$$= \lim_{n\to\infty} \frac{1}{5}\frac{(n+1)!}{n!}$$

In order to do this limit we will need to eliminate the factorials. We simply can't do the limit with the factorials in it. To eliminate the factorials we will recall from our discussion on factorials above that we can always "strip out" terms from a factorial. If we do that with the numerator (in this case because it's the larger of the two) we get,

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+1}{5} = \infty > 1$$

So, by the Ratio Test this series diverges.

Example 3: Determine if the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$ is convergent or divergent.

Solution:

Let's first get r,

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{(n+1)^2+1}\frac{n^2+1}{(-1)^n}\right|$$

$$= \lim_{n\to\infty} \frac{n^2+1}{(n+1)^2+1} = 1$$

So, as implied earlier we get r =1 which means the ratio test is no good for determining the convergence of this series. We will need to resort to another test for this series. This series is an alternating series and so let's check the two conditions from that test.

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n^2+1} = 0$$

$$b_n = \frac{1}{n^2+1} > \frac{1}{(n+1)^2+1} = b_{n+1}$$

The two conditions are met and so by the Alternating Series Test this series is

convergent.

Example 4: Determine if the series $\sum_{n=1}^{\infty} \frac{(-10)^n}{4^{2n+1}(n+1)}$ is convergent or divergent.

Solution: Check that this series is convergent.

1.7.4 Root Test

This is the last test for series convergence that we're going to be looking at. As with the Ratio Test this test will also tell whether a series is absolutely convergent or not rather than simple convergence. This test is convenient to apply when n^{th} powers occur.

Root Test

Suppose that we have the series $\sum a_n$. Define,

$$r = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} |a_n|^{\frac{1}{n}} \quad \text{then,}$$

i) If r <1 the series is absolutely convergent (and hence convergent).

ii) If r >1 the series is divergent.

iii) If r =1 the series may be divergent, convergent, or absolutely convergent.

As with the ratio test, if we get r =1 the root test will tell us nothing and we'll need to use another test to determine the convergence of the series.

As an exercise for the reader please show that $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$

Example1: Determine if the series $\sum_{n=1}^{\infty} \frac{n^n}{3^{1+2n}}$ is convergent or divergent.

Solution:

There really isn't much to these problems other than computing the limit and then using the root test. Here is the limit for this problem.

$$r = \lim_{n\to\infty}\left|\frac{n^n}{3^{1+2n}}\right|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{n}{3^{\frac{1}{n}+2}} = \infty > 1$$

So, by the Root Test this series is divergent.

Example 2: Determine if the series $\sum_{n=1}^{\infty}\left(\frac{2n+3}{3n+2}\right)^n$ is convergent or divergent.

Solution:

Now let's find the value of r

$$r = \lim_{n\to\infty}\left|\frac{2n+3}{3n+2}\right|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{2n+3}{3n+2} = \frac{2}{3} < 1$$

Hence, by the Root Test this series is convergent.

Example 3: Determine if the series $\sum_{n=1}^{\infty}\frac{2^n}{n}$ is convergent or divergent.

Solution:

Similarly,

$$r = \lim_{n\to\infty}\left|\frac{2^n}{n}\right|^{\frac{1}{n}} = \lim_{n\to\infty}\frac{2}{(n)^{\frac{1}{n}}}$$

Now let's compute $\lim_{n\to\infty}(n)^{\frac{1}{n}} = 1$

Hence, $r = \lim_{n\to\infty}\left|\frac{2^n}{n}\right|^{\frac{1}{n}} = \lim_{n\to\infty}\frac{2}{(n)^{\frac{1}{n}}} = \frac{2}{1} = 2 > 1$

There fore by the root test the given series diverges.

Example 4: Determine if the series $\sum_{n=1}^{\infty}\frac{2^n}{n^2}$ is convergent or divergent.

Exercise!!

Note: if we get r =1 from the ratio test then the root test will also give also r =1 and so there isn't any reason to try the root test on anything that gives r =1 on the ratio test.

1.8 Generalized convergence tests

By combining the theorem "If $\sum_{n=1}^{\infty}|a_n|$ converges, then $\sum_{n=1}^{\infty}a_n$ converges" with our tests for nonnegative series, we obtain convergence tests that apply to any series, nonnegative or not.

Theorem :(Generalized convergence tests)

Let $\sum_{n=1}^{\infty}a_n$ be a series.

a) Generalized comparison test: If $|a_n| \le |b_n|$ for $n \ge 1$, and if $\sum_{n=1}^{\infty}|b_n|$ converges then $\sum_{n=1}^{\infty}a_n$ converges (absolutely).

b) Generalized Limit comparison test: If $\underset{n\to\infty}{Lim}\left|\frac{a_n}{b_n}\right| = L$, where L is a positive number, and if $\sum_{n=1}^{\infty}|b_n|$ converges then $\sum_{n=1}^{\infty}a_n$ converges (absolutely).

c) Generalized ratio test: Suppose that $a_n \neq 0$ for $n \ge 1$ and that $\underset{n\to\infty}{Lim}\left|\frac{a_{n+1}}{a_n}\right| = 0$ (Possibly ∞)

If $r < 1$ then $\sum_{n=1}^{\infty}a_n$ converges (absolutely).If $r > 1$ then $\sum_{n=1}^{\infty}a_n$ diverges.

If $r = 1$, then from this test alone we cannot draw any conclusion about the convergence of the series.

d) Generalized root test: Suppose that $\lim_{n\to\infty} \sqrt[n]{|a_n|} = r$ (possibly ∞)

If $r < 1$ then $\sum_{n=1}^{\infty} a_n$ converges (absolutely).If $r > 1$ then

$\sum_{n=1}^{\infty} a_n$ diverges.

If $r = 1$, then from this test alone we cannot draw any conclusion about the convergence of the series.

Since we consider different examples previously read them in detail and have the following as an exercise:

a) Show that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges absolutely for $|x| < 1$, converges conditionally for $x = -1$, and diverges for $x = 1$ and for $|x| > 1$.

b) Determine weather the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n!}{100^n}$ converges or diverges.

Chapter Two: Power series

2.1 Definition of power series at any point.

We have spent quite a bit of time talking about series now and with only a couple of exceptions we have spent most of that time talking about how to determine if a series will

converge or not. It's now time to start looking at some specific kinds of series and we will eventually reach the point where we can talk about applications of series.

In this section we are going to start talking about power series. A *power series about a,* or just *power series in (x-a),* is any series that can be written in the form,

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + ...$$

Where *a* and c_n are numbers. The c_n's are often called the *coefficients* of the series. The first thing to notice about a power series is that it is a function of *x*. That is different from any other kind of series that we've looked at to this point. In all the prior sections we've only allowed numbers in the series and now we are allowing variables to be in the series as well. This will not change how things work however. Everything that we know about series still holds.

If the value of a is 0 then the power series becomes $\sum_{n=0}^{\infty} c_n x^n$

Example:

$$a) \sum_{n=0}^{\infty} 2(x-3)^n = 2 + 2(x-3) + 2(x-3)^2 + ... + 2(x-3)^n + ...$$

$$b) \sum_{n=0}^{\infty} 3x^n = 3 + 3x + 3x^2 + 3x^3 + ... + 3x^n + ...$$

2.2 Convergence and divergence, radius and interval of convergence

In the discussion of power series convergence is still a major question that we will be dealing with. The difference is that the convergence of the series will now depend upon the value of *x* that we put into the series. A power series may converge for some values of *x* and not for other values of *x*.

Before we get too far into power series there is some terminology that we need to get out of the way.

First, as we will see in our examples, we will be able to show that there is a number *R* so

First, as we will see in our examples, we will be able to show that there is a number *R* so

that the power series will converge for, $|x-a| < R$ and will diverge for $|x-a| > R$. This number is called the *radius of convergence* for the series. Note that the series may or

may not converge if $|x-a| = R$. What happens at these points will not change the radius of convergence.

Secondly, the interval of all *x*'s, including the end points if need be, for which the power series converges is called the *interval of convergence* of the series.

These two concepts are fairly closely tied together. If we know that the radius of convergence of a power series is *R* then we have the following.

$x-a<R<x+a$	*the power series converges*
$x<a-R \text{ and } x>a-R$	*the power series diverges*

The interval of convergence must then contain the interval $a-R<x<a+R$ since we know that the power series will converge for these values. We also know that the interval of convergence can't contain *x*'s in the ranges $x<a-R \text{ and } x>a-R$ since we know the power series diverges for these value of *x*. Therefore, to completely identify the interval of convergence all that we have to do is determine if the power series will converge for $x=a-R$ or $x=a+R$. If the power series converges for one or both of these values then we'll need to include those in the interval of validity.

Example 1: Determine the radius of convergence and interval of convergence for the

Power series $\sum_{n=1}^{\infty} \frac{(-1)^n n}{4^n}(x+3)^n$.

Solution:

We know that this power series will converge for $x = -3$, but that's it at this point.

To determine the remainder of the *x*'s for which we'll get convergence we can use any of the tests that we have discussed to this point. After application of the test that we choose to work with we will arrive at condition(s) on *x* that we can use to determine which values of *x* for which the power series will converge and which values of *x* for which the power series will diverge. From this we can get the radius of convergence and most of the interval of convergence (with the possible exception of the endpoints.

With all that said, the best tests to use here are almost always the ratio or root test. Most

of the power series that we will be looking at are set up for one or the other. In this case we will use the ratio test.

$$r = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}(n+1)(x+3)^{n+1}}{4^{n+1}}\,\frac{4^n}{(-1)^n(n)(x+3)^n}\right|$$
$$= \lim_{n\to\infty}\left|\frac{-(n+1)(x+3)}{4n}\right|$$

Before going any farther with the limit let's notice that since x is not dependent on the limit and so it can be factored out of the limit. Notice as well that in doing this well need to keep the absolute value bars on it since we need to make sure everything stays positive and x could well be a value that will make things negative. The limit is then,

$$r = |x+3|\lim_{n\to\infty}\frac{n+1}{4n}$$
$$= \frac{1}{4}|x+3|$$

So, the ratio test tells us that if r <1 the series will converge, if r >1 the series will diverge, and if r =1 we don't know what will happen. So, we have,

$$\frac{1}{4}|x+3| < 1 \qquad \Rightarrow |x+3| < 4 \qquad \text{series converges}$$

$$\frac{1}{4}|x+3| > 1 \qquad \Rightarrow |x+3| > 4 \qquad \text{series diverges}$$

We will deal with the r =1 case in a bit. Notice that we now have the radius of convergence for this power series. These are exactly the conditions required for the radius of convergence. The radius of convergence for this power series is R = 4.

Now, let's get the interval of convergence. We will get most (if not all) of the interval by solving the first inequality from above

$$-4 < x+3 < 4$$
$$-7 < x < 7$$

So, most of the interval of validity is given by $-7 < x < 1$. All we need to do is determine if the power series will converge or diverge at the endpoints of this interval. Note that these values of x will correspond to the value of x that will give r =1.

The way to determine convergence at these points is to simply plug them into the original power series and see if the series converges or diverges using any test necessary.

$x = -7$:

In this case the series is,

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{4^n}(-4)^n = \sum_{n=1}^{\infty} \frac{(-1)^n n}{4^n}(-1)^n (4)^n$$

$$= \sum_{n=1}^{\infty} (-1)^n (-1)^n n$$

$$= \sum_{n=1}^{\infty} n$$

This series is divergent by the Divergence Test since $\lim_{n\to\infty} n = \infty \neq 0$.

$x = 1$:

In this case the series is,

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{4^n}(4)^n = \sum_{n=1}^{\infty} (-1)^n n$$

This series is also divergent by the Divergence Test since $\lim_{n\to\infty} (-1)^n n$ doesn't exist.

So, in this case the power series will not converge for either endpoint. The interval of convergence is then,

$$-7 < x < 1$$

In the above example the power series did not converge for either end point of the interval. Sometimes that will happen, but do not always expect that to happen. The power x-series could converge at either both of the end points or only one of the end points.

Example 2: Determine the radius of convergence and interval of convergence for the

power series $\sum_{n=0}^{\infty} n! x^n$

Solution:

We use the Ratio Test. If we let a_n , as usual, denote the *n*th term of the series,

then $a_n = n!x^n$ .If $x \neq 0$,we have

$$r = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right|$$

$$= \lim_{n\to\infty}|(n+1)x|$$

$$= |x|\lim_{n\to\infty}(n+1) = \infty$$

By the Ratio Test, the series diverges when $x \neq 0$. Thus, the given series converges only when $x = 0$.

Example 3: Determine the radius of convergence and interval of convergence for the

power series $\sum_{n=1}^{\infty}\frac{(x-3)^n}{n}$

Solution:

Similarly we have $a_n = \frac{(x-3)^n}{n}$

And By the Ratio test $r = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(x-3)^{n+1}}{n+1}\frac{n}{(x-3)^n}\right|$

$$= \lim_{n\to\infty}\left|(x-3)\frac{n}{n+1}\right| = |x-3|\lim_{n\to\infty}\frac{n}{n+1} = |x-3|$$

By the Ratio Test, the given series is absolutely convergent, and therefore convergent, $|x-3|<1$ and divergent when $|x-3|>1$ .Now so the series converges when $2 < x < 4$ and diverges when $x < 2 \; and \; x > 4$.

The Ratio Test gives no information when $|x-3| = 1$.So we must consider $x = 2$ and $x = 4$ separately. If we put $x = 4$ in the series, it becomes $\sum_{n=1}^{\infty}\frac{1}{n}$, the harmonic series, which is divergent. If $x = 2$, the series is $\sum_{n=1}^{\infty}(-1)^n\frac{1}{n}$, which converges by the Alternating Series Test. Thus, the given power series converges for $2 \le x < 4$.

Example 4: Determine the radius of convergence and interval of convergence for the

power series $\sum_{n=1}^{\infty} \frac{x^{2n}}{(-3)^n}$

Let's leave it as an exercise to the reader

2.3 Representations of Functions as Power Series

In this section we learn how to represent certain types of functions as sums of power series by manipulating geometric series or by differentiating or integrating such a series. You might wonder why we would ever want to express a known function as a sum of infinitely many terms. We will see later that this strategy is useful for integrating functions that don't have elementary antiderivatives, for solving differential equations, and for approximating functions by polynomials

We start with an equation that we have seen before:

Recall that the geometric series is

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad provided that |r| < 1$$

Don't forget as well that if $r \geq 1$ the series diverges.

Now, if we take $a=1$ and $r = x$ this becomes,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{provided that } |x| < 1 \qquad (1)$$

Turning this around we can see that we can represent the function

$$f(x) = \frac{1}{1-x} \qquad (2)$$

with the power series ,

$$\sum_{n=0}^{\infty} x^n \qquad \text{provided that } |x| < 1 \qquad (3)$$

This provision is important. We can clearly plug any number other than $x=1$ into the function, however, we will only get a convergent power series if $x < 1$. This means the equality in (1) will only hold if $x < 1$. For any other value of x the equality won't hold.

Note as well that we can also use this to acknowledge that the radius of convergence of this power series is $R = 1$ and the interval of convergence is $x < 1$.

This idea of convergence is important here. We will be representing many functions as power series and it will be important to recognize that the representations will often only be valid for a range of x's and that there may be values of x that we can plug into the function that we can't plug into the power series representation.

In this section we are going to concentrate on representing functions with power series where the functions can be related back to (2).

Example1: Find a power series representation for the given function $g(x) = \dfrac{1}{1+x^3}$ and determine it's interval of convergence.

Solution:

What we need to do here is to relate this function back to (2). This is actually easier than it might look. Recall that the x in (2) is simply a variable and can represent anything. So, a quick rewrite of $g(x)$ gives,

$$g(x) = \frac{1}{1-(-x^3)}$$

and so the $-x^3$ in $g(x)$ holds the same place as the x in (2). Therefore, all we need to do is replace the x in (3) and we've got a power series representation for $g(x)$.

$$g(x) = \sum_{n=0}^{\infty} (-x^3)^n \qquad \text{provided } \left|-x^3\right| < 1$$

Notice that we replaced both the x in the power series and in the interval of convergence. All we need to do now is a little simplification.

$$g(x) = \sum_{n=0}^{\infty} (-1)^n x^{3n} \qquad \text{provided } \left|x^3\right| < 1 \qquad \Rightarrow |x| < 1$$

So, in this case the interval of convergence is the same as the original power series. This usually won't happen. More often than not the new interval of convergence will be different form the original interval of convergence.

Example 2: Find a power series representation for the given function $h(x) = \dfrac{2x^2}{1+x^3}$ and determine it's interval of convergence.

Solution:

This function is similar to the previous function. The difference is the numerator and at first glance that looks to be an important difference. Since (2) doesn't have an x in the numerator it appears that we can't relate this function back to that.

However, now that we've worked the first example this one is actually very simple since we can use the result of the answer from that example. To see how to do this let's first rewrite the function a little.

$$h(x) = 2x^2 \frac{1}{1+x^3}$$

Now, from the first example we've already got a power series for the second term so let's use that to write the function as,

$$h(x) = 2x^2 \sum_{n=0}^{\infty} (-1)^n x^{3n} \qquad \text{provided } |x| < 1$$

Notice that the presence of x's outside of the series will NOT affect its convergence and so the interval of convergence remains the same.

The last step is to bring the coefficient into the series and we'll be done. When we do this make sure and combine the x's as well. We typically only want a single x in a power series

$$h(x) = \sum_{n=0}^{\infty} 2(-1)^n x^{3n+2} \qquad \text{provided } |x| < 1$$

So, hopefully we now have an idea on how to find the power series representation for some functions. Admittedly all of the functions could be related back to (2) but it's a start.

2.4 Differentiation and integration of power series

We now need to look at some further manipulation of power series that we will need to do on occasion. We need to discuss differentiation and integration of power series.

The sum of a power series is a function $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ whose domain is the interval of convergence of the series. We would like to be able to differentiate and integrate such functions, and the following theorem (which we won't prove) says that we

can do so by differentiating or integrating each individual term in the series, just as we would for a polynomial. This is called *term-by-term differentiation and integration*.

Let's start with differentiation of the power series,

$$f(x)=\sum_{n=0}^{\infty}c_n(x-a)^n=c_0+c_1(x-a)+c_2(x-a)^2+c_3(x-a)^3+.....+c_n(x-a)^n+....$$

Now, we know that if we differentiate a finite sum of terms all we need to do is differentiate each of the terms and then add them back up. With infinite sums there are some subtleties involved that we need to be careful with, but are somewhat beyond the scope of this course. While, we can always just differentiate all the terms in an infinite series is it not always guaranteed to be the power series representation of the derivative of the original function.

Nicely enough for us however, it is known that if the power series representation of *f(x)* has a radius of convergence of $R > 0$ then the term by term differentiation of the power series will also have a radius of convergence of R and (more importantly) will in fact be the power series representation of $f'(x)$ provided x is in the interval of convergence of the original function.

In other words,

$$f'(x)=c_c+2c_2(x-a)+3c_3(x-a)^2+.....+nc_n(x-a)^{n-1}+.....$$

$$=\sum_{n=1}^{\infty}nc_n(x-a)^{n-1}$$

Note the initial value of this series. It has been changed from *n=0* to *n=1*. This is an acknowledgement of the fact that the derivative of the first term is zero and hence isn't in the derivative. Notice however, that since the *n=0* term of the above series is also zero, we could start the series at *n=0* if it was required for a particular problem. In general however, this won't be done in this class.

We can now find formulas for higher order derivatives as well now.

$$f''(x)=\sum_{n=2}^{\infty}n(n-1)(x-a)^{n-2}$$

$$f'''(x)=\sum_{n=3}^{\infty}n(n-1)(n-2)(x-a)^{n-3}$$

.

.

Etc

Once again, notice that the initial value of *n* changes with each differentiation in order to acknowledge that a term from the original series differentiated to zero.

Let's now briefly talk about integration. Just as with the differentiation, when we've got an infinite series we need to be careful about just integration term by term. As long as we are in the interval of convergence for the original function we can do the integration. In this case we get,

$$\int f(x)dx = c + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

Notice that we pick up a constant of integration, *C*, that is outside the series here.

Let's summarize the differentiation and integration ideas before moving on to an example

If $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ has a radius of convergence of $R > 0$ then,

$$f'(x) = \sum_{n=1}^{\infty} n(x-a)^{n-1} \quad \text{and} \quad \int f(x)dx = c + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

and both of these also have a radius of convergence of *R*.

Now, let's see how we can use these facts to generate some more power series representations of functions.

Example: Find a power series representation for the following function and determine it's interval of convergence.

$$a)\ g(x) = \frac{1}{(1-x)^2} \qquad b)\ f(x) = \tan^{-1} x \qquad c)\ h(x) = \ln(5-x)$$

Solution:

(a) To do this problem let's notice that

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right)$$

Then since we've got a power series representation for

$$\frac{1}{1-x}$$

all that we'll need to do is differentiate that power series to get a power series representation for *g(x)*.

$$\begin{aligned} g(x) &= \frac{1}{(1-x)^2} \\ &= \frac{d}{dx}\left(\frac{1}{1-x}\right) \\ &= \frac{d}{dx}\sum_{n=0}^{\infty} x^n \\ &= \sum_{n=1}^{\infty} n x^{n-1} \end{aligned}$$

Then since the original power series had a radius of convergence of $R=1$ the derivative, and hence *g(x)*, will also have a radius of convergence of $R=1$.

(b) We observe that $f'(x) = \frac{1}{(1+x^2)}$ and find the required series by integrating the power series for $\frac{1}{(1+x^2)}$

$$\begin{aligned} \tan^{-1} x &= \int \frac{1}{1+x^2}\,dx = \int (1 - x^2 + x^4 - x^6 + ...)\,dx \\ &= c + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ... \end{aligned}$$

To find *c* we put $x =$ and obtain $c = \tan^{-1} 0 = 0$.

Therefore,

$$\begin{aligned} \tan^{-1} x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ... \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \end{aligned}$$

Since the radius of convergence of the series for $\frac{1}{(1+x^2)}$ is 1, the radius of convergence of this series for $\tan^{-1} x$ is also 1.

2.5 Taylor series; Taylor polynomial and application

In the preceding section we were able to find power series representations for certain restricted class of functions. Here we investigate more general problems: Which functions have power series representations? How can we find such representations?

In the previous section we started looking at writing down a power series representation of a function. The problem with the approach in that section is that everything came down to needing to be able to relate the function in some way to

$$\frac{1}{1-x}$$

and while there are many functions out there that can be related to this function there are many more that simply can't be related to this.

So, without taking anything away from the process we looked at in the previous section, we need to do is come up with a more general method for writing a power series representation for a function.

So, for the time being, let's make two assumptions. First, lets assume that the function *f(x)* does in fact have a power series representation about $x=a$,

$$f(x)=\sum_{n=0}^{\infty}c_n(x-a)^n=c_0+c_1(x-a)+c_2(x-a)^2+c_3(x-a)^3+.....+c_n(x-a)^n+....$$

Next, we will need to assume that the function, *f(x)*, has derivatives of every order and that we can in fact find them all.

Now that we've assumed that a power series representation exists we need to determine what the coefficients, c_n, are. This is easier than it might at first appear to be. Let's first just evaluate everything at $x=a$. This gives,

$$f(a)=0$$

So, all the terms except the first are zero and we now know what c_0 is. Unfortunately, there isn't any other value of x that we can plug into the function that will allow us to quickly find any of the other coefficients. However, if we take the derivative of the function (and its power series) then plug in $x=a$ we get,

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + ... + nc_n(x-a)^{n-1} + ...$$

$$f'(a) = c_1$$

and we now know c_1.

Lets' continue with this idea and find the second derivative.

$$f''(x) = 2c_2 + 3(2)c_3(x-a) + 4(3)c_4(x-a)^2 + ...$$

$$f''(a) = 2c_2$$

So, it looks like, $c_2 = \dfrac{f''(a)}{2}$

Similarly using the third derivative gives,

$$c_3 = \frac{f'''(a)}{3(2)} = \frac{f'''(a)}{3!}$$

And also from the fourth derivative we get

$$c_4 = \frac{f^4(a)}{4(3)(2)} = \frac{f^4(a)}{4!}$$

Hopefully by this time you have seen the pattern here. It looks like, in general, we have got the following formula for the coefficients.

$$c_n = \frac{f^n(a)}{n!}$$

This even works for $n=0$ if you recall that 0! =1 and define $f^{(0)}(x) = f(x)$.

So, provided a power series representation for the function $f(x)$ about $x=a$ exists it will have the form,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$$

This is called the *Taylor Series* for $f(x)$ about $x=a$.

In the case that $a=0$, so the Taylor Series about $x=0$, we have,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(0)}{n!}(x)^n$$

This is called a *Maclaurin Series* for $f(x)$.

Before working any examples of Taylor Series we first need to address the assumption that a Taylor Series will in fact exist for a given function. Let's first get some notation

out of the way first.

We will first split the Taylor Series formula as follows,

$$\sum_{n=0}^{\infty}\frac{f^{n}(a)}{n!}(x-a)^{n}=\sum_{i=0}^{n}\frac{f^{i}(a)}{i!}(x-a)^{i}+\sum_{i=n+1}^{\infty}\frac{f^{i}(a)}{i!}(x-a)^{i}$$

$$=T_{n}(x)+R_{n}(x)$$

Where,

$T_{n}(x)=\sum_{i=0}^{n}\frac{f^{i}(a)}{i!}(x-a)^{i}$ is called the *nth degree Taylor Polynomial of f(x)* and

$R_{n}(x)=\sum_{i=n+1}^{\infty}\frac{f^{i}(a)}{i!}(x-a)^{i}$ is called the *remainder*

Note that the Taylor polynomial really is a polynomial and its degree will be at most *n*. We will see a nice application of Taylor polynomials in the next section.

Theorem: Suppose that $f(x)=T_{n}(x)+R_{n}(x).$ If, $\lim_{n\to\infty}R_{n}(x)=0$ for $|x-a|<R$, then

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{n}(a)}{n!}(x-a)^{n} \text{ on } |x-a|<R$$

In general showing that $\lim_{n\to\infty}R_{n}(x)=0$ is a somewhat difficult process and so we will be assuming that this can be done for some *R* in all of the examples that we will be looking at.

Example 1: Find the Taylor Series for $f(x)=e^{x}$ about $x=0$.

Solution:

This is actually one of the easier Taylor Series that we'll be asked to compute. To find the Taylor Series for a function we will need to determine a general formula for $f^{n}(a)$. This is one of the few functions where this is easy to do right from the start.

To get a formula for $f^{0}(a)$ all we need to do is recognize that,

$$f^{n}(x)=e^{x} \qquad \textit{for } n=0,1,2,3,4,.....$$

And so, $f^{n}(0)=e^{0}=1$ $\qquad$ *for* $n=0,1,2,3,4,.....$

Therefore, the Taylor series for $f(x) = e^x$ about $x = 0$ is,

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Example 2: Find the Taylor Series for $f(x) = x^4 e^{-3x^2}$ about $x = 0$

Solution:

For this example we will take advantage of the fact that we already have a Taylor Series for e^x about *x=0*. In this example, unlike the previous example, doing this directly would be significantly longer and more difficult.

$$x^4 e^{-3x^2} = x^4 \sum_{n=0}^{\infty} \frac{(-3x^2)^n}{n!}$$

$$= x^4 \sum_{n=0}^{\infty} \frac{(-3)^n x^{2n}}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(-3)^n x^{2n+4}}{n!}$$

Example 3: Find the Taylor Series for $f(x) = e^{-x}$ about $x = -4$.

Solution:

Finding a general formula for $f^{n}(-4)$ is fairly simple.

$f^{n}(x) = (-1)^n e^{-x}$ $\qquad$ $f^{n}(-4) = (-1)^n e^4$

The Taylor Series is then,

$$e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n e^4}{n!} (x+4)^n$$

Example 4: Find the Taylor Series for $f(x) = \sin(x)$ about *x=0*.

Solution:

First we will need to take some derivatives of the function and evaluate them at *x=0*.

$f^{(0)}(x) = \sin x$ $\qquad$ $f^{(0)}(0) = 0$

$f^{(1)}(x) = \cos x$ $\qquad$ $f^{(1)}(0) = 1$

$f^{(2)}(x) = -\sin x$ $\qquad$ $f^{(2)}(0) = 0$

$$f^{(3)}(x) = -\cos x \qquad f^{(3)}(0) = -1$$
$$f^{(4)}(x) = \sin x \qquad f^{(4)}(0) = 0$$
$$\vdots \qquad\qquad \vdots$$

So, we get a similar pattern for this one. Let's plug the numbers into the Taylor Series.

$$\sin x = \sum_{n=0}^{\infty} \frac{f^{n}(0)}{n!} x^{n}$$
$$= \frac{1}{1!}x - \frac{1}{3!}x^{3} + \frac{3}{5!}x^{5} - \frac{1}{7!}x^{7} + \ldots\ldots$$

In this case we only get terms that have an odd exponent on *x* and as with the last problem once we ignore the zero terms there is a clear pattern and formula. So renumbering the terms as we did in the previous example we get the following Taylor Series,

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)x^{2n+1}}{(2n+1)!}$$

Exercise: Find the Taylor Series for the following

a) $f(x) = \ln(x)$ about $x = 2$

b) $f(x) = \cos x$ about $x = 0$

c) $f(x) = \frac{1}{x^{2}}$ about $x = -1$

Now, that we know how to represent function as power series we can now talk about at least a couple of applications of series.

There are in fact many applications of series, unfortunately most of them are beyond the scope of this course. One application of power series (with the occasional use of Taylor Series) is in the field of Ordinary Differential Equations when finding series solutions to Differential Equations. If you are interested in seeing how that works you can check out the next course Applied Mathematics III.

Another application of series arises in the study of Partial Differential Equations. One of the more commonly used methods in that subject makes use of Fourier series.

Many of the applications of series, especially those in the differential equations fields,

rely on the fact that functions can be represented as a series. In these applications it is very difficult, if not impossible, to find the function itself. However, there are methods of determining the series representation for the unknown function.

Example: For the function $f(x) = \cos(x)$ plot the function as well as $T_2(x)$, $T_4(x)$ and $T_8(x)$ on the same graph for the interval [-4,4].

Solution: Here is the general formula for the Taylor polynomials for cosine

$$T_n(x) = \sum_{i=0}^{n} \frac{(-1)^i x^{2i}}{(2i)!}$$

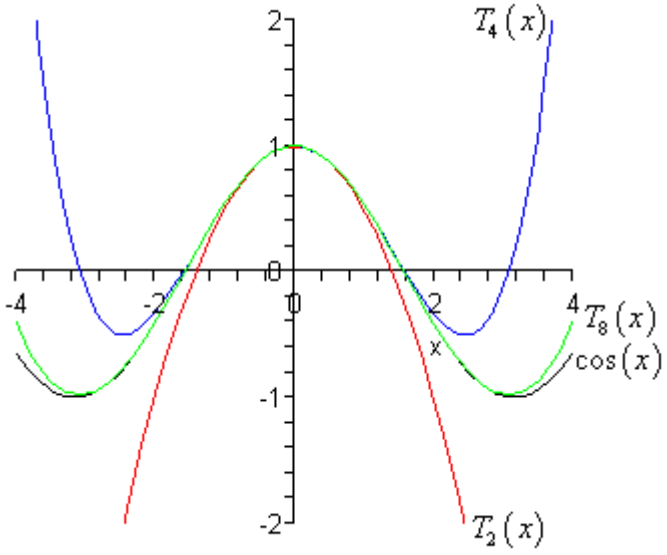
The three Taylor polynomials that we've got are then,

$$T_2(x) = 1 - \frac{x^2}{2}$$

$$T_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

$$T_8(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \frac{x^8}{40320}$$

Here is the graph of these three Taylor polynomials as well as the graph of cosine.



As we can see from this graph as we increase the degree of the Taylor polynomial it starts to look more and more like the function itself. In fact by the time we get to $T_8(x)$ the only difference is right at the ends. The higher the degree of the Taylor polynomial the

better it approximates the function.
Also the larger the interval the higher degree Taylor polynomial we need to get a good approximation for the whole interval.

Chapter Three: Orthogonal functions and Fourier series

This is going to be a short section. We just need to have a brief discussion about a couple of ideas that we will be dealing with on occasion as we move into the next topic of this chapter.

3.1 Periodic Function

The first topic we need to discuss is that of a periodic function. A function is said to be periodic with period ***T*** if the following is true,

$$f(x+T) = f(x) \quad \text{for all x}$$

Fact: If *f* and *g* are both periodic functions with period *T* then so is $f+g \;\; and \;\; fg$.

This is easy enough to prove so let's do that.

Since $f \text{ and } g$ are periodic functions by definition of periodic function we have

$f(x+T) = f(x)$ and $g(x+T) = g(x)$

Now to prove that above:

m

$(fg)(x+T) = f(x+T)g(x+T) = f(x)g(x)$

The two periodic functions that most of us are familiar are sine and cosine and in fact we'll be using these two functions regularly in the remaining sections of this chapter. So, having said that let's close off this discussion of periodic functions with the following fact,

$Cos(\omega x)$ and $\sin(\omega x)$ are periodic functions of period $T = \dfrac{2\pi}{\omega}$

3.2 Even and Odd Functions

The next quick idea that we need to discuss is that of even and odd functions.

Recall that a function is said to be even if, $f(x) = f(-x)$

And a function is said to be odd if, $f(-x) = -f(x)$

The standard examples of even functions are $f(x) = x^2$ and $g(x) = \cos(x)$ while the standard examples of odd functions are $f(x) = x^3$ and $g(x) = \sin(x)$.

The following fact about certain integrals of even/odd functions will be useful in some of our work.

1) If $f(x)$ is an even function then,

$$\int_{-L}^{L} f(x)dx = 2\int_{0}^{L} f(x)dx$$

2) If $f(x)$ is an odd function then,

$$\int_{-L}^{L} f(x)dx = 0$$

Note that this fact is only valid on a "symmetric" interval, *i.e.* an interval in the form [-L, L]. If we are not integrating on a "symmetric" interval then the fact may or may not be true.

3.3 Orthogonal Functions

The final topic that we need to discuss here is that of orthogonal functions. This idea will be integral to what we'll be doing in the remainder of this chapter and in the next chapter as we discuss one of the basic solution methods for partial differential equations.

Let's first get the definition of orthogonal functions out of the way.

Definition:

1) Two non-zero functions $f(x)$ and $g(x)$ are said to be orthogonal on $a \le x \le b$ if, $\int_a^b f(x)g(x)dx = 0$
2) A set of non-zero functions, $\{f_i(x)\}$, is said to be mutually orthogonal on $a \le x \le b$ (or just an orthogonal set if we're being lazy) if $f_i(x)$ and $f_j(x)$ are orthogonal for every $i \ne j$ .In other words,

$$\int_a^b f_i(x) f_j(x)dx = \begin{cases} 0 & i \ne j \\ c > 0 & i = j \end{cases}$$

Note that in the case of $i = j$ for the second definition we know that we'll get a positive value from the integral because, $\int_a^b f_i(x) f_i(x)dx = \int_a^b [f_i(x)]^2 dx > 0$.

Recall that when we integrate a positive function we know the result will be positive as well.

Also note that the non-zero requirement is important because otherwise the integral would be trivially zero regardless of the other function we were using.

Before we work some examples there are a nice set of trig formulas that we'll need to help us with some of the integrals.

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha-\beta)+\sin(\alpha+\beta)]$$

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)]$$

Example 1: Show that $\left\{\cos\left(\frac{n\pi x}{L}\right)\right\}_{n=0}^{\infty}$ is mutually orthogonal on $-L \le x \le L$.

Solution: This is not too difficult to do. All we really need to do is evaluate the following integral.

$$\int_{-L}^{l}\cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

Before we start evaluating this integral let's notice that the integrand is the product of two even functions and so must also be even. This means that we can use Fact 3 above to write the integral as,

$$\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 2\int_{0}^{L}\cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

There are two reasons for doing this. First having a limit of zero will often make the evaluation step a little easier and that will be the case here. We will discuss the second reason after we're done with the example.

Now, in order to do this integral we'll actually need to consider three cases

Case1: $n = m = 0$

In this case the integral is very easy and is,

$$\int_{-L}^{L} \cos(0)\cos(0)\,dx = \int_{-L}^{L} dx = 2\int_{0}^{L} dx = 2L$$

Case 2: $n = m \neq 0$

This integral is a little harder than the first case, but not by much (provided we recall a simple trig formula). The integral for this case is,

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi x}{L}\right)dx = \int_{-L}^{L} \cos^2\left(\frac{n\pi x}{L}\right)dx = 2\int_{0}^{L} \cos^2\left(\frac{n\pi x}{L}\right)dx$$

$$= \int_{0}^{L} 1 + \cos\left(\frac{2n\pi x}{L}\right)dx$$

$$= \left(x + \frac{1}{2n\pi}\sin\left(\frac{2n\pi x}{L}\right)\right)\Bigg|_{0}^{L} = L + \frac{L}{2n\pi}\sin(2n\pi)$$

Now, at the point we need to recall that n is an integer and so $\sin(2n\pi) = 0$ and our final value for the is,

$$\int_{-L}^{L} \cos^2\left(\frac{n\pi x}{L}\right)dx = 2\int_{0}^{L} \cos^2\left(\frac{n\pi x}{L}\right)dx = L$$

The first two cases are really just showing that if $n = m$ the integral is not zero (as it shouldn't be) and depending upon the value of *n* (and hence *m*) we get different values of the integral. Now we need to do the third case and this, in some ways, is the important case since we must get zero out of this integral in order to know that the set is an orthogonal set. So, let's take care of the final case.

Case 3: $n \neq m$

This integral is the "messiest" of the three that we've had to do here. Let's just start off by writing the integral down.

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 2\int_{0}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

In this case we can't combine/simplify as we did in the previous two cases. We can however, acknowledge that we've got a product of two cosines with different arguments and so we can use one of the trig formulas above to break up the product as follows,

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 2\int_{0}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

$$= \int_{0}^{L} \cos\left(\frac{(n-m)\pi x}{L}\right) + \cos\left(\frac{(n+m)\pi x}{L}\right)dx$$

j

$$= \frac{L}{(n-m)\pi}\sin((n-m)\pi) + \frac{L}{(n+m)\pi}\sin((n+m)\pi)$$

Now, we know that *n* and *m* are both integers and so $n-m$ and $n+m$ are also integers and so both of the sines above must be zero and all together we get,

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 2\int_{0}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 0$$

So, we've shown that if $n \neq m$ the integral is zero and if $n = m$ the value of the integral is a positive constant and so the set is mutually orthogonal.

Example 2: Show that $\left\{\sin\left(\frac{n\pi x}{L}\right)\right\}_{n=1}^{\infty}$ is mutually orthogonal on $-L \le x \le L$.

Let's leave this for the reader as an exercise!!! Do not jump this one since we use it repeatedly.

3.4 Fourier sine series and Fourier cosine series

3.4.1 Fourier Sine series

In this section we are going to start taking a look at Fourier series. We should point out that this is a subject that can span a whole class and what we'll be doing in this section (as well as the next couple of sections) is intended to be nothing more than a very brief look at the subject. The point here is to do just enough to allow us to do some basic solutions to partial differential equations in the next chapter. There are many topics in the study of Fourier series that we'll not even touch upon here.

So, with that out of the way let's get started, although we're not going to start off with Fourier series. Let's instead think back to our Calculus class where we looked at Taylor Series. With Taylor Series we wrote a series representation of a function $f(x)$, as a series

whose terms were powers of $x-a$ for some $x=a.$ with some conditions we were able to show that,

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{n}(a)}{n!}(x-a)^{n}$$

and that the series will converge to $f(x)$ on $|x-a|<R$ for some R that will be dependent upon the function itself.

There is nothing wrong with this, but it does require that derivatives of all orders exist at $x=a$ .Or in other words $f^{n}(a)$ exists for $n=0,1,2,3,4,.....$ Also for some functions the value of R may end up being quite small.

These two issues (along with a couple of others) mean that this is not always the best way or writing a series representation for a function. In many cases it works fine and there will be no reason to need a different kind of series. There are times however where another type of series is either preferable or required.

We are going to build up an alternative series representation for a function over the course of the next couple of sections. The ultimate goal for the rest of this chapter will be to write down a series representation for a function in terms of sines and cosines.

We will start things off by assuming that the function, $f(x),$ we want to write a series representation for is an odd function i.e. $f(-x)=-f(x)$ .Because $f(x)$ is odd it makes some sense that should be able to write a series representation for this in terms of sines only (since they are also odd functions).

What we will try to do here is writing $f(x)$ as the following series representation, called a *Fourier sine series*, on $-L\le x\le L$.

$$\sum_{n=1}^{\infty}B_{n}\sin\left(\frac{n\pi x}{L}\right)$$ Where B_n is called the coefficients.

There are a couple of issues to note here. First, at this point, we are going to assume that the series representation will converge to $f(x)\ on\ -L \le x \le L$ .Assuming that the series does converge to $f(x)$ it is interesting to note that, unlike Taylor Series, this representation will always converge on the same interval and that the interval does not depend upon the function.

The question now is how to determine the coefficients B_n , in the series.

Let's start with the series above and multiply both sides by $\sin\left(\frac{m\pi x}{L}\right)$ where *m* is a fixed integer in the range $\{1,2,3,4.....\}$. In other words we multiply both sides by any of the sines in the set of sines that we are working with here. Doing this gives,

$$f(x)\sin\left(\frac{m\pi x}{L}\right) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)$$

Now, let's integrate both sides of this from $x = -L$ to $x = L$.

$$\int_{-L}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx = \int_{-L}^{L}\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx$$

At this point we've got a small issue to deal with. We know from Calculus that an integral of a finite series (more commonly called a finite sum….) is nothing more than the (finite) sum of the integrals of the pieces. In other words for finite series we can interchange an integral and a series. For infinite series however, we cannot always do this. For some integrals of infinite series we cannot interchange an integral and a series. Luckily enough for us we actually can interchange the integral and the series in this case.

Doing this gives and factoring the constant, B_n , out of the integral gives,

$$\int_{-L}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx = \sum_{n=1}^{\infty}\int_{-L}^{L} B_n \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx$$

$$= \sum_{n=1}^{\infty} B_n \int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx$$

Note that from example 2 above we have $\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx = \begin{cases} L & if\ n = m \\ 0 & if\ n \neq m \end{cases}$

i.e. $\left\{\sin\left(\frac{n\pi x}{L}\right)\right\}_{n=1}^{\infty}$ is orthogonal on $-L \le x \le L$.

So, what does this mean for us? As we work through the various values of *n* in the series and compute the value of the integrals all but one of the integrals will be zero. The only non-zero integral will come when we have $n = m$, in which case the integral has the value of *L*. Therefore, the only non-zero term in the series will come when we have $n = m$ and our equation becomes,

$$\int_{-L}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx = B_m L$$

Finally all we need to do is dividing by *L* and we know have an equation for each of the coefficient

$$B_m = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx \qquad m = 1,2,3,.....$$

Next, note that because we're integrating two odd functions the integrand of this integral is even and so we also know that,

$$B_m = \frac{2}{L}\int_{0}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx \qquad m = 1,2,3,....$$

Summarizing all this work up the Fourier sine series of an odd function $f(x)\ on\ -L \le x \le L$ is given by,

$$f(x)=\sum_{n=1}^{\infty}B_n\sin\left(\frac{n\pi x}{L}\right)\quad\text{, Where }\quad B_n=\frac{1}{L}\int_{-L}^{L}f(x)\sin\left(\frac{n\pi x}{L}\right)dx\qquad n=1,2,3,....$$

Example1: Find the Fourier sine series for $f(x)=x\ on -L\le x\le L$.

Solution: First note that the function we're working with is in fact an odd function and so this is something we can do. There really isn't much to do here other than to compute the coefficients for $f(x)=x$.

Here is that work and note that we're going to leave the integration by parts details to you to verify. Don't forget that *n*, *L*, and π are constants!

$$B_n=\frac{1}{L}\int_{-L}^{L}x\sin\left(\frac{n\pi x}{L}\right)dx=\frac{2}{L}\int_{0}^{L}x\sin\left(\frac{n\pi x}{L}\right)dx$$

$$=\frac{2}{L}\left(\frac{L}{n^2\pi^2}\right)\left(L\sin\left(\frac{n\pi x}{L}\right)-n\pi x\cos\left(\frac{n\pi x}{L}\right)\right)\Bigg|_0^L$$

$$=\frac{2}{n^2\pi^2}\left(L\sin(n\pi)-n\pi L\cos(n\pi)\right)$$

These integrals can, on occasion, be somewhat messy especially when we use a general *L* for the endpoints of the interval instead of a specific number.

Now, taking advantage of the fact that *n* is an integer we know that

$\sin(n\pi)=0\,and\cos(n\pi)=(-1)^n$.We therefore have,

$$B_n=\frac{2}{n^2\pi^2}\left(-n\pi L(-1)^n\right)=\frac{(-1)^{n+1}2L}{n\pi}\qquad n=1,2,3,4,.....$$

The Fourier sine series is then,

$$x=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)=\frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{L}\right)$$

At this point we should probably point out that we will be doing most, if not all, of our work here on a general interval ($-L\le x\le L$ or $0\le x\le L$)instead of intervals with specific numbers for the endpoints. There are a couple of reasons for this. First, it gives a much more general formula that will work for any interval of that form which is always nice. Secondly, when we run into this kind of work in the next chapter it will also be on general intervals so we may as well get used to them now.

Now, finding the Fourier sine series of an odd function is fine and good but what if, for some reason, we wanted to find the Fourier sine series for a function that is not odd? To see how to do this we're going to have to make a change. The above work was done on the interval $-L \le x \le L$ .In the case of a function that is not odd we'll be working on the interval $0 \le x \le L$. The reason for this will be made apparent in a bit.

So, we are now going to do is to try to find a series representation for $f(x)$ on the interval $0 \le x \le L$ hat is in the form,

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

or in other words,

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

As we did with the Fourier sine series on $-L \le x \le L$ we are going to assume that the series will in fact converge to $f(x)$ and we will hold off discussing the convergence of the series in another section not on this module.

There are two methods of generating formulas for the coefficients, B_n although we'll see in a bit that they really the same way, just looked at from different perspectives.

The first method is to just ignore the fact that $f(x)$ is odd and proceed in the same manner that we did above only this time we'll take advantage of the fact that we left as an exercise in the previous section that $\left\{\sin\left(\frac{n\pi x}{L}\right)\right\}_{n=1}^{\infty}$ also forms an orthogonal set on $0 \le x \le L$ and that,

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx = \begin{cases} \frac{L}{2} & \text{if } n = m \\ 0 & \text{if } n \ne m \end{cases}$$

So, if we do this then all we need to do is multiply both sides of our series by $\sin\left(\frac{m\pi x}{L}\right)$,integral from 0 to L and interchange the integral and series to get,

$$\int_0^L f(x)\sin\left(\frac{m\pi x}{L}\right)dx = \sum_{n=1}^{\infty} B_n \int_0^L \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx$$

Now, plugging in for the integral we arrive at,

$$\int_0^L f(x)\sin\left(\frac{m\pi x}{L}\right)dx = B_m\left(\frac{L}{2}\right)$$

Upon solving for the coefficient we arrive at,

$$B_m = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{m\pi x}{L}\right)dx \quad m = 1,2,3,4,.....$$

Note that this is identical to the second form of the coefficients that we arrived at above by assuming $f(x)$ was odd and working on the interval $-L \le x \le L$. The fact that we arrived at essentially the same coefficients is not actually all the surprising as we'll see once we have looked the second method of generating the coefficients.

Before we look at the second method of generating the coefficients we need to take a brief look at another concept. Given a function, $f(x)$, we define the *odd extension* of $f(x)$ to be the new function,

$$g(x) = \begin{cases} f(x) & if\ 0 \le x \le L \\ f(-x) & if\ -L \le x \le 0 \end{cases}$$

It's pretty easy to see that this is an odd function.

$$g(-x) = -f(-(-x)) = -f(x) = -g(x) \qquad \text{for} \quad 0 < x < L$$

and we can also know that on $0 \le x \le L$ we have that $g(x) = f(x)$. Also note that if $f(x)$ is already an odd function then we in fact get $g(x) = f(x)$ on $-L \le x \le L$.

Example 2: Sketch the odd extension of the given function.

$$f(x) = L - x \qquad \text{on } 0 \le x \le L$$

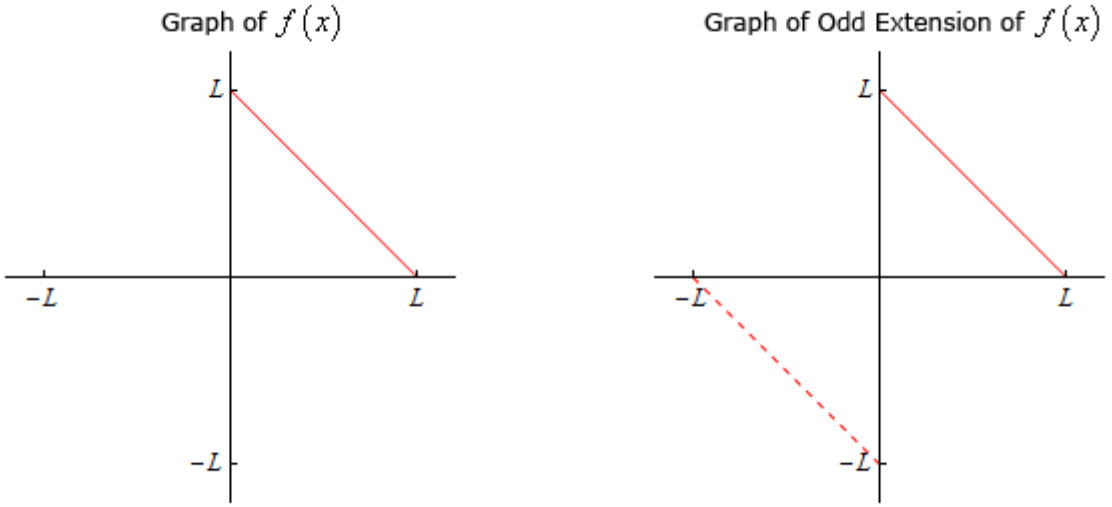
Solution: Not much to do with this other than to define the odd extension and then sketch it

Here is the odd extension of this function.

$$g(x) = \begin{cases} f(x) & \text{if } 0 \le x \le L \\ f(-x) & \text{if } -L \le x \le 0 \end{cases}$$

$$= \begin{cases} L - x & \text{if } 0 \le x \le L \\ -L - x & \text{if } -L \le x \le 0 \end{cases}$$

Below is the graph of both the function and its odd extension. Note that we've put the "extension" in with a dashed line to make it clear the portion of the function that is being added to allow us to get the odd extension.



Example 3: Find the Fourier sine series for $f(x) = L - x \qquad \text{on } 0 \le x \le L$

Solution: There really isn't much to do here other than computing the coefficients so here they are,

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx = \frac{2}{L}\int_0^L (L-x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$= \frac{2}{L}\left(-\frac{L}{n^2\pi 2}\right)\left[L\sin\left(\frac{n\pi x}{L}\right) - n\pi(x-L)\cos\left(\frac{n\pi x}{L}\right)\right]_0^L$$

$$= \frac{2}{L}\left[\frac{L^2}{n^2\pi 2}(n\pi - \sin n\pi)\right]$$

$$= \frac{2L}{n\pi}$$

In the simplification process do not forget that *n* is an integer.

So, with the coefficients we get the following Fourier sine series for this function.

$$f(x) = \sum_{n=1}^{\infty}\frac{2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\text{jj}$$

Example 4: Find the Fourier sine series for $f(x) = \pi - x$ on $0 \le x \le \pi$

Answer: $f(x) = \sum_{n=1}^{\infty}\frac{2}{n}\sin(nx)$

3.4.2 Fourier Cosine Series

In this section we're going to take a look at Fourier cosine series. We will start off much as we did in the previous section where we looked at Fourier sine series. Let's start by assuming that the function, $f(x)$ we will be working with initially is an even function [i.e. $f(-x) = f(x)$] and that we want to write a series representation for this function on $-L \le x \le L$ in terms of cosines (which are also even). In other words we are going to look for the following,

$$f(x) = \sum_{n=0}^{\infty}A_n\cos\left(\frac{n\pi x}{L}\right)$$

This series is called a *Fourier cosine series* anjd note that in this case (unlike with Fourier sine series) we are able to start the series representation at $n = 0$ since that term will not be zero as it was with sines. Also, as with Fourier Sine series, the argument of $\frac{n\pi x}{L}$ in the cosines is being used only because it is the argument that we will be running into in the next topic. The only real requirement here is that the given set of functions we're using be orthogonal on the interval we're working on.

Note as well that we are assuming that the series will in fact converge to $f(x)\ on - L \le x \le L$ at this point.

So, to determine a formula for the coefficients, A_n, we'll use the fact that $\left\{\cos\left(\frac{n\pi x}{L}\right)\right\}_{n=0}^{\infty}$ do form an orthogonal set on the interval $-L \le x \le L$ as we showed in the previous section. In that section we also derived the following formula that we will need in a bit.

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = \begin{cases} 2L & if\ n = m = 0 \\ L & if\ n = m \ne 0 \\ 0 & if\ n \ne m \end{cases}$$

We'll get a formula for the coefficients in almost exactly the same fashion that we did in the previous section. We'll start with the representation above and multiply both sides by $\cos\left(\frac{m\pi x}{L}\right)$ where *m* is a fixed integer in the range $\{1,2,3,4,.....\}$.Doing this gives,

$$f(x)\cos\left(\frac{m\pi x}{L}\right) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)$$

Next, we integrate both sides from $x = -L$ to $x = L$ and as we were able to do with the Fourier Sine series we can again interchange the integral and the series.

$$\int_{-L}^{L} f(x)\cos\left(\frac{m\pi x}{L}\right)dx = \int_{-L}^{L}\sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

$$= \sum_{n=0}^{\infty} A_n \int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

We now know that the all of the integrals on the right side will be zero except when $n = m$ because the set of cosines form an orthogonal set on the interval $-L \le x \le L$ However, we need to be careful about the value of *m* (or *n* depending on the letter you want to use). So, after evaluating all of the integrals we arrive at the following set of formulas for the coefficients.

For $m = 0$, $\int_{-L}^{L} f(x)dx = A_0(2L)$ $\Rightarrow A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx$

For $m \neq 0$, $\int_{-L}^{L} f(x)\cos\left(\frac{m\pi x}{L}\right)dx = A_m L$ $\Rightarrow A_m = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{m\pi x}{L}\right)dx$

Summarizing everything up then, the Fourier cosine series of an even function, $f(x)\ on - L \le x \le L$ is given by,

$$f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \quad \text{Where} \quad A_n = \begin{cases} \frac{1}{2L}\int_{-L}^{L} f(x)dx & n = 0 \\ \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx & n \neq 0 \end{cases}$$

Example 1: Find the Fourier cosine series for $f(x) = x^2$ on $-L \le x \le L$

Solution: We clearly have an even function here and so all we really need to do is compute the coefficients and they are liable to be a little messy because we'll need to do integration by parts twice. We'll leave most of the actual integration details to you to verify.

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx = \frac{1}{L}\int_{0}^{L} f(x)dx$$

$$= \frac{1}{L}\int_{0}^{L} x^2 dx = \frac{L^2}{3}$$

Again, $A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{2}{L}\int_{0}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$

$$= \frac{2}{L}\int_0^L x^2 \cos\left(\frac{n\pi x}{L}\right)dx$$

$$= \frac{2}{L}\left(\frac{L}{n^3\pi^3}\right)\left(2\ln \pi x \cos\left(\frac{n\pi x}{L}\right) + \left(n^2\pi^2x^2 - 2L^2\right)\sin\left(\frac{n\pi x}{L}\right)\right)\Bigg|_0^L$$

Substituting the values we get,

$$A_n = \frac{4L^2(-1)^n}{n^2\pi^2} \qquad n = 1,2,3,4,.....$$

The coefficients are then, $A_0 = \frac{L^2}{3}$ and $A_n = \frac{4L^2(-1)^n}{n^2\pi^2}$ $n = 1,2,3,4,.....$

The Fourier cosine series is then,

$$x^2 = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$
$$= \frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2(-1)^n}{n^2\pi 2}\cos\left(\frac{n\pi x}{L}\right)$$

Note that we will often strip out $n = 0$ from the series as we have done here because it will almost always be different from the other coefficients and it allows us to actually plug the coefficients into the series.

Now, just as we did in the previous section let's ask what we need to do in order to find the Fourier cosine series of a function that is not even. As with Fourier sine series when we make this change we'll need to move onto the interval $0 \le x \le L$ now instead of $-L \le x \le L$ and again we'll assume that the series will converge to $f(x)$ at this point.

We could go through the work to find the coefficients here twice as we did with Fourier sine series; however there's no real reason to. So, while we could redo all the work above to get formulas for the coefficients let's instead go straight to the second method of finding the coefficients.

In this case, before we actually proceed with this we will need to define the even extension of a function, $f(x)$ on $-L \le x \le L$ .So; given a function $f(x)$ we will define the even extension of the function as,

$$g(x) = \begin{cases} f(x) & if\ 0 \le x \le L \\ f(-x) & if\ -L \le x \le 0 \end{cases}$$

Showing that this is an even function is simple enough.

$g(-x) = f(-(-x)) = f(x) = g(x)$ for $0 < x < L$ and we can see that $g(x) = f(x)\ on\ 0 \le x \le L$ and if $f(x)$ is already an even function we get $f(x) = g(x)\ \ on -L \le x \le L.$

Example 2: Sketch the even extension of the given function.

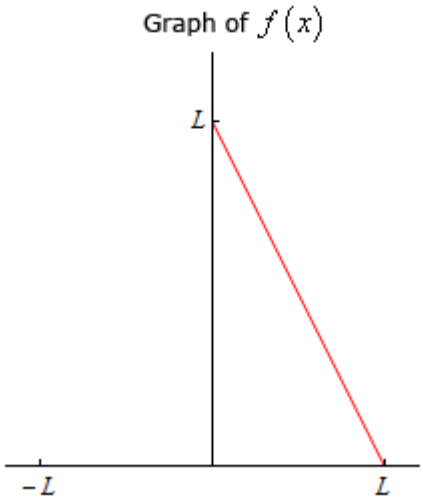
$$f(x) = L - x\ \ on\ 0 \le x \le L$$

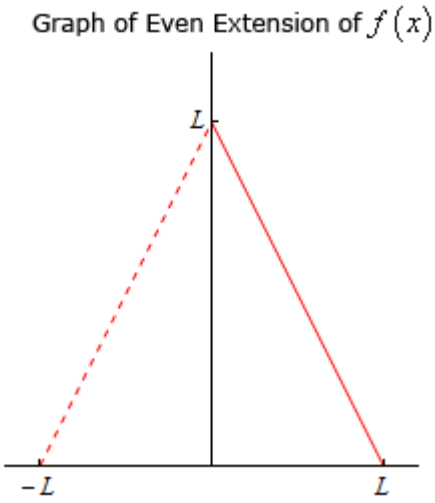
Solution: Here is the even extension of this function.

$$g(x) = \begin{cases} f(x) & if\ 0 \le x \le L \\ f(-x) & if\ -L \le x \le 0 \end{cases}$$

$$g(x) = \begin{cases} L - x & on\ 0 \le x \le L \\ L + x & on -L \le x \le 0 \end{cases}$$

Here is the graph of both the original function and its even extension. Note that we've put the "extension" in with a dashed line to make it clear the portion of the function that is being added to allow us to get the even extension





Example 3: Find the Fourier cosine series for $f(x) = L - x \;\; on \; 0 \le x \le L$

Solution: All we need to do is compute the coefficients so here is the work for that,

$$A_0 = \frac{1}{L}\int_0^L f(x)dx = \frac{1}{L}\int_0^\infty (L-x)dx = \frac{L}{2}$$

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{2}{L}\int_0^L (L-x)\cos\left(\frac{n\pi x}{L}\right)dx$$ using integration by parts we arrive at,

$$A_n = \frac{2L}{n^2\pi^2}(1+(-1)^{n+1}) \qquad n = 1,2,3,4,.....$$

The Fourier cosine series is then,

$$f(x) = \frac{L}{2} + \sum_{n=1}^{\infty}\frac{2L}{n^2\pi^2}(1+(-1)^{n+1})\cos\left(\frac{n\pi x}{L}\right)$$

Note that as we did with the first example in this section we stripped out the A_0 term before we plugged in the coefficients.

Example 4: Find the Fourier cosine series for $f(x) = 2\pi - x \;\; on \; 0 \le x \le 2\pi$

Let's leave this as an exercise for the reader.

3.5 Fourier series

In the previous two sections we have looked at Fourier sine and Fourier cosine series. It is now time to look at a Fourier series. With a Fourier series we are going to try to write a series representation for $f(x)$ on $-L \le x \le L$ in the form,

$$f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

So, a Fourier series is, in some way a combination of the Fourier sine and Fourier cosine series. Also, like the Fourier sine/cosine series we'll not worry about whether or not the series will actually converge to $f(x)$ or not at this point. For now we will just assume that it will converge (on this module we do not cover about the convergence of Fourier series)

Determining formulas for the coefficients, $A_n \;\; and \; B_n$, will be done in exactly the same manner as we did in the previous two sections. We will take advantage of the fact that

$\left\{\cos\left(\frac{n\pi x}{L}\right)\right\}_{n=0}^{\infty}$ and $\left\{\sin\left(\frac{n\pi x}{L}\right)\right\}_{n=1}^{\infty}$ are mutually orthogonal on $-L \le x \le L$ as we agreed previously. We will also need the following formulas that we derived when we proved the two sets were mutually orthogonal.

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = \begin{cases} 2L & if \;\; n = m = 0 \\ L & if \;\; n = m \ne 0 \\ 0 & if \;\; n \ne m \end{cases}$$

$$\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{m\pi x}{L}\right)dx = \begin{cases} L & if \;\; n = m \\ 0 & if \;\; n \ne m \end{cases}$$

$$\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx = 0$$

So, let's start off by multiplying both sides of the series above by $\cos\left(\frac{m\pi x}{L}\right)$ and integrate both sides from $-L$ to L and interchange the integral and summation to get,

$$\int_{-L}^{L} f(x)\cos\left(\frac{m\pi x}{L}\right)dx = \sum_{n=0}^{\infty} A_n \int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx + \sum_{n=1}^{\infty} B_n \int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{m\pi x}{L}\right)dx$$

We can now take advantage of the fact that the sines and cosines are mutually orthogonal and also using the above facts we arrive at

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx \qquad A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{m\pi x}{L}\right)dx \qquad m = 1,2,3,4,.....$$

Again we get,

$$B_m = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{m\pi x}{L}\right)dx \qquad m = 1,2,3,4,.....$$

In the previous two sections we also took advantage of the fact that the integrand was even to give a second form of the coefficients in terms of an integral from 0 to L. However, in this case we don't know anything about whether $f(x)$ will be even, odd, or more likely neither even nor odd. Therefore, this is the only form of the coefficients for the Fourier series.

Example1: Find the Fourier series for $f(x) = \begin{cases} L & if \ -L \le x \le 0 \\ 2x & if \ \ 0 \le x \le L \end{cases}$ on $-L \le x \le L$

Solution: Because of the piece-wise nature of the function the work for the coefficients is going to be a little unpleasant but let's get on with it.

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx = \frac{1}{2L}\left[\int_{-L}^{0} f(x)dx + \int_{0}^{L} f(x)dx\right]$$

$$= \frac{1}{2L}\left[\int_{-L}^{0} L dx + \int_{0}^{L} 2x dx\right]$$

$$= \frac{1}{2L}\left[L^2 + L^2\right]$$

$$= L\,\mathrm{k}$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{1}{L}\left[\int_{-L}^{0} f(x)\cos\left(\frac{n\pi x}{L}\right)dx + \int_{0}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx\right]$$

$$= \frac{1}{L}\left[\int_{-L}^{0} L\cos\left(\frac{n\pi x}{L}\right)dx + \int_{0}^{L} 2x\cos\left(\frac{n\pi x}{L}\right)dx\right]$$

At this point it will probably be easier to do each of these individually.

$$\int_{-L}^{0} L\cos\left(\frac{n\pi x}{L}\right)dx = \left(\frac{L^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\right)\Bigg|_{-L}^{0} = \frac{L^2}{n\pi}\sin(n\pi) = 0$$

Again,

$$\int_{0}^{L} 2x\cos\left(\frac{n\pi x}{L}\right)dx = \left(\left(\frac{2L}{n^2\pi^2}\right)\left(L\cos\left(\frac{n\pi x}{L}\right) + n\pi x\sin\left(\frac{n\pi x}{L}\right)\right)\right)_{0}^{L}$$

$$= \frac{2L^2}{n^2\pi^2}\left((-1)^n - 1\right)$$

So, if we put all of this together we have,

$$A_n = \frac{2L}{n^2\pi^2}\left((-1)^n - 1\right), \qquad n = 1,2,3,4,.....$$

So, we have gotten the coefficients for the cosines taken care of and now we need to take care of the coefficients for the sines.

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx = \frac{1}{L}\left[\int_{-L}^{0} f(x)\sin\left(\frac{n\pi x}{L}\right)dx + \int_{0}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx\right]$$

$$= \frac{1}{L}\left[\int_{-L}^{0} L\sin\left(\frac{n\pi x}{L}\right)dx + \int_{0}^{L} 2x\sin\left(\frac{n\pi x}{L}\right)dx\right]$$

As with the coefficients for the cosines will probably be easier to do each of these individually.

$$\int_{-L}^{0} L\sin\left(\frac{n\pi x}{L}\right)dx = \left(-\frac{L^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right)\Bigg|_{-L}^{0} = \frac{L^2}{n\pi}\left(-1+\cos(n\pi)\right) = \frac{L^2}{n\pi}\left((-1)^n - 1\right)$$

Again,

$$\int_{0}^{L} 2x\sin\left(\frac{n\pi x}{L}\right)dx = \left(\left(\frac{2L}{n^2\pi^2}\right)\left(L\sin\left(\frac{n\pi x}{L}\right) - n\pi x\cos\left(\frac{n\pi x}{L}\right)\right)\right)_{0}^{L}$$

$$= \left(\frac{2L}{n^2\pi^2}\right)\left(L\sin(n\pi) - n\pi\cos(n\pi)\right)$$

$$= \left(\frac{2L^2}{n^2\pi^2}\right)\left(-n\pi(-1)^n\right)$$

$$= -\frac{2L^2}{n\pi}(-1)^n$$

So, if we put all of this together we have,

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx = \frac{1}{L}\left[\frac{L^2}{n\pi}\left((-1)^n - 1\right) - \frac{2L^2}{n\pi}(-1)^n\right]$$

$$= -\frac{L}{n\pi}\left(1+(-1)^n\right) \qquad n = 1,2,3,4,....$$

So, after all that work the Fourier series is,

$$f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$= A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$= L + \sum_{n=1}^{\infty} \frac{2L}{n^2\pi 2}\left((-1)^n - 1\right)\cos\left(\frac{n\pi x}{L}\right) - \sum_{n=1}^{\infty} \frac{L}{n\pi}\left(1+(-1)^n\right)\sin\left(\frac{n\pi x}{L}\right)$$

Exercise: Find the Fourier series for $f(x) = \begin{cases} L & if \ -L \le x \le 0 \\ 2x & if \ \ 0 \le x \le L \end{cases}$ on $-\pi \le x \le \pi$

Example 2: Find the Fourier series for $f(x) = x$ on $-L \le x \le L$.

Solution: Let's start with the integrals for A_n,

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx = \frac{1}{2L}\int_{-L}^{L} x dx = 0$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{1}{L}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)dx = 0$$

In both cases note that we are integrating an odd function (*x* is odd and cosine is even so the product is odd) over the interval [-L, L] and so we know that both of these integrals will be zero.

Next here is the integral for B_n,

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx = \frac{1}{L}\int_{-L}^{L} x\sin\left(\frac{n\pi x}{L}\right)dx = \frac{2}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)dx$$

In this case we're integrating an even function (x and sine are both odd so the product is even) on the interval [-L, L] and so we can "simplify" the integral as shown above. The reason for doing this here is not actually to simplify the integral however. It is instead done so that we can note that we did this integral back in the Fourier sine series section and so don't need to redo it in this section. Using the previous result we get,

$$B_n = \frac{(-1)^{n+1}2L}{n\pi} \qquad n = 1,2,3,4,.....$$

In this case the Fourier series is,

$$f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)$$

Exercise: Find the Fourier series for $f(x) = x$ on $-2 \le x \le 2$.

Chapter Four: Differential calculus of Functions of several variables

4.1 Notations, Examples, Level curves and Graphs

4.1.1 Notations and Examples

There are many familiar formulas in which a given variable depends on two or more other variables. For example, the area A of a triangle depends on the base *length b*, and height *h* by the formula $A = \frac{1}{2}bh$: the volume V of a rectangular box depends on the length l, the width, and the height h by the formula $V = hwl$: and the arithmetic average x of n real numbers, $x_1, x_2, ..., x_n$ depends on those numbers by the formula

$$x = \frac{x_1 + x_2 + ... + x_n}{n}$$

Thus, we say that

A is a function of the two variables b and *h*

V is a function of the three variables /, u, and h;

x is a function of the n variables $x_1, x_2, ..., x_n$.

The terminology and notation for functions of two or more variables is similar to that for functions of one variable. For example, the expression *Z=f(x, y)* means that z is a function of x and y, in the sense that a unique value of the dependent variable *z* is determined by specifying values for the independent variables x and y.

Similarly, $w = f(x, y, z.)$ expresses *w* as a function of x, y and z.

One can think of a function f of two or more variables as a computer program that takes two or more inputs, operates on those inputs, and produces an output . In this section we will only be concerned with functions whose inputs and outputs are real numbers. One can also think of such functions in more geometric terms. For example, if $z = f(x, y)$, then we can view (x, y) as a point in the -xy-plane and think of f as a rule that associates a unique numerical value z: with the point (x, y); similarly, we can think of $w = f(x, y, z)$ as a rule that associates a unique numerical value w with a point (x, y, z) in an ,xyz -coordinate system.

Definition: A function f of two variables, x and y, is a rule that assigns a unique real number

$f(x, y)$ to each point (x, y) in some set D in the xy-plane.

Definition: A function f of three variables, x, y, and z, is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in three-dimensional space.

Example: Let $f(x, y) = 3x^2\sqrt{y} - 1$

Find $f(1,4)$, $f(0,9)$, $f(t2, t)$, $f(ab, 9b)$, and the natural domain of f.

Solution: By substitution

$$f(1,4) = 3(1)2\sqrt{4} - 1 = 5$$

$$f(0,9) = 3(0)2\sqrt{9} - 1 = -1$$

$$f(t^2, t) = 3(t^2)^2\sqrt{t} - 1 = 3t^4\sqrt{t} - 1$$

$$f(ab, 9b) = 3(ab)2\sqrt{9b} - 1 = 9a2b2\sqrt{b} - 1$$

The domain of functions of several variables consists of all points in space or(in plane for functions of two variables) for which the formula is meaningful.

Example 1: $f(x,y) = \sqrt{9-4x^2-y^2}$

The domain of f is the region in the xy plane bounded by the ellipse $4x^2+y^2=9$, because the square root is defined only for none negative numbers.

Example 2: If g(x, y, z) =$\sqrt{x^2+y^2+z^2}$ then the domain of g consists of all points (x, y, z) in space, because $x^2+y^2+z^2 \geq 0$ for all (x, y , z).

The following are some of functions of several variables with there domains and representations

1. $f(x,y) = xy \; for \; x \geq 0 \; and \; y \geq 0$ area the rectangle
2. $f(x,y,z) = xyz \; for \; x \geq 0 \; y \geq 0 \; and \; z \geq 0$ volume of rectangular parallelepiped
3. f(x ,y ,z) = 2xy +2yx +2xz for surface area of a parallelepiped rectangular for $x \geq 0$ $y \geq 0$ and $z \geq 0$

Combinations of Functions of Several Variables

The sum, product, and quotient of two functions f and g of several variables are defined as follows.

$$(f+g)(x,y) = f(x,y) + g(x,y)$$
$$(f-g)(x,y) = f(x,y) - g(x,y)$$
$$(fg)(x,y) = f(x,y)g(x,y)$$
$$(\frac{f}{g})(x,y) = \frac{f(x,y)}{g(x,y)}$$

The formula for functions of thee variables is analogous. The domain of f+g, f-g , and fg consists of all points simultaneously in the domain of f and in the domain of g, whereas

the domain of $\frac{f}{g}$ consists of all points simultaneously in the domain of f and in the domain of g at which g does not assume the value 0.

4.1.2 Level curves and graphs

The graph of a function of two variables is the collection of points $(x, y, f(x, y))$ for which (x, y) is in the domain of f. It is customary to let $z = f(x, y)$; the graph of f consists of all points

(x, y, z) Such that $z = f(x, y)$.

In sketching the graph of the function f of two variables, it is often helpful to determine the intersection of the graph of f with planes of the form $z = c$. We call each such intersection the trace of the graph of f in the plane $z = c$. Thus the trace of the graph of f in the plane z=c is the collection of points (x, y, c) such that $f(x, y) = c$.

Definition**:** The of points (x, y) inn the xy plane such that $f(x, y) = c$ is called a level Curve of f.

We identify the level curve with the equation $f(x, y) = c$ and call the equation a level curve of f.

Example: Describe the graph of the function in an xyz coordinate system and the level curves

associated with them

a) $f(x, y) = 1 - x - {}^{1}/_{2}\, y$

Solution:

By definition, the graph of the given function is the graph of the equation

Z=$1 - x - {}^{1}/_{2}\, y$

which is a plain A triangular portion of the plain can be sketched by plotting the intersection with the coordinates axes and joining them ((0,0,1), (0,2,1) and (1,0,0)) with line segment.

For any value of c the level curve $f(x,y) = c$ is the line with equation

$x + {}^{1}/_{2}\, y = c - 1$

b) $f(x,y) = \sqrt{1 - x^2 - y^2}$

Solution:

By definition, the graph of the given function is the graph of the equation

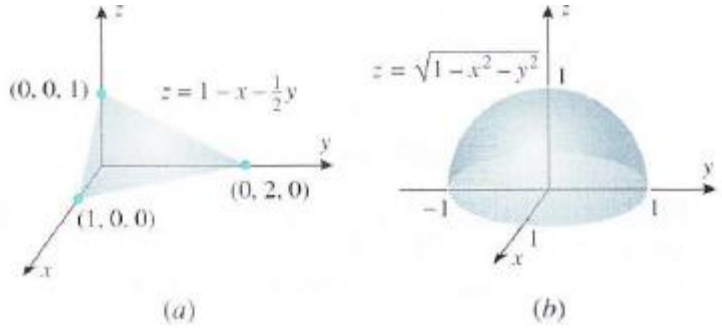
$$z = \sqrt{1 - x^2 - y^2} \qquad (1)$$

After squaring both sides, this can be rewritten as

$x^2+y^2+z^2=1$

which represent a sphere of radius 1, center with the origin. Since (1) implies the added condition $z \geq 0$ the graph represents the upper hemisphere.

In this case the level curve $f(x,y) = c$ is a circle if $0 \leq c < 1$



1.5 (a) (b)

Exercise: Identify the type of the level curve $f(x,y) = c$ sketch it.

a) $f(x, y)=x^2+4y^2$; c=1,4

b) $f(x, y)=x^2-y$; c=-2,2

c) $f(x, y)=x^2-y^2$;c=-1,0,1

4.2 Limit and Continuity

In this section we will introduce the notion of limit and continuity for functions of two or more variables. Our objective is to develop the basic concepts accurately and to obtain results needed in the later section.

4.2.1 Limit

If D is a set of points in 2 space, then a point (a, b) is called an interior point of D if there is some circular disk with positive radius, centered at (a, b), and containing only points in D.

Definition: Let *f* be defined on the interior of a circle centered at the point (*a*, *b*), except possibly

at (*a*, *b*) itself. We say that $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every ε > 0 there exists a δ > 0

such that $|f(x,y) - L| < \varepsilon$ whenever $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$.

and say that $\lim_{(x,y)\to(a,b)} f(x,y) = L$ exist.

Similarly let f be defined thought a set containing a ball at (a, b, c) except possibly at (a, b, c) it self. Then we say

$$\lim_{(x,y,z)\to(a,b,c)} f(x,y,z) = L$$

if for every ε > 0 there exists a δ > 0 such that

$|f(x,y,z) - L| < \varepsilon$ Whenever $0 < \sqrt{(x-a)^2 + (y-a)^2 + (z-b)^2} < \delta$

We say $\lim_{(x,y)\to(a,b)} f(x,y) = L$ exist

Example1: Show that

$$\lim_{(x,y,)\to(a,b)} x = a \quad \text{and} \quad \lim_{(x,y)\to(a,b)} y = b$$

Solution: Let ε > 0, Observe that

$\sqrt{(x-a)^2} \le \sqrt{(x-a)^2 + (y-b)^2}$

Therefore, if we let δ=ε it follows that

$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$, then $|x-a| = \sqrt{(x-a)^2} < \delta = \varepsilon$

This proves that $\lim_{(x,y,)\to(a,b)} x = a$. the second limit is established analogously.

Example2: Show that

$\lim_{(x,y,z)\to(a,b,c)} y = b$

Solution: Let ε > 0. Since

$$\sqrt{(y-b)^2} \leq \sqrt{(x-a)^2+(y-b)^2+(z-c)^2}$$

We can let δ=ε and deduced that

$0 < \sqrt{(x-a)^2+(y-b)^2+(z-c)^2} < \delta,$

then $|y-b| = \sqrt{(y-b)^2} < \delta = \varepsilon$

Thus the limit is verified.

Corollary results for limits of combinations of functions:

If $\lim_{(x,y)\to(a,b)} f(x,y)$ and $\lim_{(x,y)\to(a,b)} g(x,y)$ exists, then

- The limit of a sum or difference is the sum or difference of the limits.

$$\lim_{(x,y)\to(a,b)} \left(f(x,y) \pm g(x,y) \right) = \lim_{(x,y)\to(a,b)} f(x,y) \pm \lim_{(x,y)\to(a,b)} g(x,y)$$

- The limit of a product is the product of the limits

$$\lim_{(x,y)\to(a,b)} \left(f(x,y)\, g(x,y) \right) = \left[\lim_{(x,y)\to(a,b)} f(x,y) \right] \left[\lim_{(x,y)\to(a,b)} g(x,y) \right]$$

- The limit of a quotient is the quotient of the limits (except where the denominator is zero)

- $\lim_{(x,y)\to(a,b)} \left(\dfrac{f(x,y)}{g(x,y)} \right) = \dfrac{\lim_{(x,y)\to(a,b)} f(x,y)}{\lim_{(x,y)\to(a,b)} g(x,y)}$ provided that $\lim_{(x,y)\to(a,b)} g(x,y) \neq 0$

- The limit of a polynomial always exists and is found simply by substitution.

$$\lim_{(x,y)\to(a,b)} \left(P_n(x,y) \right) = P_n(a,b)$$

Example3: Show that

$$\lim_{(x,y)\to(-1,2)} \frac{x^3+y^3}{x^2+y^2} = \frac{7}{5} \text{ and } \lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} = 0$$

Solution: Since

$$\lim_{(x,y,)\to(-1,2)} x = -1 \qquad \text{and} \qquad \lim_{(x,y,)\to(-1,2)} y = 2$$

The product formula yields

$$\lim_{(x,y,)\to(-1,2)} x^3 = -1 \qquad \text{and} \qquad \lim_{(x,y,)\to(-1,2)} y^3 = 8$$

$$\lim_{(x,y,)\to(-1,2)} x^2 = 1 \qquad \text{and} \qquad \lim_{(x,y,)\to(-1,2)} y^2 = 4$$

Hence the sun and the quotient formulas combined to yield

$$\lim_{(x,y)\to(-1,2)} \frac{x^3+y^3}{x^2+y^2} = \frac{\lim_{(x,y)\to(-1,2)}(x^3+y^3)}{\lim_{(x,y)\to(-1,2)}(x^2+y^2)}$$

$$= \frac{\lim_{(x,y)\to(-1,2)} x^3 + \lim_{(x,y)\to(-1,2)} y^3)}{\lim_{(x,y)\to(-1,2)} x^2 + \lim_{(x,y)\to(-1,2)} y^2)}$$

$$= \frac{-1+8}{1+4} = \frac{7}{5}$$

Using the same procedure one can show that,

$$\lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} = 0$$

Exercise:

1. Evaluate $\lim_{(x,y)\to(2,1)} \dfrac{2x^2y+3xy}{5xy^2+3y}$
2. Evaluate $\lim_{(x,y)\to(e,1)} ln\dfrac{x}{y}$

Disproving limits:

- For a limit to exist, the function must approach that limit for every possible path of (x, y) approaching (a, b). Thus, it is usually very hard to prove a limit exists, and easier to show a limit does not exist.
- So, if a function $f(x, y)$ approaches L_1 as (x, y) approaches (a, b) along a path P_1 and $f(x, y)$ approaches $L_2 \neq L_1$ as (x, y) approaches (a, b) along a different path P_2, then $\lim_{(x,y)\to(a,b)} f(x,y)$ *does not exist*.

- Some simple paths to try are the lines along $x = a$, $y = b$, or any other line through the point.

Example: Show the following limit does not exist: $\lim_{(x,y)\to(1,0)} \frac{y}{x+y-1}$

Solution: The point (x, y) may approach (0,0) in infinitely many paths and if the limit exits it is independent of the path followed. Among these paths we cane take the y-axes and the line $y = x - 1$

Case 1: when (x, y) may approach (0,0) along y-axis

$$\text{Lim}_{(x,y)\to(1,0)} \frac{y}{x+y-1}$$

$$= \lim_{y\to 0} \frac{y}{y-1} = 0$$

Case2: when (x, y) may approach (0,0) along the line $y = x - 1$

$$\text{Lim}_{(x,y)\to(1,0)} \frac{y}{x+y-1}$$

$$= \lim_{x\to 1} \frac{x-1}{x+x-1-1}$$

$$= \lim_{x\to 1} \frac{x-1}{2(x-1)} = \frac{1}{2}$$

Thus the limit is not the same in different paths and by uniqueness of limit

$\text{Lim}_{(x,y)\to(1,0)} \frac{y}{x+y-1}$ doesn't exist

Exercise: Show the following limit does not exist

a) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$

b) $\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4}$

4.2.2 Continuity

Definition**:**

a) Continuity of a Function of Two Variables

Suppose $f(x, y)$ is defined in the interior of a circle centered at the point (a, b). We say

that *f* is continuous at (*a*, *b*) if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$. If *f*(*x*, *y*) is not continuous

at (*a*, *b*), then we call (*a*, *b*) a discontinuity of ***f.***

b) Continuity of a Function of Three Variables

Suppose *f*(*x*, *y*) is defined in the interior of a sphere centered at the point (*a*, *b*, *c*). We

say that *f* is continuous at (*a*, *b*, *c*) if $\lim_{(x,y,z)\to(a,b,c)} f(x,y,z) = f(a,b,c)$. If *f*(*x*, *y*, *z*) is

not continuous at (*a*, *b*, *c*), then we call (*a*, *b*, *c*) a discontinuity of *f*.

If f is continuous at each point of a region R in the xy-plane, then we say that f is continuous on R; and if f is continuous at every point in the xy- plane, then we say that f is continuous everywhere. In addition, we will say that f is a continuous function if it is continuous at each point of its domain.

Theorem:

a) If g and h are continuous functions of one variable, then f(x, y) =g(x) h(y) is a continuous function of x and y.
b) If g is a continuous function of one variable and h is continuous function of two variables, then their composition f(x,y)=g(h(x,y)) is a continuous function of x and y.

Example: Show that $f(x,y) = 3x^2y^5$ and $f(x,y) = \sin(3x^2y^5)$ are continuous functions.

Solution:

The function $f(x,y) = 3x^2y^5$ is continuous because it is the product of the continuous functions g(x) $= 3x^2$ and h(y) = y5, and the function $f(x,y) = \sin(3x^2y^5)$ is continuous because it is the composition of the continuous function sin x and the continuous function$3x^2y^5$.

The above theorem can be summarized by

. A composition of continuous functions is continuous.

. A sum, difference, or product of continuous functions is continuous.

. A quotient of continuous functions is continuous, except where the denominator is zero.

Example: Evaluate $\lim_{(x,y)\to(-1,2)} \frac{xy}{x^2+y^2}$

Solution:

Since $f(x,y) = \frac{xy}{x^2+y^2}$ is continuous at (-1,2) it follows from the definition of continuity

$$\lim_{(x,y)\to(-1,2)} \frac{xy}{x^2+y^2} = \frac{(-1)(2)}{(-1)^2+(2)^2} = \frac{-2}{5}$$

Example: Since the function $f(x,y) = \frac{x^3y^2}{1-xy}$ is a quotient of continuous functions, it is continuous except where $1 - xy = 0$. Thus, $f(x,y)$ is continuous every where except on the hyperbola $xy = 1$.

Exercise: Use limit laws and continuity properties to evaluate the limit

1) $\lim_{(x,y)\to(1,3)}(4xy^2 - x)$
2) $\lim_{(x,y)\to(0,0)} ln(1 + x^2y^3$
3) $\lim_{(x,y)\to(-1,2)}(\frac{xy^3}{x+y})$
4) $\lim_{(x,y)\to(1/2,\pi)}(xy^2 \sin xy)$

4.3 Partial Derivatives, Higher order partial derivatives

4.3.1Partial Derivatives

If $z = f(x.y)$, then one can inquire how the value of z. changes if x is held fixed and y is allowed vary or if t is held fixed and x is allowed to vary. For example, the ideal gas law in physics states that under appropriate conditions the pressure exerted by a gas is a function of the volume of the gas and its temperature. Thus, a physicists study gasses might be interested in the rate of change of the pressure if the volume is held fixed and the temperature is allowed to vary or if the temperature is held fixed and the temperature is allowed to vary. In this section we will develop the mathematical tools for studding rates of change that involves to or more independent variables.

Definition**:** Let f be a function of two variables and let (x_0, y_0) be in the domain of f.

The partial derivative of f with respect to x at (x_0, y_0) is defined by

$$f_x(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0+h, y_0) - f(x_0+, y_0)}{h}$$

provided that the limit exists.

The partial derivative of f with respect to y at (x_0, y_0) is defined by

$$f_y(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0, y_0 + h) - f(x_0+, y_0)}{h}$$

provided that the limit exists.

Example: Let $f(x, y) = x^2 + 3xy + y - 1$. Find the values of $\frac{\partial f}{\partial x} = f_x$ and $\frac{\partial f}{\partial y} = f_y$ at the point (4, -5).

Solution :

To find $\frac{\partial f}{\partial x}$, we treat y as a constant and differentiate with respect to x.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(x^2 + 3xy + y - 1\right) = 2x + 3y$$

The value of $\frac{\partial f}{\partial x}$ at (4, -5) is 2(4) + 3(-5) = -7.

To find $\frac{\partial f}{\partial y}$, we treat x as a constant and differentiate with respect to y:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial x}\left(x^2 + 3xy + y - 1\right) = 3x + 1.$$

The value of $\frac{\partial f}{\partial y}$ at (4, -5) is 3(4) + 1 = 13

Example: Find the partial derivatives of

$$f(x, y) = 2x^3y^2 + 2y + 4x$$

Solution :

Treating y as a constant and differentiating with respect to x, we obtain

$$f_x(x,y) = 6x^2y^2 + 4$$

Treating x as a constant and differentiating with respect to y, we obtain

$$f_y(x,y) = 4x^3y + 2$$

Exercise:

a) Let $f(x,y) = 24xy - 6x^2y$. Find f_x and f_y at (1, 2).

b) Let $Z = x^2 \cos y$. Find $\frac{\partial z}{\partial x}$ *and* $\frac{\partial z}{\partial y}$.

Note that these two partial derivatives are sometimes called the first order partial derivatives. Just as with functions of one variable we can have derivatives of all orders.

4.3.2 Higher order partial derivatives

Just as we had higher order derivatives with functions of one variable we will also have higher order derivatives of functions of more than one variable. However, this time we will have more options since we do have more than one variable.

Consider the case of a function of two variables, $f(x,y)$ since both of the first order partial derivatives are also functions of *x* and *y* we could in turn differentiate each with respect to *x* or *y*. This means that for the case of a function of two variables there will be a total of four possible second order derivatives. Here they are and the notations that we'll use to denote them.

- The partial derivative with respect to *x* of $\frac{\partial f}{\partial x}$ is $\frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right]$.

  Alternative notations: $\frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right] = \frac{\partial^2 f}{\partial x^2} = f_{xx}$

- The partial derivative with respect to *y* of $\frac{\partial f}{\partial y}$ is $\frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right]$.

  Alternative notations: $\frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right] = \frac{\partial^2 f}{\partial y^2} = f_{yy}$

- The partial derivative with respect to *x* of $\frac{\partial f}{\partial y}$ is $\frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]$.

  Alternative notations: $\frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]=\frac{\partial^2 f}{\partial x \partial y}=\left(f_y\right)_x=f_{yx}$. This is a mixed second-order partial derivative.

- The partial derivative with respect to *y* of $\frac{\partial f}{\partial x}$ is $\frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right]$.

- Alternative notations: $\frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right]=\frac{\partial^2 f}{\partial y \partial x}=\left(f_x\right)_y=f_{xy}$. This is also a mixed second-order partial derivative.

Note as well that the order that we take the derivatives in is given by the notation for each these. If we are using the subscripting notation, *e.g* .f_{xy}, then we will differentiate from left to right. In other words, in this case, we will differentiate first with respect to *x* and then with respect to *y*. With the fractional notation, *e.g* .$\frac{\partial^2 f}{\partial y \partial x}$,*is the opposite*. In these cases we differentiate moving along the denominator from right to left. So, again, in this case we differentiate with respect to *x* first and then *y*.

Example: Find all the second order derivatives for $f(x,y)=\cos(2x)-x^2e^{5y}+3y^2$

Solution:

We should first need the first order derivatives

$$f_x(x,y)=-2\sin(2x)-2xe^{5y}$$
$$f_y(x,y)=-5x^2e^{5y}+6y$$

Now, let's get the second order derivatives.

$$f_{xx}(x,y)=-4\cos(2x)-2e^{5y}$$
$$f_{xy}(x,y)=-10xe^{5y}$$
$$f_{yx}(x,y)=-10xe^{5y}$$
$$f_{yy}(x,y)=-25x^2e^{5y}+6$$

Example: Let$f(x,y)=\sin(xy^2)$. Find all the second partial derivatives of f

Solution: The first partial derivatives are given by

$$f_x(x,y) = y^2\cos(xy^2) \text{ and } f_y(x,y) = 2xy\cos(xy^2)$$

We obtain the second partials by computing the partial derivatives of the first partials:

$$f_{xx}(x,y) = -4y^4\sin(xy^2)$$
$$f_{xy}(x,y) = 2y\cos(xy^2) - 2xy^3\sin(xy^2)$$
$$f_{yx}(x,y) = 2y\cos(xy^2) - 2xy^3\sin(xy^2)$$
$$f_{yy}(x,y) = 2x\cos(xy^2) - -4x^2y^2\sin(xy^2)$$

Exercise: Let

$$f(x,y) = \begin{cases} \frac{x^3y - xy^3}{x^2+y^2}, for\ (x,y) \neq (0,0) \\ 0,\ for \qquad (x,y) = (0,0) \end{cases}$$

Show that $f_{xy}(0,0) \neq f_{yx}(0,0)$

Theorem: Let f be a function of two variables, and assume that $f_{xy}(x,y)$ and $f_{yx}(x,y)$ are continuous at (a,b). Then

$$f_{xy}(a,b) = f_{yx}(a,b)$$

Exercise: 1) Verify the above theorem for $f(x,y) = xe^{-x^2y^2}$

2) Compute all four second-order partial derivatives for

a) $f(x,y) = 3x^2 + 4y^2 - 2x^2y + 7xy^2 - 9$

b) $f(x,y) = \cos(x)\sin(y) - e^{xy}$

3) Compute a) f_{xyx} and f_{xyxy} for $f(x,y) = \cos(x)\sin(y) - e^{xy}$

b) f_x, f_{xy}, and f_{xyz} for $f(x,y,z) = \sqrt{xyz} + x^2y^3z^4$

4.4 Directional derivatives and gradient

To this point we've only looked at the two partial derivatives $f_x(x,y)$ and $f_y(x,y)$. Recall that these derivatives represent the rate of change of *f* as we vary *x* (holding *y* fixed) and as we vary *y* (holding *x* fixed) respectively. We now need to discuss how to find the rate of change of *f* if we allow both *x* and *y* to change simultaneously.

Definition: The rate of change of $f(x,y)$ in the direction of the unit vector $\vec{u} = (a,b)$ is called the directional derivative and is denoted by

$$D_{\vec{u}}(x,y) = \lim_{h\to 0} \frac{f(x+ah, y+bh) - f(x,y)}{h}$$

Theorem: Let f be differentiable at (x_0, y_0).Then f has directional derivative at (x_0, y_0) in every direction, moreover, if $\vec{u} = a_1\, i + a_2\, j$ is a unit vector, then

$$D_{\vec{u}} f(x_0, y_0) = f_x(x_0, y_0)a_1 + f_y(x_0, y_0)a_2$$

Example: Let $f(x, y) = 6 - 3x^2 - y^2$, and $\vec{u} = \left(1/\sqrt{2}\right) i - \left(1/\sqrt{2}\right) j$. Find $D_{\vec{u}} f(1,2)$

Solution Notice that $\vec{u}$ is a unit vector

First let us calculate the partial derivatives of f

$$f_x(x, y) = -6x \text{ ,and } f_y(x, y) = -2y$$

Then $D_{\vec{u}} f(1,2) = f_x(1,2)\left(1/\sqrt{2}\right) + f_y(1,2)\left(-1/\sqrt{2}\right)$

$$= (-6)\left(1/\sqrt{2}\right) + (-4)\left(-1/\sqrt{2}\right) = -\sqrt{2}$$

In the expression$D_{\vec{u}}(x_0, y_0)$, $\vec{u}$ represents a unit vector. The directional derivative in the direction of an arbitrary vector $\vec{a}$ is defined to be $D_{\vec{u}} f(x_0, y_0)$ where

$$\vec{u} = \frac{\vec{a}}{\|\vec{a}\|}$$

Example: Let f(x,y) = xy^2 and let $\vec{a} = i - 2j$. Find the directional derivative of f at $(-3,1)$ in the direction of $\vec{a}$.

Solution:

Here $\|\vec{a}\| = \sqrt{1+4} = \sqrt{5}$

So, $D_{\vec{u}} f(-3,1)$ where

$\vec{u} = \frac{\vec{a}}{\|\vec{a}\|} = \frac{1}{\sqrt{5}}$(i-2j)

And $f_x(x, y) = y^2$and $f_y(x, y) = 2xy$

It follows that

$$D_{\vec{u}} f(-3,1) = f_x(-3,1)\frac{1}{\sqrt{5}} + f_y(-3,1)\frac{(-2)}{\sqrt{5}}$$

$$= 1\frac{(1)}{\sqrt{5}} + (-6)\frac{(-2)}{\sqrt{5}} = \frac{13}{\sqrt{5}}$$

Exercise: Find each of the directional derivatives

a) $D_{\vec{u}} f(2,0)$ where $f(x,y) = xe^{xy} + y$ and $\vec{u}$ is the unit vector in the direction of $\theta = \frac{2\pi}{3}$

b) $D_{\vec{u}} f(x,y,z)$ where $f(x,y,z) = x^2 z + y^3 z^2 - xyz$ in the direction of $\vec{v} = (-1,0,3)$

There is another form of the formula that we used to get the directional derivative that is a little nicer and somewhat more compact. It is also a much more general formula that will encompass both of the formulas above.

Let's start with the second one and notice that we can write it as follows,

$$\begin{aligned} D_{\vec{u}} f(x,y,z) &= f_x(x,y,z)a_1 + f_y(x_0,y_0)a_2 + f_z(x,y,z)a_3 \\ &= (f_x, f_y, f_z) \cdot (a_1, a_2, a_3) \end{aligned}$$

In other words we can write the directional derivative as a dot product and notice that the second vector is nothing more than the unit vector $\vec{u}$ that gives the direction of change. Also, if we had used the version for functions of two variables the third component wouldn't be there, but other than that the formula would be the same.

Now let's give a name and notation to the first vector in the dot product since this vector will show up fairly regularly throughout this course. The **gradient of *f*** or **gradient vector of *f*** is defined to be,

$$\nabla f(x,y,z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \qquad \text{Or}$$

$$\nabla f(x,y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = f_x(x,y)\mathbf{i} + f_y(x,y)\mathbf{j},$$

The definition is only shown for functions of two or three variables; however there is a natural extension to functions of any number of variables that we'd like.

With the definition of the gradient we can now say that the directional derivative is given by,

$$D_{\mathbf{u}} f(x,y,z) = \nabla f(x,y,z) \cdot \mathbf{u}$$

Example: Find the gradient of the function $f(x,y) = xcos(xy)$ at $(1,-\pi)$

Solution: By definition $\nabla f(1,-\pi) = f_x(1,-\pi)\ i + f_y(1,-\pi)\ j$

But $f_x(x,y) = \cos(xy) - xysin(xy)$ and $f_y(x,y) = -x^2 sin(xy)$

Then $f_x(1,-\pi) = \cos(-\pi) + \pi sin(-\pi) = -1$ and $f_y(1,-\pi) = 0$

Hence $\nabla f(1,-\pi) = -i$

Exercise:

1. For the function $f(x, y) = x^2 y - 4y^2$, find $D_{\hat{u}} f(2,-1)$ in the direction of the vector $\langle 2, 5\rangle$ using the limit definition and the gradient shortcut.

2. 2. Find the derivative of the function $f(x, y, z) = xy + yz + zx$ at the point (1,-1, 2) in the direction $\mathbf{A} = 3i + 6j - 2k$

3. Let $f(x, y) = \sin x + e^{xy}$
   a. Compute $\nabla f(x, y)$
   b. Compute $\nabla f(0, 1)$
   c. Compute $D_{\hat{u}} f(0, 1)$, where $\hat{u} = \left\langle -\frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle$.

Theorem: (The Directional Derivative in Terms of the Gradient)

If *f*(*x*, *y*) is differentiable at the point (*a*, *b*), then:

(i). The maximum rate of change of *f* at (*a*, *b*) is $\|\nabla f(a,b)\|$, occurring in the direction of the gradient.

(ii). The minimum rate of change of *f* at (*a*, *b*) is -$\|\nabla f(a,b)\|$, occurring in the direction opposite of the gradient.

(iii). The rate of change of *f* at (*a*, *b*) is 0 in directions orthogonal to the gradient.

(iv). The gradient $\nabla f(a,b)$ is orthogonal to the level curve $f(x, y) = f(a, b)$.

Example: Let $f(x,y) = 6 - 3x^2 - y^2$

a) In which direction is f increasing most rapidly at $(1,2)$.

b) Find the maximum change of f at $(1,2)$

Solution:

a) by the above theorem the maximum change occurs in the direction of the gradient at that point. Then,

$$f_x(x,y) = -6x \text{ and, } f_y(x,y) = -2y$$

This implies that $\nabla f(1,2) = f_x(1,2)\, i + f_y(1,2)\, j$

$$\nabla f(1,2) = (-6)(1)i + (-2)(2)j = -6\, i - 4\, j$$

So f increases most rapidly at $(1,2)$ in the direction of $-6\, i - 4\, j$

b) maximum change of f at $(1,2)$ is $\|\nabla f(1,2)\| = \sqrt{(-6)^2 + (-4)^2} = \sqrt{52}$

Exercise:

a) Find the gradient of the function $f(x, y) = \ln(x^2 + y^2)$ at the point (1,1)

b) A man stands at the point $(1, 2, 192)$ on a hill whose elevation is given by $z = 200 - 4x^2 - y^2$. In what direction should he begin to walk in order to climb the hill most rapidly? Assuming he walks in this direction, what will be his rate of ascent initially?

4.5 Total differential and Tangent planes

4.5.1Tangent planes

Suppose a surface S has equation $z = f(x, y)$, where f has continuous first partial derivatives, and let $P(x_o, y_o, z_o)$ be a point on S. As in the preceding section, let C_1 and C_2 be the curves obtained by intersecting the vertical planes $y = y_o$ and $x = x_o$ with the surface S. Then the point P lies on both C_1 and C_2. Let T_1 and T_2 be the tangent lines to the curves C_1 and C_2 at the point P. Then the tangent plane to the surface S at the point P

is defined to be the plane that contains both tangent lines T_1 and T_2 . Therefore, you can think of the tangent plane to S at P as consisting of all possible tangent lines at P to curves that lie on S and pass through P. The tangent plane at P is the plane that most closely approximates the surface near the point.

We know Equation any plane passing through the point $P(x_o, y_o, z_o)$ with a normal $N = A\,i + B\,j + C\,k$ has an equation of the form

$$A(x - x_o) + B(y - y_o) + C(z - z_o) = 0$$

By dividing this equation by C and letting $a = -{}^{A}/_{C}$ and $b = -{}^{B}/_{C}$ we can write it in the form

$$(z - z_o) = a(x - x_o) + b(y - y_o)$$

Theorem: Let $P(x_o, y_o, z_o)$ be any point on the surface $z = f(x, y)$. If $f(x, y)$ is differentiable at (x_o, y_o), then the surface has tangent plane at P, and this plane has equation

$$f_x(x_o, y_o)(x - x_o) + f_y(x_o, y_o)(y - y_o) - (z - z_o) = 0$$

Example: Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1,1,3)$

Solution: Let $f(x, y) = 2x^2 + y^2$. Then

$$f_x(x, y) = 4x \quad \text{and,} \quad f_y(x, y) = 2y$$

$$f_x(1,1) = 4 \quad \text{and,} \quad f_y(1,1) = 2$$

Then by the above theorem

$$4(x - 1) + 2(y - 1) - (z - 3) = 0$$

$$z = 4x - 2y - 3$$

Example: Find an equation for the tangent plane to the surface $z = x^2 y$ at the point $(2,1,4)$

Solution: Since $f(x, y) = x^2 y$, it follows that

$$f_x(x, y) = 2xy \quad \text{and,} \quad f_y(x, y) = x^2$$

So that $x = 2$ and $y = 1$

$$f_x(2,1) = 4 \quad \text{and,} \quad f_y(2,1) = 4$$

Therefore, the tangent plane has the equation

$$4(x-2) + 4(y-2) - (z-4) = 0 \text{ or } 4x + 4y - z = 8$$

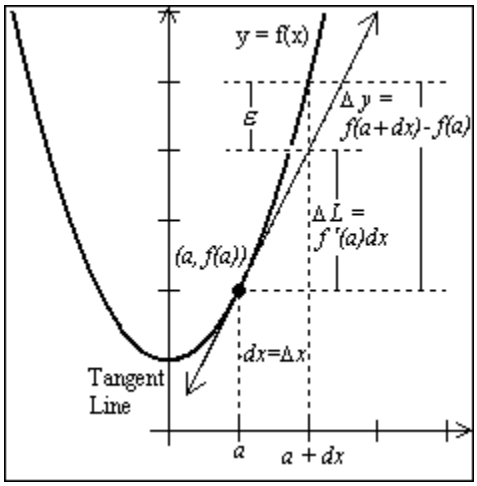
4.5.2 Total differential

Let $y = f(x)$ be a differentiable function. The differential $\boldsymbol{dx}$ is an independent variable. The differential $\boldsymbol{dy}$ is: $dy = f'(x)dx$.

Geometrically, dy is the vertical change along the tangent line of a curve when $x = a$ changes by an amount $dx = \Delta x$. The quantity

$\Delta y = f(a + dx) - f(a)$ is called the increment in $\boldsymbol{y}$.

In the limit $\Delta x \rightarrow 0$, $\Delta y \rightarrow dy$. The error $\varepsilon \rightarrow 0$ as $\Delta x \rightarrow 0$ as well



Analogously, if $z = f(x,y)$ is a function of two variables, we will define dz to be the change in z along the tangent plane at (x_o, y_o, z_o) to the surface $z = f(x,y)$ produced by changes dx and dy in x and y respectively. This is in contrast to

$$\Delta z = f(x_o + \Delta x, y_o + \Delta y) - f(x_o, y_o) \qquad (1)$$

which represents the change in z along the surface produced by changes Δx and Δy in x and y. To derive a formula for dz let $P(x_o, y_o, z_o)$ be a fixed point on the surface

$z = f(x,y)$.If we assume$f$ to be differentiable at (x_o, y_o), then the surface has a tangent plane at P,given by the equation

$$f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) - (z-z_o) = 0 \qquad (2)$$

or

$$z = z_o + f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) \qquad (3)$$

It follows from (3) that the tangent plane has height z_o at $x = x_o$ and $y - y_o$ and it has height

$$z = z_o + f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) \qquad (4)$$

when $x = x_o + dx, y = y_o + dy$ Thus, the change dz in the height of the tangent plane (x,y) Varies from (x_o, y_o) to $(x_o + dx, = y_o + dy)$ is obtained by subtracting z_o from expression (4). This yield

$$dz = f_x(x_o,y_o)dx + f_y(x_o,y_o)dy \quad \text{at } (x_o,y_o) \text{ or} \qquad (5)$$

$$dz = f_x(x,y)dx + f_y(x,y)dy \qquad (6)$$

We call $dz$the total differential of z or total differential of f

Definition**:** Total Differential of a Function of Two Variables is given by

$$\boxed{dz = f_x(x,y)dx + f_y(x,y)dy}$$

Example :

If $z = f(x,y) = x^2 + 3xy - y^2$ find the differential dz

Solution: by the definition

$$dz = f_x(x,y)dx + f_y(x,y)dy$$
$$= (2x + 3y)dx + (3x - 2y)dy$$

4.6 Application, Tangent plane approximation of values of a function

Recall the definition**:**

A function f of two variables is differentiable at (x_o, y_o) if there exist a disk D centered at (x_o, y_o) and functions $\varepsilon_1$and ε_1 of two variables such that

$$f(x,y) - f(x_o,y_o) = f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) + \varepsilon_1(x,y)(x-x_o) + \varepsilon_2(x,y)(y-y_o)$$

for (x, y) in D (1)

Where $\lim_{(x,y)\to(x_o,y_o)} \varepsilon_1(x,y) = 0$ and $\lim_{(x,y)\to(x_o,y_o)} \varepsilon_2(x,y) = 0$ (2)

Since the limits in (2) are 0 the two numbers as $f(x,y) - f(x_o,y_o)$and $f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o)$ are approximately equal when (x, y) close to (x_o, y_o)

$$f(x,y) - f(x_o,y_o) \approx f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) \quad (3)$$

Or equivalently

$$f(x,y) \approx f(x_o,y_o) + f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) \quad (4)$$

From section 3.5 equation (3) any point (x, y, z)on the plane tangent to the graph of f at $(x_o, y_o, f(x_o, y_o))$ satisfies$(z_o = f(x_o, y_o))$

$$z = f(x_o,y_o) + f_x(x_o,y_o)(x-x_o) + f_y(x_o,y_o)(y-y_o) \quad (5)$$

Since the right side of (4) and (5) are identical , we can use z from (5) to approximate $f(x, y)$ if (x, y)close to (x_o, y_o). The use of the tangent plane to approximate $(x, y, f(x, y))$on the graphs of f is called tangent plane approximation.

In order to emphasize that we will consider only points (x, y) that are close to (x_o, y_o), we replace x by $x_o + h$ and y by $y_o + k$ in (4) which becomes

$$f(x_o + h, y_o + k) \approx f(x_o,y_o) + f_x(x_o,y_o)h + f_y(x_o,y_o)k \quad (6)$$

Example:

a) Find the approximation of $f(x,y) = \sqrt{x^2+y^2}$ at a point (x_o, y_o)

b) Use a) to approximate $f(3.04,3.98) = \sqrt{(3.04)^2 + (3.98)^2}$

Solution:

a) $f_x(x,y) = \frac{x}{\sqrt{x^2+y^2}}$ and, $f_y(x,y) = \frac{y}{\sqrt{x^2+y^2}}$

Then using the above discussion

$$\sqrt{x^2+y^2} \approx \frac{x_o}{\sqrt{x_o^2+y_o^2}}(x-x_o) + \frac{y_o}{\sqrt{x_o^2+y_o^2}}(y-y_o)$$

b) Appling formula (6),since 3.04 is close to 3and 3.98 is close to 4

take $x_o = 3$, $y_o = 4$,$h = 0.04$,and $k = -0.02$

$$\sqrt{(3.04)^2 + (3.98)^2} = \sqrt{3^2+4^2} + \frac{3}{5}(0.04) + \frac{4}{5}(-0.02) = 5.008$$

4.7 The Chain Rule and Implicit differentiation

4.7.1The Chain Rule

We have been using the standard chain rule for functions of one variable throughout the last couple of sections. It's now time to extend the chain rule out to more complicated situations. Before we actually do that let's first review the notation for the chain rule for functions of one variable.

The notation that's probably familiar to most people is the following.

$$F(x) = f(g(x)) \quad , F'(x) = f'\big(g(x)\big)g'(x)$$

As with many topics in multivariable calculus, there are in fact many different formulas depending upon the number of variables that we're dealing with. So, let's start this discussion off with a function of two variables, $z = f(x,y)$.

From this point there are still many different possibilities that we can look at. We will be looking at two distinct cases prior to generalizing the whole idea out.

Case 1: $z = f(x,y)$, $x = g(t)$, $y = h(t)$ and compute $\frac{dz}{dt}$

In this case we are going to compute an ordinary derivative since *z* really would be a function of *t* only if we were to substitute in for *x* and *y*.

The chain rule for this case is

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial x}\frac{dy}{dt}$$

So, basically what we're doing here is differentiating *f* with respect to each variable in it and then multiplying each of these by the derivative of that variable with respect to *t*. The final step is to then add all this up.

Example: Compute $\frac{dz}{dt}$ for each of the following

a) $z = xe^{xy}, x = t^2, y = t^{-1}$

b) $z = x^2y^3 + ycosx, x = \ln(t^2), y = \sin(4t)$

Solution:

a) Using the formula

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial x}\frac{dy}{dt} = (e^{xy} + yxe^{xy})(2t) + x^2e^{xy}(-t^{-2})$$

$$2t(e^{xy} + yxe^{xy}) - t^{-2}x^2e^{xy}$$

Substituting the values of x and *y* we have

$$\frac{dz}{dt} = 2t(e^t + te^t) - t^{-2}t^4e^t = 2te^t + t^2e^t$$

Notice that we completed the solution by writing the answer in terms of t, since we desired $\frac{dz}{dt}$

Case 2: $z = f(x,y)$, $x = g(s,t)$, $y = h(s,t)$ and compute $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

c) $z = x^2y^3 + ycosx, x = \ln(t^2), y = \sin(4t)$

Using the formula $\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial x}\frac{dy}{dt}$

$$\frac{dz}{dt} = (2xy^3 \pm ysinx)\left(\frac{2}{t}\right) + (3x^2y^2 + cosx)(4\cos(4t))$$

$$= \frac{4\sin^3(4t)\ln t^2 - 2\sin(4t)\sin(\ln t^2)}{t} + 4\cos(4t)\,(3\sin^2(4t)[\ln t^2]^2 + \cos(\ln t^2))$$

Now let's take a look at the second

Case 3: $z = f(x,y)$, $x = g(s,t)$, $y = h(s,t)$ and compute $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

In this case if we were to substitute in for *x* and *y* we would get that *z* is a function of *s* and *t* and so it makes sense that we would be computing partial derivatives here and that there would be two of them.

Here is the chain rule for both of these cases.

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial x}\frac{\partial y}{\partial s} \qquad \text{and,} \qquad \frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial x}\frac{\partial y}{\partial t}$$

Example: Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ for $z = e^{2r}\sin(3\theta), r = st - t^2, \theta = \sqrt{s^2 + t^2}$

Solution:

Here is the chain rule for $\frac{\partial z}{\partial s}$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial r}\frac{\partial r}{\partial s} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial s}$$

$$= (2e^{2r}\sin(3\theta))(t) + (3e^{2r}\cos(3\theta))\frac{s}{\sqrt{s^2+t^2}}$$

$$= t\left(2e^{2(st-t^2)}\sin\left(3\sqrt{s^2+t^2}\right)\right) + \frac{3se^{2(st-t^2)}\cos\left(3\sqrt{s^2+t^2}\right)}{\sqrt{s^2+t^2}}$$

Now the chain rule for $\frac{\partial z}{\partial t}$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial r}\frac{\partial r}{\partial t} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial t}$$

$$= (2e^{2r}\sin(3\theta))(s - 2t) + (3e^{2r}\cos(3\theta))\frac{t}{\sqrt{s^2+t^2}}$$

$$= (s-2t)\left(2e^{2(st-t^2)}\sin\left(3\sqrt{s^2+t^2}\right)\right)+\frac{3te^{2(st-t^2)}\cos\left(3\sqrt{s^2+t^2}\right)}{\sqrt{s^2+t^2}}$$

General version of the chain rule.

Suppose that *z* is a function of *n* variables, $x_1, x_2, \ \ldots\ x_n$, and that each of these variables are in turn functions of *m* variables, $t_1, t_2, \ \ldots\ t_m$. Then for any variable $t_i, i = 1,2,3,\ldots,m$ we have the following,

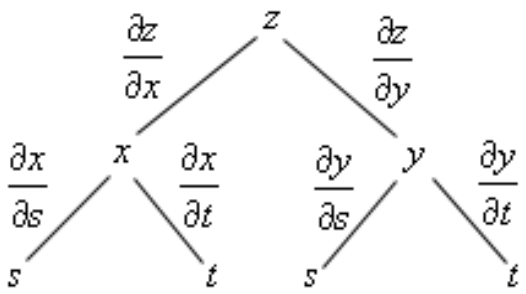
$$\frac{\partial z}{\partial t_i} = \frac{\partial z}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial z}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \ \ldots + \frac{\partial z}{\partial x_n}\frac{\partial x_n}{\partial t_i}$$

That's a lot to remember. There is actually an easier way to construct all the chain rules that we've discussed in the section or will look at in later examples. We can build up a **tree diagram** that will give us the chain rule for any situation. To see how these work let's go back and take a look at the chain rule for $\frac{\partial z}{\partial s}$ given that $z = f(x,y)$, $x = g(s,t)$, $y = h(s,t)$.

We already know what this is, but it may help to illustrate the tree diagram if we already know the answer. For reference here is the chain rule for this case,

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

Here is the tree diagram for this case.



We start at the top with the function itself and the branch out from that point. The first set of branches is for the variables in the function. From each of these endpoints we put down a further set of branches that gives the variables that both *x* and *y* are a function of. We connect each letter with a line and each line represents a partial derivative as shown. Note that the letter in the numerator of the partial derivative is the upper "node" of the tree and the letter in the denominator of the partial derivative is the lower "node" of the tree.

Example: Use a tree diagram to write down the chain rule for the given derivatives.

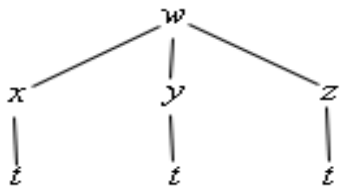
a) $\frac{dw}{dt}$ for $w = f(x,y,z), x = g_1(t), y = g_2(t),$ and $z = g_3(t)$

b) $\frac{dw}{dr}$ for $w = f(x,y,z), x = g_1(s,t,r), y = g_2(s,t,r),$ and $z = g_3(s,t,r)$

Solution:

a) $w = f(x,y,z), x = g_1(t), y = g_2(t),$ and $z = g_3(t)$

So, we'll first need the tree diagram so let's get that.

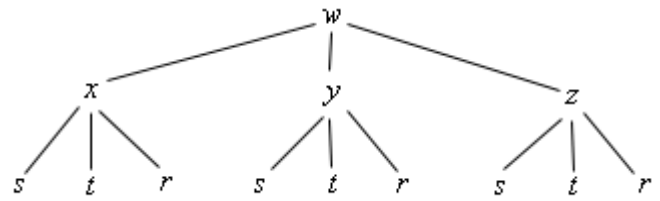


From this is looks like the chain rule for this case should be,

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

b) $w = f(x,y,z), x = g_1(s,t,r), y = g_2(s,t,r),$ and $z = g_3(s,t,r)$

Here is the tree diagram for this situation.



From this it looks like the derivative will be,

$$\frac{dw}{dr} = \frac{\partial f}{\partial x}\frac{dx}{dr} + \frac{\partial f}{\partial y}\frac{dy}{dr} + \frac{\partial f}{\partial z}\frac{dz}{dr}$$

Exercise:

a) Let $w = xcosyz^2, x = sint, y = t^2$ and $z = e^t$

Find $dw/_{dt}$

b) Let $w = \sqrt{x} + y^2z^3, x = 1 + u^2 + v^2, y = uv,$ and $z = 3u.$

Find $\frac{dw}{du}$ and $\frac{dw}{dv}$

c) Suppose that $w(x, y), x = g(u, v), y = h(u, v), u = j(t), v = k(t).$

Find a formula for $dw/_{dt}$

4.7.2 Implicit differentiation

Through the chain rule we can more completely describe implicit differentiation which you have studied in applied I

Example: Suppose that$w = f(x, y) and\ y = g(x)$. Find a formula for $dw/dx.$

Solution: drawing appropriate diagram shows that

$$\frac{dw}{dx} = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y}\frac{dy}{dx}$$

(In this formula, $\partial w/_{\partial x}\ refres\ to\ the\ partial\ derivative\ of\ z = w(x, y)$ with respect to x, whereas $dw/_{dx}$ refers to the derivative of $z = f(x, g(x))$ with respect to x.)

Now suppose that f is a function of two variables that has partial derivatives, and assume that the equation $f(x,y) = 0$ defines a differentiable function $y = g(x)$ of x so that $f\big(x, g(x)\big) = 0$.if $w = f(x,y)$, then by assumption

$$\frac{dw}{dx} = \frac{d}{dx}\big(f\big(x, g(x)\big) = \frac{d}{dx}(0) = 0$$

$$0 = \frac{dw}{dx} = {}^{\partial w}/_{\partial x} + {}^{\partial w}/_{\partial y}\frac{dy}{dx}$$

Which in turn implies $\frac{dy}{dx} = -\frac{{}^{\partial w}/_{\partial x}}{{}^{\partial w}/_{\partial y}}$ (*)

Example: Let $x^3 + y^3 = 2xy$. Find $\frac{dy}{dx}$

Solution: let $w = x^3 + y^3 = 2xy$. Then

${}^{\partial w}/_{\partial x} = 3x^2 - 2y$ and ${}^{\partial w}/_{\partial y} = 3y^2 - 2x$

Equation (*) above implies

$$\frac{dy}{dx} = \frac{{}^{\partial w}/_{\partial x}}{{}^{\partial w}/_{\partial y}} = \frac{-(3x^2 - 2y)}{(3y^2 - 2x)} = \frac{2y - 3x^2}{3y^2 - 2x}$$

Exercise :

a) Find $\frac{dy}{dx}$ for $xcos(3x) + x^3y^5 = 3x - e^{xy}$

b) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $x^2\sin(2y - 5z) = 1 + ycos(6zx)$

4.8 Relative extreme of function of two variables

In this section we are going to be looking at identifying relative minimums and relative maximums. Recall as well that we will often use the word extrema to refer to both minimums and maximums.

The definition of relative extrema for functions of two variables is identical to that for functions of one variable we just need to remember now that we are working with functions of two variables. So, for the sake of completeness here is the definition of relative minimums and relative maximums for functions of two variables.

Definition:

1. A function $f(x,y)$ has a *relative minimum* at the point(a,b) if $f(x,y) \geq f(a,b)$ for all points (x,y) in some region around (a,b)
2. A function $f(x,y)$ has a *relative maximum* at the point(a,b) if $f(x,y) \leq f(a,b)$ for all points (x,y) in some region around (a,b)

Note that this definition does not say that a relative minimum is the smallest value that the function will ever take. It only says that in some region around the point $(a,b)$the function will always be larger than $f(a,b)$. Outside of that region it is completely possible for the function to be smaller. Likewise, a relative maximum only says that around $(a,b)$the function will always be smaller than $f(a,b)$. Again, outside of the region it is completely possible that the function will be larger.

Definition: The point (a,b) is a *critical point* (or a stationary point) of $f(x,y)$.

Provided one of the following is true,

1. $\nabla f(a,b) = \mathbf{0}$(this is equivalent to saying that $f_x(a,b) = 0$and $f_y(a,b) = 0$)
2. $f_x(a,b)$ and /or $f_y(a,b)$ doesn't exist.

Theorem: If the point (a,b) is a relative extrema of the function $f(x,y)$ then(a,b) is also a critical point of $f(x,y)$ and $\nabla f(a,b) = \mathbf{0}$

Note that this does NOT say that all critical points are relative extrema. It only says that relative extrema will be critical points of the function. To see this let's consider the function

$$f(x,y) = xy$$

The two first order partial derivatives are,

$$f_x(x,y) = y \quad f_y(x,y) = x$$

The only point that will make both of these derivatives zero at the same time is $(0,0)$ and also $(0,0)$ is a critical point for the function. But if we start at the origin and move into either of the quadrants where both *x* and *y* are the same sign the function increases. However, if we start at the origin and move into either of the quadrants where *x* and *y* have the opposite sign then the function decreases. In other words, no matter what region you take about the origin there will be points larger than $f(0,0) = 0$ and points smaller than $f(0,0) = 0$.

Therefore, there is no way that $(0,0)$can be a relative extrema.

Critical points that exhibit this kind of behavior are called *saddle points*.

Example: Let$f(x,y) = 3 - x^2 + 2x - y^2 - 4y$. Find the critical point of f

Solution: The partial derivatives of f exist at every point in the domain of f, so relative extreme values can occur only at points at which both partial derivatives are 0. The partial derivatives are

$f_x(x,y) = -2x + 2$ and $f_y(x,y) = -2y - 4$

$f_x(x,y) = 0$ only if $x = 1$ and $f_y(x,y) = 0$ only if $y = -2$.

This means that $f_x(x,y) = f_x(x,y) = 0$ only if $(x,y) = (1,-2)$

And those $(1,-2)$ is the only critical point of f

Theorem: Suppose that (a,b) is a critical point of $f(x,y)$ and that the second order partial derivatives are continuous in some region that contains(a,b). Define,

$$D = D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - \left[f_{xy}(a,b)\right]^2$$

We then have the following classifications of the critical point.

1. If $D > 0$ and $f_{xx}(a,b) > 0$ then there is a relative minimum at (a,b)
2. If $D > 0$ and $f_{xx}(a,b) < 0$ then there is a relative maximum at (a,b)
3. If $D < 0$ hen the point(a,b) is a saddle point.
4. If $D = 0$ hen the point(a,b) may be a relative minimum, relative maximum or a saddle point. Other techniques would need to be used to classify the critical point.

Note that if $D > 0$ the both$f_{xx}(a,b)$ and $f_{yy}(a,b)$will have the same sign and so in the first two cases above we could just as easily replace $f_{xx}(a,b)$ with $f_{yy}(a,b)$.

Example: Find and classify all the critical points of $f(x,y) = 4 + x^3 + y^3 - 3xy$

Solution:

We first need all the first order (to find the critical points) and second order (to classify the critical points) partial derivatives so let's get those

$$f_x(x,y) = 3x^2 - 3y \text{ and } f_y(x,y) = 3y^2 - 3x$$

$$f_{xx}(x,y) = 6x \quad ,f_{yy}(x,y) = 6y \text{ and } f_{xy}(x,y) = -3$$

Now let's first find the critical points. Critical points will be solutions to the system of equations,

$$f_x(x,y) = 3x^2 - 3y = 0$$

$$f_y(x,y) = 3y^2 - 3x = 0$$

This is a non-linear system of equations and these can, on occasion, be difficult to solve. However, in this case it's not too bad. We can solve the first equation for *y* as follows,

$$3x^2 - 3y = 0 \text{ implies } y = x^2$$

Plugging this into the second equation gives,

$$3(x^2)^2 - 3x = 3x(x^3 - 1) = 0$$

From this we can see that we must have $x = 0$ or $x = 1$ Now use the fact that $y = x^2$ to get the critical points.$(0,0)$ and $(1,1)$

So, we get two critical points. All we need to do now is classify them. To do this we will need *D*. Here is the general formula for *D*.

$$D = D(x,y) = f_{xx}(x,y)f_{yy}(x,y) - [f_{xy}(x,y)]^2$$

$$= (6x)(6y) - (-3)^2$$

$$= 36xy - 9$$

To classify the critical points all that we need to do is plug in the critical points and use the fact above to classify them

$$D = D(0,0) = -9 < 0$$

So, for $(0,0)$ D is negative and so this must be a saddle point.

$D(1,1) = D(1,1) = 36 - 9 = 27 > 0$ and $f_{xx}(1,1) = 6 > 0$

For $(1,1)$ D is positive andf_{xx} is positive and so we must have a relative minimum.

Example: Find and classify all the critical points for $f(x,y) = 3x^2y + y^3 - 3y^2 + 2$

Solution:

As with the first example we will first need to get all the first and second order derivatives.

$f_x(x,y) = 6xy - 6x$ and, $f_y(x,y) = 3x^2 + 3y^2 - 6y$

$$f_{xx}(x,y) = 6y - y \quad ,f_{yy}(x,y) = 6y - 6 \text{ and}$$

$f_{xy}(x,y) = 6x$

We'll first need the critical points. The equations that we will need to solve this time are,

$$6xy - 6x = 0$$

$$3x^2 + 3y^2 - 6y = 0$$

These equations are a little trickier to solve than the first set, but once you see what to do they really aren't terribly bad.

First, let's notice that we can factor out a 6*x* from the first equation to get,

$$6x(y - 1) = 0$$

So, we can see that the first equation will be zero if $x = 0 \; or \; y = 1$

To find the critical points we can plug these (individually) into the second equation and solve for the remaining variable.

for $x = 0$

$3y^2 - 6y = 3y(y - 2)$ implies, $y = 0, y = 2$

for $y = 1$

$3x^2 - 3 = 3(x^2 - 1) = 0$ implies, $x = -1, x = 1$

Thus, we have the critical points $(0,0)$ and $(0,2)$ when $x = 0$ and

$(1,1)$ and $(-1,1)$ when $y = 1$

Now all we need to do is classify the critical points. To do this we'll need the general formula for *D*.

$$D(x,y) = f_{xx}(x,y)f_{yy}(x,y) - \left[f_{xy}(x,y)\right]^2$$

$$D(x,y) = (6y-6)(6y-6) - (6x)^2 = (6y-6)^2 - 36x^2$$

$(0,0)$	$D(0,0) = 36 > 0$	$f_{xx}(0,0) = -6 < 0$
$(0,2)$	$D(0,2) = 36 > 0$	$f_{xx}(0,2) = 6 > 0$
$(1,1)$	$D(1,1) = -36 < 0$	
$(-1,1)$	$D(-1,1) = -36 < 0$	

So, it looks like we have the following classification of each of these critical points.

$(0,0)$: Relative Maximum

$(0,2)$: Relative Minimum

$(1,1)$: Saddle point

$(-1,1)$: Saddle point

Exercise:

a) Let $f(x,y) = y^2 - x^2$. Show that the origin is the only critical point but that $f(0,0)$ is not the relative extreme value of f

b) Let $f(x,y) = x^2 - 2xy + \frac{1}{3}y^3 - 3y$. Determine at which points f have relative extreme values and at which point f has saddle points

4.9 Largest and smallest values of a function on a given set

In this section we are going to extend the work from the previous section. In the previous section we were asked to find and classify all critical points as relative minimums,

relative maximums and/or saddle points. In this section we are want to optimize a function, that is identify the absolute minimum and/or the absolute maximum of the function, on a given region in $\boldsymbol{R^2}$. Note that when we say we are going to be working on a region in $\boldsymbol{R^2}$ we mean that we're going to be looking at some region in the xy-plane**.**

Definition**:**

1. A region in $\boldsymbol{R^2}$is called closed if it includes its boundary. A region is called open if it doesn't include any of its boundary points.
2. A region in $\boldsymbol{R^2}$ is called bounded if it can be completely contained in a disk. In other words, a region will be bounded if it is finite.

Extreme value theorem

If $f(x,y)$ is continuous in some closed, bounded set d in $\boldsymbol{R^2}$ then there are points in D,(x_1,y_1) and (x_2,y_2) so that $f(x_1,y_1)$ is the absolute maximum and $f(x_2,y_2)$ is the absolute minimum of the function in D.

Note that this theorem does NOT tell us where the absolute minimum or absolute maximum will occur. It only tells us that they will exist. Note as well that the absolute minimum and/or absolute maximum may occur in the interior of the region or it may occur on the boundary of the region.

Finding absolute extrema

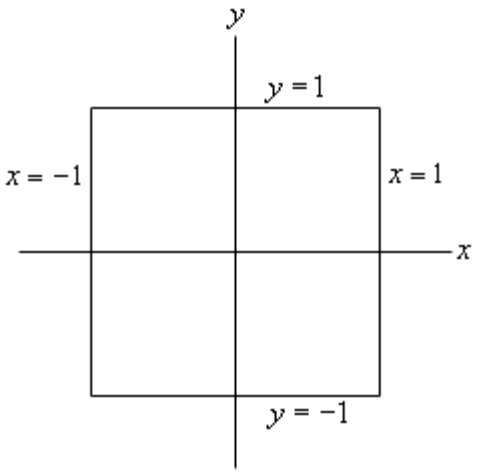
1. Find all the critical points of the function that lie in the region *D* and determine the function value at each of these points.

2. Find all extrema of the function on the boundary. (This usually involves the Applied I approach for this work.)

3. The largest and smallest values found in the first two steps are the absolute minimum and the absolute maximum of the function.

Example: Find the absolute minimum and absolute maximum of $f(x,y) = x^2 + 4y^2 - 2x^2y + 4$ on the rectangle given by $-1 \le x \le 1$ and $-1 \le y \le 1$

Solution: Let's first get a quick picture of the rectangle for reference purposes



The boundary of this rectangle is given by the following conditions

right side: $x = 1, 1 - \le y \le 1$

left side: $x = -1, -1 \le y \le 1$

upper side: $y = 1, -1 \le x \le 1$

lower side: $y = -1, -1 \le x \le 1$

These will be important in the second step of our process. We'll start this off by finding all the critical points that lie inside the given rectangle. To do this we'll need the two first order derive

$$f_x(x,y) = 2x - 4xy \qquad f_y(x,y) = 8y - 2x^2$$

Note that since we aren't going to be classifying the critical points we don't need the second

order derivatives. To find the critical points we will need to solve the system,

$$2x - 4xy = 0$$
$$8y - 2x^2 = 0$$

We can solve the second equation for *y* to get,

$$y = \frac{x^2}{4}$$

Plugging this into the first equation gives us,

$$2x - 4x\left(\frac{x^2}{4}\right) = 2x - x^3 = x(2 - x^2) = 0$$

This tells us that we must have $x = 0$ or $x = \pm\sqrt{2} = \pm 1.414\ldots$ Now, recall that we only want critical points in the region that we're given. That means that we only want critical points for which $-1 \le x \le 1$. The only value of *x* that will satisfy this is the first one so we can ignore the last two for this problem. Note however that a simple change to the boundary would include these two so don't forget to always check if the critical points are in the region (or on the boundary since that can also happen).

Plugging *x* = 0 into the equation for *y* gives us,

$$y = \frac{0^2}{4} = 0$$

The single critical point, in the region (and again, that's important), is $(0,0)$. We now need to get the value of the function at the critical point.

$$f(0,0) = 4$$

Eventually we will compare this to values of the function found in the next step and take the largest and smallest as the absolute extrema of the function in the rectangle. Now we have reached the long part of this problem. We need to find the absolute extrema of the function along the boundary of the rectangle. What this means is that we're going to need to look at what the function is doing along each of the sides of the rectangle listed above.

Let's first take a look at the right side. As noted above the right side is defined by

$$x = 1, -1 \le y \le 1$$

Notice that along the right side we know that x =1. Let's take advantage of this by defining a new function as follows,

$$g(y) = f(1,y) = 1^2 + 4y^2 - 2(1^2)\ y + 4 = 5 + 4y^2 - 2y$$

Now, finding the absolute extrema of $f\ (x,y)$ along the right side will be equivalent to finding the absolute extrema of $g(y)$ in the range $-1 \le y \le 1$. Hopefully you recall how to do this from Applied Mathematics I. We find the critical points of $g(y)$ in the range $-1 \le y \le 1$ and then evaluate $g(y)$ at the critical points and the end points of the range of y's

Let's do that for this problem.

$$g'(y) = 8y - 2 \text{ implies } y = \tfrac{1}{4}$$

This is in the range and so we will need the following function evaluations.

$$g(-1) = 11 \qquad g(1) = 7 \qquad g\left(\tfrac{1}{4}\right) = 4.75$$

Notice that, using the definition of $g(y)$ these are also function values for $f\ (x,y)$.

$$g(-1) = f(1,-1) = 11$$
$$g(1) = f(1,1) = 7$$
$$g\left(\tfrac{1}{4}\right) = f\left(1,\tfrac{1}{4}\right) = 4.75$$

We can now do the left side of the rectangle which is defined by,

$$x = -1, -1 \le y \le 1$$

Again, we will define a new function as follows,

$$g(y) = f(-1,y) = (-1)^2 + 4y^2 - 2(-1)^2\ y + 4 = 5 + 4y^2 - 2y$$

Notice however that, for this boundary, this is the same function as we looked at for the right side. This will not always happen, but since it has let's take advantage of the fact that we've already done the work for this function. We know that the critical point is $y = \tfrac{1}{4}$ and we know that the function value at the critical point and the end points are,

$$g(-1) = 11 \qquad g(1) = 7 \qquad g\left(\tfrac{1}{4}\right) = 4.75$$

The only real difference here is that these will correspond to values of $f\ (x,y)$ at different points than for the right side.

$$g(-1) = f(-1,-1) = 11$$
$$g(1) = f(-1,1) = 7$$
$$g\left(\tfrac{1}{4}\right) = f\left(-1,\tfrac{1}{4}\right) = 4.75$$

We can now look at the upper side defined by,

$y = 1, -1 \le x \le\ 1$

We'll again define a new function except this time it will be a function of *x*.

$$h(x) = f\ (x,1) = x^2 + 4(1^2) - 2x^2(1) + 4 = 8 - x^2$$

We need to find the absolute extrema of $h(x)$ on the range $-1 \le x \le\ 1$. First find the critical point(s).

$$h'(x) = -2x \text{ implies } x = 0$$

The value of this function at the critical point and the end points is,

$$h(-1) = 7 \qquad h(1) = 7 \qquad h(0) = 8$$

and these in turn correspond to the following function values for $f\ (x,y)$

$$h(-1) = f(-1,1) = 7$$
$$h(1) = f(1,1) = 7$$
$$h(0) = f(0,1) = 8$$

Note that there are several "repeats" here. The first two function values have already been computed when we looked at the right and left side. This will often happen.

Finally, we need to take care of the lower side. This side is defined by,

$$y = -1, -1 \le x \le\ 1$$

The new function we'll define in this case is,

$h(x) = f(x,-1) = x^2 + 4(-1)^2 - 2x^2(-1) + 4 = 8 + 3x^3$

The critical point for this function is,

$h'(x) = 6x$ implies, $x = 0$

The function values at the critical point and the endpoint are,

$$h(-1) = 11 \qquad h(1) = 11 \qquad h(0) = 8$$

and the corresponding values for $f\ (x, y)$ are,

$$h(-1) = f(-1, -1) = 11$$
$$h(1) = f(1, -1) = 11$$
$$h(0) = f(0, -1) = 8$$

The final step to this (long…) process is to collect up all the function values for $f\ (x, y)$ that

We have computed in this problem. Here they are,

$$f(0) = 4$$
$$f(1, -1) = 11$$
$$f(1,1) = 7$$
$$f\left(1, \tfrac{1}{4}\right) = 4.75$$
$$f(-1, -1) = 11$$
$$f(-1,1) = 7$$
$$f\left(-1, \tfrac{1}{4}\right) = 4.7$$
$$f(0,1) = 8$$
$$f(0, -1) = 8$$

The absolute minimum is at $(0,0)$ since gives the smallest function value and the absolute maximum occurs at $(1, -1)$ and $(-1, -1)$ since these two points give the largest function value.

Exercise:

1. Find the extreme values of f
   a. $f(x, y) = x^2 - y^2$, D is a disk $x^2 + y^2 \leq 1$

b. $f(x,y) = 2sinx + 3cosy$; D is the square region with vertex $(0, -\pi/2), (\pi, -\pi/2), (\pi, \pi/2), (0, \pi/2)$

2. Show that if f has critical point at (x_o, y_o) and if $f_{xx}(x_o, y_o) > 0$ *and* $f_{yy}(x_o, y_o) < 0$, then f has saddle point at (x_o, y_o).

4.10 Extreme Values under constraint Condition: Lagrange's multiplier

In the previous section we optimized (*i.e.* found the absolute extrema) a function on a region that contained its boundary. Finding potential optimal points in the interior of the region isn't too bad in general, all that we needed to do was find the critical points and plug them into the function. However, as we saw in the examples finding potential optimal points on the boundary was often a fairly long and messy process.

In this section we are going to take a look at another way of optimizing a function subject to given constraint(s). The constraint(s) may be the equation(s) that describe the boundary of a region although in this section we won't concentrate on those types of problems since this method just requires a general constraint and doesn't really care where the constraint came from.

Let us consider the problem of finding an extreme value of a function f of two variables subjected to a constraint of the form $g(x,y) = k$; that is, we seek extreme values of f on the level curve $g(x,y) = k$ (rather than on the entire domain of f). If has an extreme value on the level curve at the point (x_o, y_o), then under certain conditions there exists a number λ such that

$$\nabla f(x_o, y_o) = \lambda \nabla g(x_o, y_o)$$

Theorem: Let f and g be differentiable at (x_o, y_o). Let C be the level curve $g(x,y) = k$ that contains (x_o, y_o), Assume that C is smooth, and that (x_o, y_o) is not an end point of the curve. If $\nabla g(x_o, y_o) \neq 0$ and if f has an extreme value on C at (x_o, y_o), then the is a number λ such that

$$\nabla f(x_o, y_o) = \lambda \nabla g(x_o, y_o) \qquad (*)$$

λ is called a *Lagrange multiplier* for f and g

The method determining extreme values by means of Lagrange multipliers proceed as follows:

1. Assume that f has an extreme value on the level curve $g(x, y) = k$
2. Solve the equations

$$\text{Constraint} \qquad g(x, y) = k$$

$$\nabla f(x_o, y_o) = \lambda \nabla g(x_o, y_o) = \begin{cases} f_x(x, y) = \lambda g_x(x, y) \\ f_y(x, y) = \lambda g_y(x, y) \end{cases} \qquad (1)$$

3. Calculate the values of f at each point (x, y) that arises in step 2, and at each end point (if any) of the curve. If f has maximum value in the level curve $g(x, y) = k$ it would be the largest of the values computed; if f has a minimum value on the level curve, it will be the smallest of the values computed.

Example: let$f(x, y) = x^2 + y^3$, Find the extreme values of f on the ellipse $x^2 + 2y^2 = 1$

Solution: Let

$$g(x, y) = x^2 + 2y^2$$

So the constraint is $g(x, y) = x^2 + 2y^2 = 1$. Since

$$\nabla f(x, y) = 2x\, i + 12y^2\, j \text{ and } \nabla g(x, y) = 2x\, i + 4y\, j$$

The equation we will use to find x and y are

$$\text{Constraint} \qquad x^2 + 2y^2 = 1 \qquad (2)$$

$$\nabla f(x_o, y_o) = \lambda \nabla g(x_o, y_o) = \begin{cases} 2x = \lambda 2x \\ 12y^2 = \lambda 4y \end{cases} \qquad (3)$$

By (3) either $x = 0$ or $\lambda = 1$. If $x = 0$ then the (2) implies $2y^2 = 1$ so that$y = \frac{1}{\sqrt{2}}\ or\ y = -\frac{1}{\sqrt{2}}$. If $\lambda = 1$ then (3) becomes$12y^2 = 4y$, which means that $y = 0\ \ or\ y = \frac{1}{3}$.

By (2)

If $y = 0$, then $x^2 + 2(0)^2 = 1$, so $x = 1\ or\ x = -1$

If $y = \frac{1}{3}$, then $x^2 + 2(\frac{1}{3})^2 = 1$, so $x = \frac{\sqrt{7}}{3}\ or\ x = -\frac{\sqrt{7}}{3}$

Thus the only possible extreme values of occur at

$$\left(0, \frac{1}{\sqrt{2}}\right), \left(0, -\frac{1}{\sqrt{2}}\right), (1,0), (-1,0), \left(\frac{\sqrt{7}}{3}, \frac{1}{3}\right), and\ \left(-\frac{\sqrt{7}}{3}, \frac{1}{3}\right)$$

$$f\left(0, \frac{1}{\sqrt{2}}\right) = \sqrt{2} \qquad f\ (1,0) = 1 = f\ (-1,0)$$

$$f\left(0, -\frac{1}{\sqrt{2}}\right) = -\sqrt{2} \qquad f\left(\frac{\sqrt{7}}{3}, \frac{1}{3}\right) = f\left(-\frac{\sqrt{7}}{3}, \frac{1}{3}\right) = \frac{25}{27}$$

We conclude that the maximum value $\sqrt{2}$ of f occurs at $\left(0, \frac{1}{\sqrt{2}}\right)$ and the minimum value of $f = -\sqrt{2}$ occurs at $\left(0, -\frac{1}{\sqrt{2}}\right)$

Example: Find the maximum and minimum of $f\ (x, y) = 5x - 3y$ subject to the constraint

x^2+y^2=136.

Solution:

Constraint $x^2 + y^2 = 1$

$$\nabla f(x_o, y_o) = \lambda \nabla g(x_o, y_o) = \begin{cases} 5 = \lambda 2x \\ -3 = \lambda 2y \end{cases}$$

Notice that, as with the last example, we can't have λ = 0 since that would not satisfy the first two equations. So, since we know that $\lambda \neq 0$ we can solve the first two equations for *x* and *y* respectively. This gives,

$$x = \frac{5}{2\lambda} \qquad y = -\frac{3}{2\lambda}$$

Plugging these into the constraint gives,

$$\frac{25}{4\lambda^2} + \frac{9}{4\lambda^2} = \frac{17}{2\lambda^2} = 136$$

We can solve this for λ $\lambda = \pm\frac{1}{4}$

Now, that we know λ we can find the points that will be potential maximums and/or minimums.

If $\lambda = -\frac{1}{4}$ we get

$$x = -10 \qquad y = 6$$

and if $\lambda = \frac{1}{4}$ we get

$$x = 10 \qquad y = -6$$

Thus the only possible extreme values of occur at $(-10,6)$ and $(10,-6)$

$f(-10,6) = -68$ minimum at $(-10,6)$

$f(10,-6) = 68$ Maximum at $(10,-6)$

Exercise:

1. Find the maximum and minimum values of $f(x,y) = 4x^2 + 10y^2$ on the disk $x^2 + y^2 \leq 4$.

2. Find the maximum and minimum of f $f\ (x,y,z) = 4y - 2z$ subject to the constraints

   $2x - y - z = 2$ and $x^2 + y^2 = 1$.

Method of Lagrange Multipliers for function of three variables

1. Solve the following system of equations.

$$\nabla f(x,y,z) = \lambda\, \nabla g(x,y,z) \begin{cases} f_x = \lambda g_y \\ f_y = \lambda \mathrm{g}_y \\ f_z = \lambda \mathrm{g}_z \end{cases} \text{ and}$$

Constraint $g(x,y,z) = k$

2. Plug in all solutions, (x,y,z), from the first step into $f\ (x,y,z)$ and identify the minimum and maximum values, provided they exist.

Example: Find the minimum distance from a point on the surface $xy + 2xz = 5\sqrt{5}$ to the origin.

Solution: Let the distance function is represented by

$$f_1(x,y,z) = \sqrt{x^2 + y^2 + z^2}$$

Represents the distance from (x,y,z) to the origin and seek the minimum value of f_1 on the surface. However the computation evolves the use of Lagrange Multiplier.

Note that the functions $f_1(x,y,z) = \sqrt{x^2 + y^2 + z^2}$ and $f(x,y,z) = x^2 + y^2 + z^2$ have extreme values at identical point. Therefore, it would be easer to use f rather than f_1 let the constraint $g(x,y,z) = xy + 2xz = 5\sqrt{5}$

Hence, solve the following system of equations

$$\nabla f(x,y,z) = \lambda\, \nabla g(x,y,z) \begin{cases} 2x = \lambda(y+2z) & (1) \\ 2y = \lambda x & (2) \\ 2z = 2\lambda x & (3) \end{cases}$$

and

$$\text{Constraint} \quad g(x,y,z) = xy + 2xz = 5\sqrt{5} \qquad (4)$$

Now $\lambda \neq 0$ since $x = y = z = 0$ is not on the surface (constraint).Then

$$x = \frac{2y}{\lambda} \qquad (5)$$

Substituting $x = \frac{2y}{\lambda}$ in (4)

$$2z = 2\lambda\left(\frac{2y}{\lambda}\right) \text{ implies } z = 2y \qquad (6)$$

If $y = 0$ the by (5) and (6) we have $x = y = z = 0$ so would not hold, then $y \neq 0$

Using (5) and (6) to substitute for x we find

$$2\left(\frac{2y}{\lambda}\right) = (y + 2(2y))\lambda$$

Implies $\lambda^2 = \frac{4}{5}$, $\lambda = \pm\frac{2}{\sqrt{5}}$ (7)

Using (5) and (6) to substitute for x and z in (4)

$$5\sqrt{5} = \left(\frac{2y}{\lambda}\right) y + 2\left(\frac{2y}{\lambda}\right)(2y) = \frac{10}{\lambda} y^2 \qquad (8)$$

From (8) we see that λ is positive i.e. $\lambda = \frac{2}{\sqrt{5}}$ and the (7) implies $y = -1 \;\; or \;\; y = 1$

If $y = 1$ then $x = \frac{2y}{\lambda} = \sqrt{5}$ and $z = 2y = 2$

If $y = -1$ then $x = \frac{2y}{\lambda} = -\sqrt{5}$ and $z = 2y = -2$

Consequently the only points on the surface $xy + 2xz = 5\sqrt{5}$ that can have a minimum distance from the origin are $(\sqrt{5}, 1,2) and\ (-\sqrt{5}, -1, -2)$ since by assumption the minimum distance exist and the minimum value of f_1 on the surface

Hence,

$$f_1(\sqrt{5}, 1,2) = f(-\sqrt{5}, -1, -2) = \sqrt{10}$$

the minimum distance from a point on the surface to the origin is $\sqrt{10}$

Exercise:

1. Find the dimensions of the box with largest volume if the total surface area is $64cm^2$
2. Find the maximum and minimum values of $f\ (x, y, z) = xyz$ subject to the Constraint $x + y + z = 1$. Assume that $x, y, z \geq 0$.
3. Find the maximum and minimum of $f\ f\ (x, y, z) = 4y - 2z$ subject to the constraints
   $2x - y - z = 2$ and $x^2 + y^2 = 1$.

Chapter Five: Multiple Integral

5.1 Double Integrals and their Evaluation by Iterated integrals

Recall that the definition of definite integral of function of one variable.

Definition**:** For any function *f* defined on the interval [*a*, *b*] and ||*P*|| (the norm of the partition) defined as the maximum of all the intervals on [*a*, *b*] (i.e., ||*P*|| = max{Δx_i}), the definite integral of *f* on [*a*, *b*] is:

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(c_i)\Delta x_i \ ,$$

Provided the limit exists and is the same for all values of the evaluation points $c_i \in [x_{i-1}, x_i]$ for $i = 1, 2 \ldots n$. In this case, we say *f* is *integrable* on [*a*, *b*].

In this section we want to integrate a function of two variables, $f\ (x,y)$. With functions of one variable we integrated over an interval (*i.e.* a one-dimensional space) and so it makes some sense then that when integrating a function of two variables we will integrate over a region of R^2(two dimensional space).

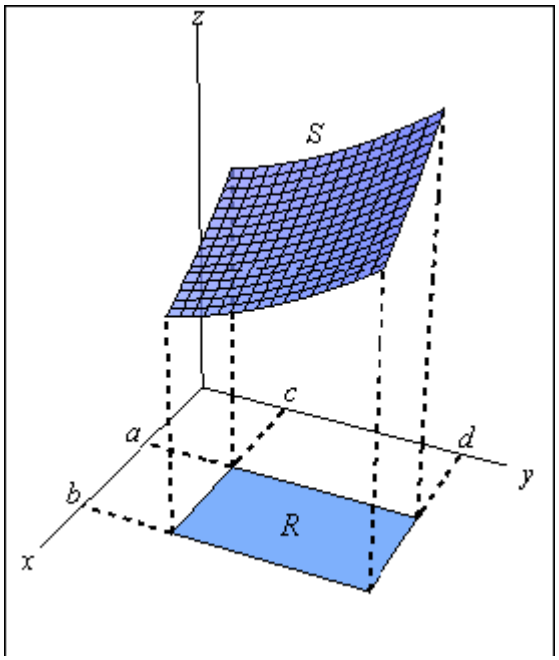
We will start out by assuming that the region in n R^2 is a rectangle which we will denote as follows,

, $R = [a,b]x[c,d]$

This means that the ranges for *x* and *y* are $a \le x \le b \text{ and } c \le y \le d.$

Also, we will initially assume that $f\ (x,y)$ ≥0 although this doesn't really have to be the case.

Let's start out with the graph of the surface *S* give by graphing $f\ (x,y)$ over the rectangle *R*.



Definition**:** Double Integral of a Function of Two Variables over a Rectangular Region

For any function *f*(*x*, *y*) defined on the rectangle $R = \{(x, y) \mid a \le x \le b, c \le y \le d\}$, and ||*P*|| (the norm of the partition) defined as the maximum diagonal of any rectangle in the partition, the *double integral* of *f* over *R* is defined as:

$$\iint_R f(x,y)\,dA = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(u_i, v_i)\,\Delta A_i\,,$$

Provided the limit exists and is the same for all choices of the evaluation points $(u_i, v_i) \in R$ for $i = 1, 2, \ldots, n$. In this case, we say *f* is integrable over *R*.

If f is non negative and integrable on *R*, then the volume **V** of the solid region between the graph of f and R is given by

$$V = \iint_R f(x,y)dA$$

Definition**:**

a. A plane region **R** is *vertically simple* if there are two continuous functions g_1 and g_2 on an interval $[a,b]$ such that $g_1(x) \leq g_2(x)$ for $a \leq x \leq b$ and such that R is the region between the graph of g_1 and g_2 on $[a,b]$. In this case we say that R is the *vertically simple region* between the graphs of g_1 and g_2 on $[a,b]$.
b. A plane region R is *horizontally simple* if there are two continuous functions h_1 and h_2 on an interval $[c,d]$ such that $h_1(y) \leq h_2(y)$ for $c \leq x \leq d$ and such that R is the region between the graph of h_1 and h_2 on $[c,d]$. In this case we say that R is the *horizontally simple region* between the graphs of h_1 and h_2 on $[c,d]$.
c. A plane region R is simple if it is both vertically simple and horizontally simple

Example: Let R be the region between the graph of $y = x^2$ and$y = x + 6$.

Show that r is simple.

Solution: First we determine where the two graphs intersect

Observe that

$y = x^2$ and $y = x + 6$ only if

Only if $y = x^2 = x + 6$

Which means that$x = -2 \ \ or \ \ x = 3$. The graph intersects at the point $(-2,4)$ and $(3,9)$. Therefore, if $g_1(x) = x^2$ and $g_2(x) = x + 6$, then if $g_1 \leq g_2$ on $[-2,3]$, so R is the vertically simple region between the graphs of $g_1 \ \ and \ \ g_2$.

To prove R is horizontally simple, we notice that R is composed of two portions, one for which $0 \leq y \leq 4$ and the other for which $4 \leq y \leq 9$

Thus (x,y) is in R provided that either

$0 \leq y \leq 4$ and $-\sqrt{y} \leq x \leq \sqrt{y}$

$4 \leq y \leq 9$ and $y - 6 \leq x \leq \sqrt{y}$

Or consequently

$$h_1(y) = \begin{cases} -\sqrt{y} & for 0 \leq y \leq 4 \\ y - 6 & for\ 4 \leq y \leq 9 \end{cases} \text{ and } h_2(y) = \sqrt{y}$$

Then, $h_1 \leq h_2$ on $[0,9]$, so R is horizontally simple region between the graphs of $h_1$and h_2 on $[0,9]$. Since R is both vertically and horizontally simple, R is simple by definition.

To compute a double integral, we can think of taking thin slices of the solid parallel to the *y*-*z* plane and integrating the function while holding *x* constant to find *A*(*x*) (i.e., perform a partial integration), and then integrating the volume of all the thin slices to find the total volume. This is called an *iterated integral*. Likewise, we could take thin slices parallel to the *x*-*z* plane to find *A*(*y*), and then integrate the volume of those slices to find total volume. In either case, we get the same answer.

Theorem: Let f be continuous on a region R in the xy plane

a. If R is vertically simple region between the graph of g_1 and g_2 on $[a, b]$, then $f$is integrable on R, and

$$\iint_R f(x,y)dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)dxdy$$

b. If R is vertically simple region between the graph of h_1 and h_2 on $[c, d]$, then $f$is integrable on R, and

$$\iint_R f(x,y)dA = \int_a^b \int_{h_1(x)}^{h_2(x)} f(x,y)dydx$$

Example: Let R be a rectangular region bounded by the lines $x = -1, x = 2, y = 0$ and $y = 2$. Find $\iint_R x^2 ydA$

Solution:

The graph of R is vertically simple region between the graphs of

$y = 0$ and $y = 2$ for $-1 \leq x \leq 2$

Therefore, $\iint_R f(x,y)dA = \int_{-1}^{2} \int_0^2 x^2 ydydx$

To evaluate the iterated integral, we first compute $\int_0^2 x^2 ydy$ for each x in$-1 \leq x \leq 2$

We obtain$\int_0^2 x^2 ydy = x^2 \int_0^2 y\, dy = x^2 \left(\frac{1}{2}y^2\right)(2-0) = 2x^2$

Because x is held constant when we integrate with respect to y. we conclude

$$\iint_R f(x,y)dA = \int_{-1}^{2}\int_{0}^{2} x^2 y\,dy\,dx = \int_{-1}^{2} 2x^2\,dx$$

$$= \frac{2}{3}((2)^3 - (-1)^3) = 6$$

Example: Evaluate $I = \iint_R \left(6x + 2y^2\right)dA$

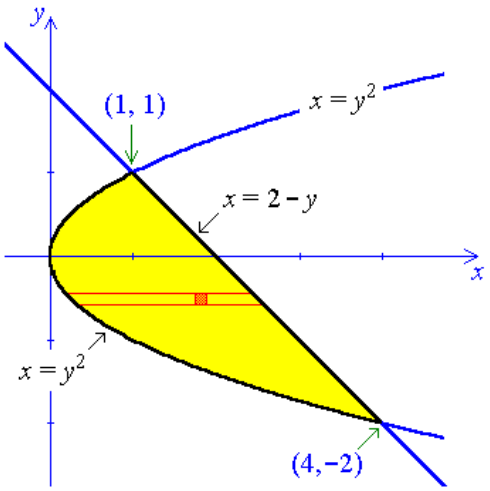
where R is the region enclosed by the parabola $x = y^2$ and the line $x + y = 2$.

Solution:

The upper boundary changes form at $x = 1$.The left boundary is the same throughout R.

The right boundary is the same throughout R.

Therefore choose horizontal strips.



$$I = \int_{-2}^{1}\int_{y^2}^{2-y} \left(6x + 2y^2\right)dx\,dy$$

$$I = \int_{-2}^{1} \left[3x^2 + 2xy^2\right]_{x=y^2}^{x=2-y} dy$$

$$= \int_{-2}^{1} \left(\left(3(2-y)^2 + 2(2-y)y^2\right) - \left(3y^4 + 2y^4\right)\right)dy$$

$$= \int_{-2}^{1} \left(\left(12 - 12y + 3y^2\right) + \left(4y^2 - 2y^3\right) - 5y^4\right)dy$$

$$= \int_{-2}^{1} \left(12 - 12y + 7y^2 - 2y^3 - 5y^4\right) dy$$

$$= \left[12y - 6y^2 + \frac{7}{3}y^3 - \frac{1}{2}y^4 - y^5\right]_{-2}^{+1}$$

$$= \left(12 - 6 + \frac{7}{3} - \frac{1}{2} - 1\right) - \left(-24 - 24 - \frac{56}{3} - 8 + 32\right)$$

Therefore,

$$I = \frac{99}{2}$$

Exercise: Find the volume V of the solid region D bounded above by the paraboloid $z = 4 - x^2 - y^2$ and below by the xy plane (hint $V = \iint_R (4 - x^2 - y^2) dA$)

The area A of a plane region R by

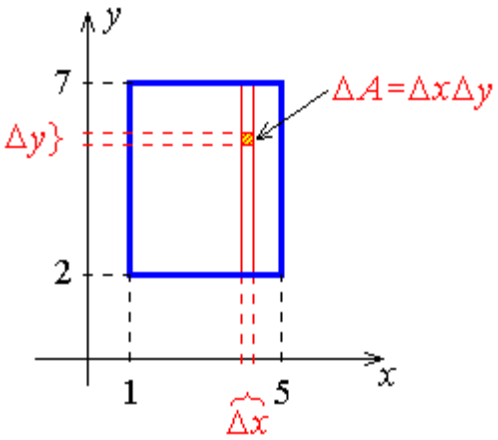
$$A = \iint_R 1 dA$$

Where R is the region between the graphs of two continuous functions g_1 and g_2 ob $[a, b]$ such that $g_1 \leq g_2$

$$A = \iint_R 1 dA = \int_a^b \int_{g_1(x)}^{g_2(x)} 1 dy dx = \int_a^b (g_1 - g_2) dx$$

Example: Find the area shown below

Solution:



$$\text{Area of strip} \approx \left(\sum_{y=2}^{7} \Delta y\right)\Delta x$$

$$\text{Total Area} \approx \sum_{x=1}^{5}\left(\left(\sum_{y=2}^{7} \Delta y\right)\Delta x\right)$$

As $\Delta x \to 0$ and $\Delta y \to 0$, the summations become integrals:

$$(\text{Total Area}) \;\to\; A = \int_{x=1}^{x=5}\left(\int_{y=2}^{y=7} 1\,dy\right)dx$$

The inner integral has no dependency at all on *x*, in its limits or in its integrand.

It can therefore be extracted as a "constant" factor from inside the outer integral.

$$\Rightarrow \quad A = \left(\int_{y=2}^{y=7} 1\,dy\right)\int_{x=1}^{x=5} 1\,dx$$

$$= [y]_2^7[x]_1^5 = (7-2)\times(5-1) = 5\times 4 = \underline{\underline{20\,\text{m}^2}}$$

Example: Suppose that the surface density on the rectangle is $\sigma = x^2 y$. Find the mass of the rectangle.

The element of mass is

$$\Delta m = \sigma \Delta A = \sigma \Delta x\, \Delta y$$

$$\to \quad m = \int_1^5\int_2^7 \sigma\, dy\, dx = \int_1^5\int_2^7 x^2 y\, dy\, dx$$

$$= \int_1^5 x^2 \left(\int_2^7 y\, dy \right) dx = \left(\int_2^7 y\, dy \right) \int_1^5 x^2\, dx$$

$$= \left[\frac{y^2}{2} \right]_2^7 \cdot \left[\frac{x^3}{3} \right]_1^5 = \frac{49-4}{2} \times \frac{125-1}{3} = 15 \times 62$$

Therefore, the mass of the rectangle is m = 930 kg

Theorem: Linear Combinations of Double Integrals

Let the function $f(x, y)$ and $g(x, y)$ be integrable over the region $R \in \mathbb{R}^2$, and let c be any constant. Then the following hold:

i. $\iint_R cf(x,y)\,dA = c\iint_R f(x,y)\,dA$

ii. $\iint_R \left[f(x,y) + g(x,y) \right] dA = \iint_R f(x,y)\,dA + \iint_R g(x,y)\,dA$

iii. if $R = R_1 \cup R_2$, where R_1 and R_2 are non overlapping regions, then

$$\iint_R f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA .$$

Exercise:

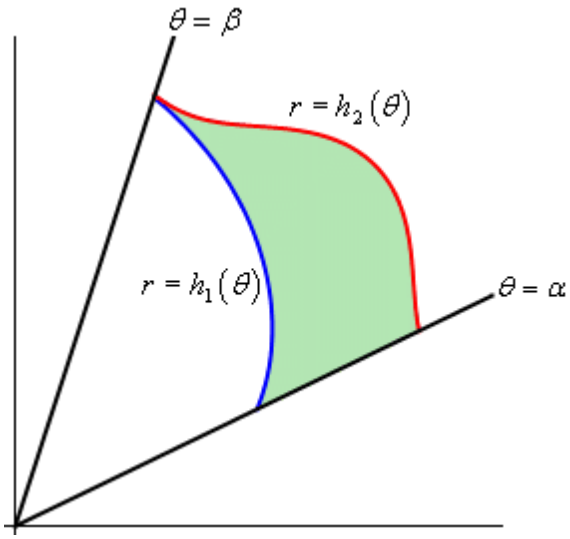
1. Compute the double integral of the function $f(x,y) = x^2 + y^2$ over the region bounded by the curves $y = 1 - x^2$ and $y = x^2 - 1$ in the x-y plane.
2. Compute the double integral of the function $f(x,y) = 6 - x + 2y$ over the region bounded by the curves $x = y^2$ and $y = 2 - x$ in the x-y plane

5.2 Double Integrals in Polar coordinates

To this point we have seen quite a few double integrals. However, in every case we've seen to this point the region R could be easily described in terms of simple functions in Cartesian coordinates. In this section we want to look at some regions that are much easier to describe in terms of polar coordinates. For instance, we might have a region that is a disk, ring, or a portion of a disk or ring. In these cases using Cartesian coordinates could be somewhat cumbersome.

So, if we could convert our double integral formula into one involving polar coordinates we would be in pretty good shape. The problem is that we can't just convert the dx and the dy into a

dr and a $d\theta$. In computing double integrals to this point we have been using the fact that $dA = dxdy$ and this really does require Cartesian coordinates to use. Once we've moved into polar coordinates $dA \neq dr\ d\theta$ and so we're going to need to determine just what dA is under polar coordinates. After some manipulations it is found that $dA = rdrd\theta$



Theorem: Suppose that h_1 and h_2 are continuous on $[\alpha, \beta]$, where $0 \leq \beta - \alpha \leq 2\pi$, and that

$0 \leq h_1(\theta) \leq h_2(\theta)$ for $\alpha \leq \theta \leq \beta$, Let R be the region between the polar graphs of

$r = h_1(\theta)$ and $r = h_2(\theta)$ for $\alpha \leq \theta \leq \beta$

If f is continuous on R, then

$$\iint_R f(x,y)dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta) r dr d\theta$$

In the event f is nonnegative on R, the volume V of the region between the graph of f and R is given by

$$V = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta) r\,dr\,d\theta$$

and the area A of R is given by

$$A = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} r\,dr\,d\theta$$

Example: Evaluate the integral by converting into polar coordinates.

$\iint_R 2xy\,dA$, R is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant.

Solution:

$\iint_R 2xy\,dA$, R is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant.

First let's get *R* in terms of polar coordinates. The circle of radius 2 is given by $r = 2$ and the circle of radius 5 is given by $r = 5$. We want the region between them so we will have the following inequality for *r*.

$$2 \le r \le 5$$

Also, since we only want the portion that is in the first quadrant we get the following range of

θ 's.

$$0 \le \theta \le \frac{\pi}{2}$$

Now that we've got these we can do the integral.

$$\begin{aligned}
\iint_R 2xy\,dA &= \int_0^{\frac{\pi}{2}} \int_2^5 2(r\cos\theta)(r\sin\theta) r\,dr\,d\theta \\
&= \int_0^{\frac{\pi}{2}} \int_2^5 r^3 \sin(2\theta)\,dr\,d\theta \\
&= \int_2^{\frac{\pi}{2}} \frac{1}{4} r^4 \sin(2\theta) \Big/_2^5 d\theta \\
&= \frac{-609}{8} \cos(2\theta) \Big/_0^{\frac{\pi}{2}} \\
&= \frac{609}{4}
\end{aligned}$$

Example: Let D be the solid region bounded above by the paraboloid $z = 4 - x^2 - y^2$ and below by the xy plane. Find the volume V of D.

Solution: The region R over which the integral is to be taken is bounded by the circle $x^2 + y^2 = 4$(Intersection of the surface with xy plane), whose equation in polar coordinate is r=2: therefore D can be described as the region between the graph of the paraboloid and the disk$x^2 + y^2 \leq 4$,

As a result

$$V = \iint_R (4 - x^2 - y^2)dA = \int_0^{2\pi} \int_0^2 (4 - r^2) r dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 (4r - r^3) dr d\theta = \int_0^{2\pi} (2r^2 - \frac{r^4}{4}) d\theta$$

$$= \int_0^{2\pi} 4 d\theta = 8\pi$$

Exercise:

1. Determine the volume of the region that lies under the sphere $x^2 + y^2 + z^2 = 9$, above the plane $z = 0$ and inside the cylinder $x^2 + y^2$ =5.
2. Determine the area of the region that lies inside $r = 3 + 2sin\theta$ and outside r=2.
3. Find the volume of the region that lies inside de $z = x^2 + y^2$and below the plane $z = 16$.

5.3 Surface Area

Definition**:** Let R be a vertically or horizontally simple region, and let f have continuous partial derivatives on R. If Σ is the graph of f on R, then the surface area **S** of Σ is defined by

$$S = \iint_R \sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}\,dA$$

Example: Find the surface area of that portion of the surface $z = f(x,y) = \sqrt{4 - x^2}$ that lies above the rectangle R in the xy plane whose coordinates satisfy $0 \leq x \leq 1$ and $0 \leq y \leq 4$

Solution:

the surface is a portion of the cylinder $x^2 + z^2 = 4$ It follows From the formula that the surface area is

$$S = \iint_R \sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}\,dA$$

Where

$$f_x(x,y) = \frac{-x}{\sqrt{4-x^2}} \qquad f_y = 0 \quad \text{for } (x,y) \text{ in R}$$

then $$S = \iint_R \sqrt{(\frac{-x}{\sqrt{4-x^2}})^2 + (0)^2 + 1}\,dA$$

$$= \int_0^4 \int_0^1 \frac{2}{\sqrt{4-x^2}}\,dx\,dy$$

$$= 2\int_0^4 \left[sin^{-1}\left(\frac{1}{2}x\right)\right]_0^1 dy = 2\int_0^4 \frac{\pi}{6}dy = \frac{4\pi}{3}$$

Example: Find the surface area of the portion of the paraboloid $x^2 + y^2 = z$ below the plane $z = 1$.

Solution:

The trace of the paraboloid $z = x^2 + y^2$ in the plane $z = 1$ projects on to a circle

$x^2 + y^2 = 1$ in the xy plane , and the portion of the paraboloid that lies below the plane $z = 1$ projects onto the region R that is enclosed by this circle. Thus, it follows from formula that the surface area is

$$S = \iint_R \sqrt{(f_x(x,y))^2 + (f_y(x,y))^2 + 1}\,dA$$

Where

$$f_x(x,y) = 2x \qquad f_y = 2y \quad \text{for } (x,y) \text{ in R}$$

Then $$S = \iint_R \sqrt{(2x)^2 + (2y)^2 + 1}\,dA$$

The expression $4x^2 + 4y^2 + 1 = 4(x^2 + y^2) + 1$ in the integrand suggests that we evaluate the integral in polar coordinates. We substituted $x = r\cos\theta$ and $y = r\sin\theta$ in the integrand, replace dA by $rdrd\theta$, and find the limits of integration by expressing the region R in polar coordinates. This yield

$$\begin{aligned} S &= \int_0^{2\pi}\int_0^1 \sqrt{4r^2 + 1}\,r\,dr\,d\theta \\ &= \int_0^{2\pi}\left[\frac{1}{12}(4r^2 + 1)^{3/2}\right]_0^1 d\theta \\ &= \int_0^{2\pi}\left[\frac{1}{12}(5\sqrt{5} - 1)\right] d\theta \\ &= \frac{1}{6}\pi(5\sqrt{5} - 1) \end{aligned}$$

Example: Let R be the rectangular region bounded by the lines

$x = 0, x = 3, \quad y = 0, \ y = 2$

and let $f(xy) = \frac{2}{3}x^{3/2}$. Find the surface area S of the portion of the graph of f that lies over R

Solution: First let us find the partial derivatives

$$f_x(x,y) = x^{1/2} \qquad f_y = 0 \quad \text{for } (x,y) \text{ in R}$$

Then, using the formula

$$S = \iint_R \sqrt{(x^{1/2})^2 + (0)^2 + 1}\,dA$$

$$S = \int_0^3 \int_0^2 \sqrt{x^2 + 1}\,dy\,dx$$

$$S = 2\int_0^3 \sqrt{x^2 + 1}\,dx = \frac{4}{3}(x+1)^{3/2}\Big|_0^3 = \frac{28}{3}$$

5.4 Triple Integrals in Cartesian, Cylindrical and Spherical Coordinates

5.4.1 Triple Integrals in Cartesian coordinates

Now that we know how to integrate over a two-dimensional region we need to move on to integrating over a three-dimensional region. We used a double integral to integrate over a two dimensional region and so it shouldn't be too surprising that we'll use a triple **integral** to integrate over a three dimensional region. The notation for the general triple integrals is,

$$\iiint_D f(x,y,z)\,dV$$

Let's start simple by integrating over the box,

$D = [a,b]x[c,d]x[r,s]$

Note that when using this notation we list the *x*'s first, the *y*'s second and the *z*'s third. The triple integral in this case is,

$$\iiint_D f(x,y,z)\,dV = \int_r^s \int_c^d \int_a^b f(x,y,z)\,dx\,dy\,dz$$

Note that we integrated with respect to *x* first, then *y*, and finally *z* here, but in fact there is no reason to the integrals in this order. There are 6 different possible orders to do the integral in and which order you do the integral in will depend upon the function and the

order that you feel will be the easiest. We will get the same answer regardless of the order however.

Example: Evaluate the following integral.

$$\iiint_D 8xyzdV,$$

$D = [2,3]x[1,2]x[0,1]$

Solution:

Just to make the point that order doesn't matter let's use a different order from that listed above.

We'll do the integral in the following order.

$$\iiint_D 8xyzdV = \int_1^2 \int_2^3 \int_0^1 8xyzdzdxdy$$

$$= \int_1^2 \int_2^3 [4xyz]_0^1 dxdy$$

$$= \int_1^2 \int_2^3 4xydxdy$$

$$= \int_1^2 [2x^2y]_2^3 dy$$

$$= \int_1^2 10ydy = 15$$

Theorem: Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on vertically or horizontally simple region R in the xy plane, and let f be continuous on D. Then

$$\iiint_D f(x,y,z)dV = \iint_R \left(\int_{F_1(x,y)}^{F_2(x,y)} f(x,y,z)dz\right)dA$$

We evaluate $\int_{F_1(x,y)}^{F_2(x,y)} f(x,y,z)dz$ by integrating with respect to z while both x and y are held fixed, thus obtaining a number depending on x and y. If is the vertically simple region between the graphs of $g_1(x)$ and $g_2(x)$ on $[a,b]$, we evaluate the double integral over R by using

$$\iiint_D f(x,y,z)dV = \int_a^b [\int_{g_1(x)}^{g_2(x)} (\int_{F_1(x,y)}^{F_2(x,y)} f(x,y,z)dz)dy)]dx$$

In contrast, if R is the horizontally simple region between the graphs of $h_1(y)$ and $h_2(y)$ on $[a,b]$, we evaluate the double integral over R by using

$$\iiint_D f(x,y,z)dV = \int_c^d [\int_{h_1(y)}^{h_2(y)} (\int_{F_1(x,y)}^{F_2(x,y)} f(x,y,z)dz)dx)]dy$$

Example: Let R be the triangular region in the xy plane between the graphs of $y = 0$ and $y = x$ for $0 \le x \le 1$, and let D be the solid region between the graphs of the surfaces $z = -y^2$ and $z = x^2$ for all (x,y) in R

Evaluate $\iiint_D (x+1)dV$

Solution: By the above discussions we have

$$\iiint_D (x+1)dV = \int_0^1 [\int_0^x (\int_{-y^2}^{x^2} (x+1)dz)dx)]dy$$

$$= \int_0^1 [\int_0^x ((x+1)z\Big|_{-y^2}^{x^2} dydx$$

$$= \int_0^1 [\int_0^x ((x+1)(x^2+y^2)dydx$$

$$= \int_0^1 x^3y + x^2y + \frac{y^3}{3})\Big|_0^x dx$$

$$\frac{4}{3}\int_0^1 (x^4 + x^3)\ dx = \frac{3}{5}$$

5.4.1.1 Volume calculated as a triple integral

Triple integrals have many physical interpretations, Some of which we will consider in the next section. However, in the special case where $f(x, y, z) = 1$ the general formula yields

$$V = \iiint_D \ 1dV, \text{ volume of the solid}$$

Example: Use triple integral to find the volume of the solid with in the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$

Solution: The lower surface of the solid is the plane $z = 1$ and the upper surface is the plane

$x + z = 5$ or, equivalently $z = 5 - x$ thus

$$V = \iiint_D \ 1dV = \iint_R \ [\int_1^{5-x} dz]dA$$

For the double integral over R, we will integrate with respect to y first.

$$V = \int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dzdydx = \int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} [z]\Big|_1^{5-x} dydx$$

$$= \int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} [4-x]\, dydx = \int_{-3}^{3} (4-x)[y]\Big|_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-3}^{3} (8 - 2x\sqrt{9-x^2})dx$$

$$= 8\int_{-3}^{3} \sqrt{9-x^2})dx - \int_{-3}^{3} 2x\sqrt{9-x^2})dx$$

$$= 36\pi$$

Example: Find the volume of the solid enclosed between the paraboloids

$$z = 5x^2 + 5y^2 \text{ and } z = 6 - 7x^2 + y^2$$

Solution: The projection of D on the xy plane R is obtained by solving the given equations simultaneously to determine where the paraboloids intersect. We obtain

$$5x^2 + 5y^2 = 6 - 7x^2 + y^2$$

Or $2x^2 + y^2 = 1$ (*)

which tells us that the paraboloids intersect in a curve on the elliptic cylinder given by (*).

The projection of this intersection on the xy plane is an ellipse with this same equation. Therefore,

Volume $V = V = \iiint_D 1dV = \iint_R [\int_{5x^2+5y^2}^{6-7x^2+y^2} dz]dA$

$$= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{5x^2+5y^2}^{6-7x^2+y^2} dzdydx$$

$$= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (6 - 12x^2 - 6y^2)dydx$$

$$= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} 6(1 - 2x^2)y - 2y^3] \Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dx$$

$$= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1 - 2x^2)^{3/2}dx = \frac{8}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} cos^4\theta d\theta$$

(Letting $x = {}^{1}/_{\sqrt{2}} sin\theta$)

$$= \frac{3\pi}{2}$$

Exercise:

1. Let D be the solid region in the first octant between the graphs of $z = x^2 + 2y + 1$ and $z = y + 2$

   Find the volume V of the solid

2. Find the volume V of the solid bounded by the surface $z = \sqrt{y}$ and the planes

$$x + y = 1, x = 0, z = 0 \text{and}$$

3. Find the volume V of the solid bounded by the surface $y = x^2$ and the planes

$$x + z = 4, \quad \text{and } , z = 0$$

5.4.1.2 Triple integral in Cylindrical Coordinates

In this section we want do take a look at triple integrals done completely in Cylindrical Coordinates. Recall that cylindrical coordinates are really nothing more than an extension of polar coordinates into three dimensions. The following are the conversion formulas for cylindrical coordinates.

$$x = rcos\theta \qquad y = rsin\theta \qquad z = z$$

In order to do the integral in cylindrical coordinates we will need to know what dV will become in terms of cylindrical coordinates. We will be able to show in the Change of Variables section of this chapter that,

$$dV = rdzdrd\theta$$

Theorem: Let D be the solid region between the graphs of two continuous functions F_1 and F_2 on region R, where R is the plane region between the polar graphs of $h_1 \ and \ h_2$ on $[\alpha, \beta]$, with $0 \le \beta - \alpha \le 2\pi \quad and \quad h_1(\theta) \le h_2(\theta)$ for a $\alpha \le \theta \le \beta$. If f be continuous on D. Then

$$\iiint_D f(x,y,z)dV = \int_\alpha^\beta [\int_{h_1(\theta)}^{h_2(\theta)} (\int_{F_1(rcos\theta\ ,rsin\theta)}^{F_2(rcos\theta\ ,rsin\theta)} rf(rcos\theta, rsin\theta, z)dz)dr)]d\theta$$

Integration by means of cylindrical coordinate is especially effective when expressions containing $x^2 + y^2$ appears in the integrand or in the limits of integration and the region over which the integration taken is easily described by polar coordinates.

Example: Evaluate $\iiint_D \ ydV$ where *D* is the region that lies below the plane $z = x + 2$ above the xy-plane and between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$

Solution:

There really isn't too much to do with this one other than do the conversions and then evaluate the integral.

We'll start out by getting the range for *z* in terms of cylindrical coordinates.

$$0 \le z \le x + 2 \qquad \text{implies} \qquad 0 \le z \le r\cos\theta + 2$$

Remember that we are above the xy −plane and so we are above the plane *z* = 0

Next, the region *D* is the region between the two circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ in the xy plane and so the ranges for it are,

$$0 \le \theta \le 2\pi \qquad\qquad 1 \le r \le 2$$

Here is the integral

$$\iiint_D \, ydV = \int_0^{2\pi} \int_1^2 \int_0^{r\cos\theta + 2} (r\sin\theta) r\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_1^2 r^2 \sin\theta\,(r\cos\theta + 2)\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_1^2 \left(\frac{1}{2} r^3 \sin(2\theta) + 2r^2 \sin\theta\right) dr\,d\theta$$

$$= \int_0^{2\pi} \left(\frac{1}{8} r^4 \sin(2\theta) + \frac{2}{3} r^3 \sin\theta\right)\Big|_1^2 d\theta$$

$$= \int_0^{2\pi} \left(\frac{15}{8} \sin(2\theta) + \frac{14}{3} r^3 \sin\theta\right) d\theta$$

$$= \left(\frac{-15}{8} \cos(2\theta) - \frac{14}{3} r^3 \cos\theta\right)\Big|_0^{2\pi}$$

$$= 0$$

Example: Covert $\int_{-1}^{1} \int_{0}^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz\,dz\,dx\,dy$ into an integral in cylindrical coordinates.

Solution:

Here are the ranges of the variables from this iterated integral.

$$-1 \le y \le 1$$

$$0 \le x \le \sqrt{1-y^2}$$

$$x^2 + y^2 \le z \le \sqrt{x^2 + y^2}$$

The first two inequalities define the region *R* and since the upper and lower bounds for the *x*'s are $x = \sqrt{1-y^2}$ and $x = 0$ we know that we've got at least part of the right half a circle of radius 1 centered at the origin. Since the range of *y*'s is $-1 \le y \le 1$ we know that we have the complete right half of the disk of radius 1 centered at the origin. So, the ranges for *R* in cylindrical coordinates are,

$$\frac{-\pi}{2} \le \theta \le \frac{\pi}{2}$$

$$0 \le r \le 1$$

All that's left to do now is to convert the limits of the *z* range, but that's not too bad.

$$r^2 \le z \le r$$

On a side note notice that the lower bound here is an elliptic paraboloid and the upper bound is a cone. Therefore *D* is a portion of the region between these two surfaces.
The integral is,

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-y^2}}\int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz\,dz\,dx\,dy = \int_{-\pi/2}^{\pi/2}\int_{0}^{1}\int_{r^2}^{r} r(r\cos\theta)(r\sin\theta)z\,dz\,dr\,d\theta$$

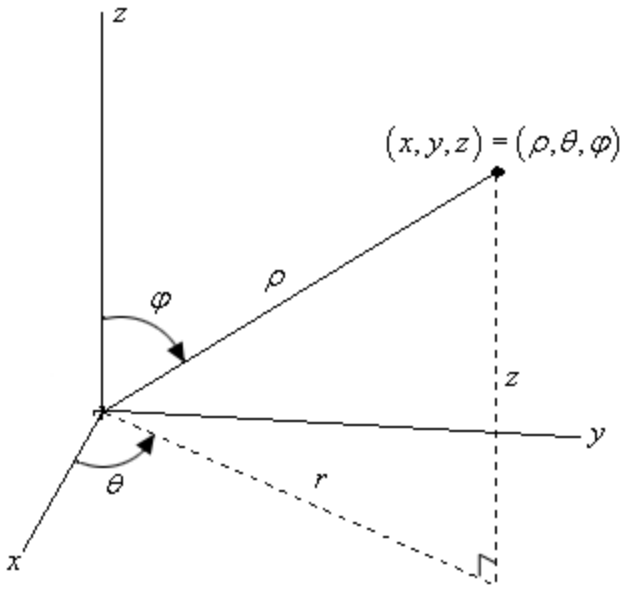
$$= \int_{-\pi/2}^{\pi/2}\int_{0}^{1}\int_{r^2}^{r} r^3 z\cos\theta\sin\theta\,dz\,dr\,d\theta$$

Exercise: Verify the formula $V = \frac{4}{3}\pi a^3$ for the volume of a sphere of radius *a*

5.4.1.3Triple integral in Spherical Coordinates

In the previous section we looked at doing integrals in terms of cylindrical coordinates and we now need to take a quick look at doing integrals in terms of spherical coordinates.

First, we need to recall just how spherical coordinates are defined. The following sketch shows the relationship between the Cartesian and spherical coordinate systems.



Here are the conversion formulas for spherical coordinates.

$x = \rho \sin\varphi \cos\theta \qquad y = \rho \sin\varphi \sin\theta \qquad z = \rho\cos\varphi \qquad x^2 + y^2 + z^2 = \rho^2$

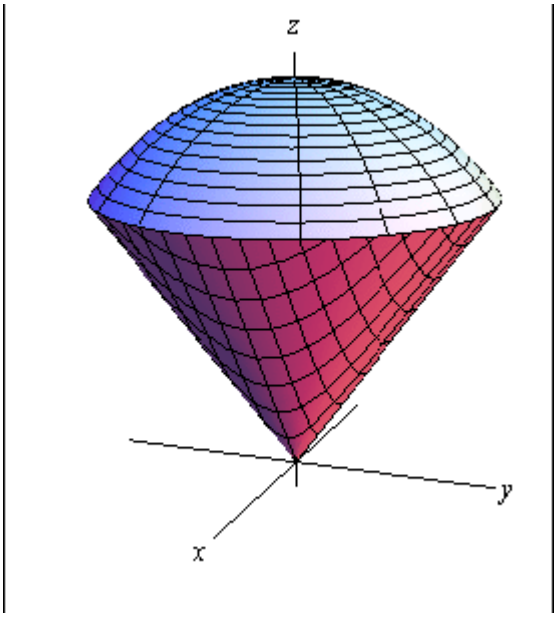
We also have the following restrictions on the coordinates.

$$\rho \geq 0 \qquad 0 \leq \varphi \leq \pi$$

For our integrals we are going to restrict *D* down to a spherical wedge. This will mean that we are going to take ranges for the variables as follows,

$$a \leq \rho \leq b$$
$$\alpha \leq \theta \leq \beta$$
$$\delta \leq \varphi \leq \gamma$$

Here is a sketch of a spherical wedge in which the lower limit for both ρ and φ are zero for reference purposes. Most of the wedges we'll be working with will fit into this pattern.



From this sketch we can see that *E* is really nothing more than the intersection of a sphere and a cone.

Theorem: Let α and β be real numbers with $\alpha \leq \beta \leq \alpha + 2\pi$. Let $h_1, h_2, \mathrm{F}_1,$ and F_2 be continuous functions with $0 \leq h_1 \leq h_2 \leq \pi$ and $0 \leq \mathrm{F}_1 \leq \mathrm{F}_2$.Let D be the solid region consisting of all the points in space whose spherical coordinates (ρ, φ, θ)ssatisfy

$$\alpha \leq \varphi \leq \beta$$

$$h_1(\varphi) \leq \theta \leq h_2(\varphi)$$

$$\mathrm{F}_1(\theta, \varphi) \leq \rho \leq \mathrm{F}_2(\theta, \varphi)$$

If f is continuous in D, then

$$\iiint_D f(x,y,z)dV =$$
$$\int_\alpha^\beta \int_{h_1(\varphi)}^{h_2(\varphi)} \int_{F_1(\theta,\varphi)}^{F_2(\theta,\varphi)} \rho^2 sin\theta f(\rho sin\varphi\, cos\theta, \rho sin\varphi\, sin\theta, \rho cos\varphi)\, d\rho d\theta d\varphi$$

Example: Evaluate $\iiint_D 16z\,dV$ where *D* is the upper half of the sphere $x^2 + y^2 + z^2 = 1$

Solution:

Since we are taking the upper half of the sphere the limits for the variables are,

$$0 \le \rho \le 1$$
$$0 \le \theta \le 2\pi$$
$$0 \le \varphi \le \frac{\pi}{2}$$

The integral is then,

$$\begin{aligned}\iiint_D 16z\,dV &= \int_0^{\pi/2}\int_0^{2\pi}\int_0^1 \rho^3 sin\varphi(16\rho cos\varphi)\,d\rho\,d\theta d\varphi \\ &= \int_0^{\pi/2}\int_0^{2\pi}\int_0^1 8\rho^3 \sin(2\varphi)\,d\rho\,d\theta d\varphi \\ &= \int_0^{\pi/2}\int_0^{2\pi} 2\sin(2\varphi)\,d\theta d\varphi \\ &= \int_0^{\pi/2} 4\pi\sin(2\varphi)\,d\varphi \\ &= -2\pi\cos(2\varphi)\Big|_0^{\pi/2} \\ &= 4\pi\end{aligned}$$

Example: Convert $\int_0^3\int_0^{\sqrt{9-y^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}}(x^2+y^2+z^2)\,dz\,dx\,dy$ into spherical coordinates.

Solution:

Let's first write down the limits for the variables.

$$0 \le y \le 3$$

$$0 \le x \le \sqrt{9-y^2}$$

$$\sqrt{x^2+y^2} \le z \le \sqrt{18-x^2-y^2}$$

The range for *x* tells us that we have a portion of the right half of a disk of radius 3 centered at the origin. Since we are restricting *y*'s to positive values it looks like we will

have the quarter disk in the first quadrant. Therefore since *R* is in the first quadrant the region, *D*, must be in the first octant and this in turn tells us that we have the following range for θ (since this is the angle around the *z*-axis).

$$0 \le \theta \le \frac{\pi}{2}$$

Now, let's see what the range for *z* tells us. The lower bound, $z = \sqrt{x^2 + y^2}$ is the upper half of a cone. At this point we don't need this quite yet, but we will later. The upper bound,

$z = \sqrt{18 - x^2 - y^2}$, is the upper half of the sphere,

$$x^2 + y^2 + z^2 = 18$$

and so from this we now have the following range for ρ

$$0 \le \rho \le \sqrt{18} = 3\sqrt{2}$$

Now all that we need is the range for φ . There are two ways to get this. One is from where the cone and the sphere intersect. Plugging in the equation for the cone into the sphere gives,

$$\begin{aligned}(\sqrt{x^2 + y^2})^2 + z^2 &= 18\\ z^2 + z^2 &= 18\\ z &= 9\end{aligned}$$

Note that we can assume *z* is positive here since we know that we have the upper half of the cone and/or sphere. Finally, plug this into the conversion for *z* and take advantage of the fact that we know that $\rho = 3\sqrt{2}$ since we are intersecting on the sphere. This gives,

$$\begin{aligned}\rho\cos\varphi &= 3\\ 3\sqrt{2}\cos\varphi &= 3\\ \cos\varphi &= \frac{1}{\sqrt{2}} \qquad \text{implies } \varphi = \frac{\pi}{4}\end{aligned}$$

So, it looks like we have the following range,

$$0 \le \varphi \le \frac{\pi}{4}$$

So, recalling that $\rho^2 = x^2 + y^2 + z^2$, the integral is then,

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} (x^2 + y^2 + z^2)\,dz\,dx\,dy = \int_0^{\pi/4} \int_0^{\pi/2} \int_0^{3\sqrt{2}} (\rho^4)\,sin\varphi\,d\rho\,d\theta\,d\varphi$$

Exercise:

1. Let D be the solid region between the spheres $\rho = 1 \quad and \quad \rho = 2$.

   Evaluate $\iiint_D z^2\,dV$

2. Find the volume V of the solid region D between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 9$, and above the upper nappe of the cone $z^2 = 3(x^2 + y^2)$

References:

1) Calculus with Analytic Geometry, third edition, Robert Ellis and Denny Gulik
2) Advanced Engineering Mathematics, Fifth edition, Peter V.O'Neil.
3) Advanced Engineering Mathematics, Fifth edition, Erwin Kreyszing
4) Calculus with Analytic Geometry, third edition, Stewart
5) Calculus complete I,II,III, Paul Dawkins
6) Calculus, sixth edition, Howard Anton